



Hybrid ARIMAX-LSTM Model for Predicting Price Volatility of Vietnam's Rice Exports

Bachelor of International Business Thesis

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Although every effort has been made to ensure the accuracy and academic quality of this thesis, certain limitations remain due to data availability, methodological assumptions, and the uncertainty inherent in forecasting.

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Authors of the Thesis.

EXECUTIVE SUMMARY

Rice is one of the world's most important agricultural commodities, and fluctuations in rice prices can affect food security, international trade, producer income, and market decision-making. As a major rice-exporting country, Vietnam faces increasing challenges in anticipating export price movements under changing international market conditions. Therefore, this study develops and evaluates a Hybrid ARIMAX-LSTM framework for forecasting the FOB price of Vietnam's 5% broken white rice.

The study employs 268 monthly observations of Vietnam's and Thailand's 5% broken white rice FOB prices from December 2003 to March 2026. Vietnam's rice price (VNRICE) serves as the target variable, while Thailand's rice price (THRICE) is the sole exogenous variable. ARIMAX first captures the linear dynamics of Vietnam's rice price and predictive information from the Thai benchmark price. Its residuals are then modeled using LSTM as a nonlinear correction component, and both components are recombined to produce the final forecast.

The results show that both price series are non-stationary at level but become stationary after first-order differencing. ARIMAX (1,1,1) is selected as the baseline model, while Thailand's rice price provides statistically significant predictive information for changes in Vietnam's rice price. Residual diagnostics indicate no significant remaining serial autocorrelation, supporting the use of ARIMAX in the subsequent hybrid procedure.

Forecasting performance is evaluated over a 38-month out-of-sample period using MAE, RMSE, MAPE, and SMAPE, with additional comparison against Random Walk benchmarks. ARIMAX records an MAE of 11.873 USD/ton, RMSE of 15.759 USD/ton, MAPE of 2.488%, and SMAPE of 2.483%. The Hybrid ARIMAX-LSTM model reduces these errors to 11.352 USD/ton, 15.611 USD/ton, 2.381%, and 2.388%, respectively, corresponding to improvements of approximately 4.38%, 0.94%, 4.32%, and 3.86%. Both ARIMAX-based models also outperform the Random Walk benchmarks, indicating that historical price dynamics and Thailand's benchmark price provide useful forecasting information beyond a simple persistence-based forecast.

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ACF	Autocorrelation Function
ADF	Augmented Dickey-Fuller Test
AI	Artificial Intelligence
AIC	Akaike Information Criterion
ANN	Artificial Neural Network
AR	Autoregressive
ARIMA	Autoregressive Integrated Moving Average
ARIMAX	Autoregressive Integrated Moving Average with Exogenous Variables
BIC	Bayesian Information Criterion
DL	Deep Learning
EU	European Union
FAO	Food and Agriculture Organization of the United Nations
FOB	Free on Board
GARCH	Generalized Autoregressive Conditional Heteroskedasticity
GBM	Gradient Boosting Machine
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
MA	Moving Average
MAE	Mean Absolute Error
MAPE	Mean Absolute Percentage Error

Abbreviation	Full Form
SMAPE	Symmetric Mean Absolute Percentage Error
ML	Machine Learning
MLE	Maximum Likelihood Estimation
PACF	Partial Autocorrelation Function
RF	Random Forest
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
SARIMAX	Seasonal Autoregressive Integrated Moving Average with Exogenous Variables
SVM	Support Vector Machine
THRICE	Thailand's 5% Broken White Rice FOB Price
USDA	United States Department of Agriculture
USDA FAS	United States Department of Agriculture - Foreign Agricultural Service
USD	United States Dollar
VAR	Vector Autoregression
VECM	Vector Error Correction Model
VFA	Vietnam Food Association
VNRICE	Vietnam's 5% Broken White Rice FOB Price
WB	World Bank
WTO	World Trade Organization
XGBoost	Extreme Gradient Boosting

CHAPTER 1. INTRODUCTION

This chapter establishes the overall foundation of the study by presenting the research background and highlighting the growing importance of agricultural price forecasting in the context of increasing volatility in international rice markets. It emphasizes the strategic role of Vietnam's rice exports and the need for more accurate forecasting approaches capable of capturing both linear and nonlinear price dynamics. Based on the identified research gap, the chapter outlines the research objectives, research questions, scope, and significance of the study. In addition, it introduces the rationale for integrating ARIMAX with LSTM to forecast Vietnam's 5% broken white rice export prices, thereby establishing a clear direction for the theoretical, methodological, and empirical analyses developed in the subsequent chapters.

1.1. Background of Research

In recent years, agricultural price forecasting has increasingly become an important topic in economic research, international trade, and supply chain management due to the essential role of agricultural commodities in ensuring food security and promoting global economic growth (FAO, 2004; Sun et al., 2023). Agricultural prices directly affect farmers' income, production costs, food inflation, import-export activities, and the policy-making process of countries that heavily depend on agriculture (FAO, 2004). In the context of increasingly volatile global markets influenced by climate change, geopolitical conflicts, logistics disruptions, and protectionist trade policies, the need to accurately forecast agricultural prices and understand their fluctuations has become increasingly important for governments, businesses, and stakeholders across the supply chain (Sun et al., 2023; Theofilou et al., 2025).

Along with the development of big data, artificial intelligence, and computational capabilities, research on agricultural price forecasting worldwide has shifted significantly from traditional statistical models to machine learning, deep learning, and hybrid models. According to Sun et al. (2023), current forecasting studies can generally be classified into three main groups: traditional models, intelligent models, and hybrid models. Among these, recent research trends have focused on integrating multiple models to improve forecasting performance in nonlinear and highly volatile data environments.

Previously, many studies mainly employed traditional time-series and econometric models such as ARIMA, ARIMAX, VAR, and GARCH due to their advantages in economic interpretability and their ability to effectively capture linear trends in data (Hyndman and Athanasopoulos, 2021). However, these models often face limitations in capturing nonlinear relationships, structural breaks, and abnormal market shocks (Sun et al., 2023; Nayak et al., 2024). In contrast, deep learning models such as Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) demonstrate stronger capabilities in learning long-term dependencies and nonlinear structures in time-series data, particularly in highly volatile environments (Sun et al., 2023; Manogna et al., 2025). Nevertheless, deep learning models generally provide lower economic interpretability compared to traditional econometric models. Therefore, recent research trends have increasingly shifted toward hybrid frameworks that combine the strengths of econometric models and deep learning architectures (Sun et al., 2023; Mahawan et al., 2022; Anggraeni et al., 2019).

In global agricultural trade, rice continues to play a strategic role in food security and international commerce. According to the Food and Agriculture Organization of the United Nations (FAO), rice is the staple food for more than half of the world's population and, together with wheat and maize, provides approximately 60% of total global food energy (FAO, 2004). Therefore, fluctuations in rice prices not only affect trade activities but also have significant impacts on food inflation, social stability, and food security policies in many countries.

For Vietnam, rice remains one of the country's strategic agricultural export commodities, playing a significant role in economic growth and international trade. According to the WTO Center VCCI (2024), Vietnam's rice exports in 2024 reached record highs in both export volume and export value, reflecting the increasingly important role of the rice sector in international agricultural trade. However, the international rice market in recent years has experienced substantial volatility due to the impacts of the COVID-19 pandemic, global logistics disruptions, and export policies implemented by competing countries. In particular, India's ban on non-basmati white rice exports in 2023 caused Asian rice prices to surge sharply and forced many Vietnamese exporters to renegotiate and adjust their selling prices (Reuters, 2023). Meanwhile, India's easing of export restrictions in 2024 led to a significant decline in Asian rice prices and intensified competitive pressure in the international market

(WTO Center VCCI, 2024). These fluctuations indicate that the rice export market has become increasingly difficult to forecast and more uncertain than in previous years (World Bank, 2026).

In this study, the research group focuses on Vietnam's 5% broken white rice price because it is a benchmark price widely used in international rice trade and regularly monitored by the World Bank and the USDA (World Bank, 2026; USDA FAS, 2026). In addition, Thailand's 5% broken white rice price is considered an important exogenous variable due to its direct competitive relationship with Vietnam's rice export prices in the international market (USDA FAS, 2026; Mahawan et al., 2022).

Price volatility is a widely recognized characteristic of agricultural commodity markets, where prices may exhibit substantial fluctuations over time in response to changes in supply and demand conditions, production shocks, trade policies, and broader market uncertainty (Gilbert and Morgan, 2010; Sun et al., 2023). In the agricultural price literature, the term is often used to describe the degree of observed price fluctuation over time, although in financial econometrics volatility may also be defined more formally in terms of time-varying conditional variance. Within the scope of this study, price volatility is considered in the former, descriptive sense, referring to the historical fluctuations and movements of Vietnam's rice export price series. Accordingly, the empirical analysis focuses on modeling and forecasting the level and movement of the FOB price series through the ARIMAX and Hybrid ARIMAX-LSTM frameworks, rather than estimating a separate conditional variance process. This interpretation is consistent with the study's forecasting-oriented objective and the use of historical price dynamics as the primary basis for short-term prediction (Hyndman and Athanasopoulos, 2021; Sun et al., 2023).

Although forecasting price fluctuations has been widely studied around the world, most previous studies have primarily focused on traditional statistical models such as ARIMA or GARCH. More recent studies have begun applying hybrid models that combine econometric approaches and deep learning techniques in agricultural price forecasting. For example, Anggraeni et al. (2019) developed a hybrid ARIMAX-ANN model to forecast rice prices in Indonesia and reported superior performance compared to individual models. Similarly, Mahawan et al. (2022) proposed a hybrid ARIMAX-LSTM model for forecasting Thailand's rice export price and demonstrated improved forecasting accuracy in highly volatile data environments.

However, in Vietnam in general and at FPT University Can Tho in particular, the number of studies directly applying the ARIMAX combined with Long Short-Term Memory (LSTM) model in forecasting rice export prices remains relatively limited, especially for benchmark FOB price series and regionally competitive exogenous variables. Most existing studies mainly focus on traditional ARIMA models or production forecasting rather than developing hybrid frameworks capable of simultaneously capturing both the linear and non-linear structures of time-series data (Sun et al., 2023).

In this study, the ARIMAX model is selected because of its ability to process time-series data and incorporate exogenous variables while maintaining high economic interpretability. However, since rice export price data often exhibit nonlinear characteristics and abnormal market shocks, the study further employs the Long Short Term Memory (LSTM) model to enhance the ability to learn nonlinear patterns and long-term dependencies. Based on this, a hybrid ARIMAX-LSTM framework is proposed to combine the strengths of both approaches in order to improve forecasting performance in the context of increasingly volatile international rice markets (Mahawan et al., 2022; Anggraeni et al., 2019).

Based on this research gap, the present study is conducted to develop a forecasting model for the export price volatility of Vietnam's 5% broken white rice by integrating the ARIMAX model with Long Short-Term Memory (LSTM). Within the conceptual scope established above, export price volatility is interpreted as the observed movements and fluctuations of the rice export price series over time. The study is expected to contribute to improving forecasting accuracy in the context of increasingly complex and uncertain international rice markets.

1.2. Research Objectives

This study aims to forecast the volatility trend of Vietnam's 5% broken white rice export prices through 2027. The findings are expected to provide a scientific basis for market forecasting activities, policy formulation, and the sustainable development of Vietnam's rice export sector.

ROI: To analyze the volatility trend of Vietnam's 5% broken white rice FOB export prices during the period from December 2003 to March 2026.

RO2: To develop and evaluate the ARIMAX model and the hybrid ARIMAX-LSTM model for forecasting Vietnam's 5% broken white rice export prices through March 2027.

RO3: To propose appropriate solutions aimed at improving forecasting effectiveness and enhancing the stability of Vietnam's rice export market.

1.3. Research Questions

This study is conducted to address the following main research questions:

RQ1: How did Vietnam's 5% broken white rice FOB export prices fluctuate during the period from December 2003 to March 2026?

RQ2: How do the ARIMAX model and the hybrid ARIMAX-LSTM model differ in forecasting performance for Vietnam's 5% broken white rice export prices through March 2027?

RQ3: What solutions and implications can be proposed to improve forecasting effectiveness and enhance the stability of Vietnam's rice export market?

1.4. Research Scope

Temporal Scope

This study utilizes monthly time-series data from December 2003 to March 2026 to analyze historical price volatility and fluctuations and to develop forecasting models for Vietnam's rice export prices. The final analytical dataset consists of 268 monthly observations over the research period.

The forecasting scope of the study focuses on the short term, specifically through March 2027, in order to ensure consistency with the characteristics of time-series models and to maintain the reliability of the forecasting results over the selected horizon. In addition, the study also considers forecasting uncertainty, as the level of forecasting error and uncertainty generally increases over longer forecasting horizons.

Spatial Scope

This study is conducted at the national level and focuses on the benchmark FOB price of Vietnam's 5% broken white rice in the international export market. The unit of analysis is Vietnam's rice export market at the overall national scale.

The study does not provide separate forecasts for individual provinces, enterprises, or specific import markets such as the Philippines, Indonesia, or the European Union (EU). Therefore, the findings should be interpreted as reflecting the general trend of Vietnam's rice export prices in the international market rather than the price fluctuations of individual importing markets.

Content Scope

This study focuses on forecasting the FOB export price of Vietnam's 5% broken white rice as a benchmark indicator representing Vietnam's mainstream rice export market. The primary dependent variable of the study is the FOB export price of Vietnam's 5% broken white rice.

In addition, the study employs Thailand's FOB export price of 5% broken white rice as the sole exogenous variable, given Thailand's role as one of Vietnam's major competitors in the international rice market and the predictive relevance of Thailand's rice price movements to regional rice market dynamics.

The study concentrates on the benchmark price series of FOB 5% broken white rice rather than analyzing individual rice varieties, rice categories, or specific segments within the 5% broken rice category such as ST25, Jasmine, Japonica, or premium Dai Thom rice. This approach is mainly due to limitations in publicly available data as well as the need to ensure the consistency of the time-series dataset over time.

1.5. Research Significance

Academic Significance

From an academic perspective, this study contributes additional empirical evidence regarding the applicability and forecasting performance of hybrid models that combine econometrics and deep learning in the field of agricultural export price forecasting. By developing and comparing the standalone ARIMAX model with the Hybrid ARIMAX-LSTM model using the same real-world dataset, the study provides further evidence on the potential of hybrid forecasting approaches to capture both linear and nonlinear patterns in agricultural price data under volatile market conditions.

In addition, the study contributes to the body of literature on agricultural price forecasting, agricultural economics, and the application of artificial intelligence in time-series forecasting in the context of Vietnam's rice export market. The research also helps clarify

the potential of hybrid frameworks to integrate the economic interpretability and exogenous-variable modeling capability of ARIMAX with the nonlinear learning capability of LSTM.

Practical Significance

From a practical perspective, the findings of this study may support rice export enterprises in market forecasting, business planning, and price risk management in the context of increasingly volatile international rice markets.

For government authorities, the study may provide additional quantitative references for market monitoring, export policy formulation, and agricultural trade management.

Furthermore, the research findings may provide useful information for stakeholders in the agricultural supply chain, such as exporters, distributors, and related organizations, when assessing future rice price movements and adapting their planning and decision making to fluctuations in the international rice market.

1.6. Thesis Outline

The content of this thesis is organized into five main chapters as follows:

CHAPTER 1: INTRODUCTION

1.1. Background of Research

1.2. Research Objectives

1.3. Research Questions

1.4. Research Scope

1.5. Research Significance

1.6. Thesis Outline

CHAPTER 2: LITERATURE REVIEW

2.1. Definitions and Key Concepts

2.2. Vietnam's Rice Export Market and Price Dynamics

2.3. Theoretical Foundations of Time Series Forecasting

2.4. Forecasting Models and Methods

2.5. Empirical Evidence on Agricultural Price Forecasting

2.6. Research Gap

2.7. Conceptual Framework

CHAPTER 3: METHODOLOGY

3.1. Research Design

3.2. Data Collection

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CHAPTER 4: ANALYSIS AND FINDINGS

4.1. Descriptive Analysis of Vietnam's and Thailand's 5% Broken Rice FOB Prices (2003-2026)

4.2. Forecasting Results for Vietnam's Broken White Rice FOB Price through March 2027

CHAPTER 5: CONCLUSION AND RECOMMENDATIONS

5.1. Conclusion

5.2. Research Contributions

5.3. Policy Recommendations

5.4. Research Limitations

5.5. Directions for Future Research

CHAPTER 2. LITERATURE REVIEW

This chapter establishes the theoretical and empirical foundation of the study by reviewing key concepts related to agricultural commodities, international rice trade, export price dynamics, and time-series forecasting. It provides an overview of Vietnam's rice export market and examines the characteristics and volatility of benchmark rice prices within the regional and international context. The chapter further discusses the theoretical foundations and development of forecasting approaches, ranging from traditional econometric models to machine learning, deep learning, and hybrid forecasting methods, with particular emphasis on ARIMAX and LSTM. In addition, relevant empirical studies on agricultural price forecasting are critically reviewed to identify existing methodological limitations and research gaps. Based on these foundations, the chapter develops the conceptual framework for the proposed Hybrid ARIMAX-LSTM model, providing the theoretical basis for the methodology and empirical analysis presented in the subsequent chapters.

2.1. Definitions and Key Concepts

2.1.1. Agricultural Commodities

In agricultural economics and international trade, agricultural commodities are products derived from agricultural production activities that are produced for consumption, processing, and commercial exchange in both domestic and international markets. Unlike subsistence - oriented agricultural production, commercial agricultural commodities are typically integrated into supply chains and are influenced by market demand and supply conditions, international prices, logistics costs, and cross-border trade activities (FAO, 2024). In the context of globalization and economic integration, agricultural commodities not only play a vital role in economic growth and the livelihoods of millions of people but also hold strategic importance for food security, social stability, and sustainable development in many countries (FAO, 2024).

However, agricultural commodity markets are generally more volatile than many industrial commodity markets due to their simultaneous exposure to multiple factors, including demand and supply fluctuations, climatic conditions, production costs, logistics expenses, exchange rate movements, international trade environments, and government intervention policies (FAO, 2024; Gilbert and Morgan, 2010). According to Sun et al. (2023), agricultural prices often exhibit nonlinear characteristics, non-stationarity, and structural breaks, making

analysis and forecasting more challenging than in many other commodity markets. These characteristics have made agricultural price forecasting an increasingly important area of research in economics, international trade, and supply chain management (Sun et al., 2023; Lim and Zohren, 2021).

In agricultural commodity research, price volatility is commonly associated with the extent to which observed prices fluctuate over time, reflecting the instability and uncertainty arising from changing market conditions (Gilbert and Morgan, 2010). This descriptive interpretation differs from the more specific concept of conditional volatility in financial econometrics, where volatility is represented as a time-varying conditional variance of a return or price-change process (Engle, 1982; Bollerslev, 1986). Consistent with the agricultural price-forecasting context of this study, price volatility is therefore interpreted as observed fluctuations and movements in the rice export price series over time. The empirical analysis consequently focuses on modeling and forecasting the level and movement of the observed FOB price series rather than estimating a separate conditional variance process.

For export-oriented agricultural commodities, sensitivity to international market fluctuations is even greater because prices are directly influenced by global demand, international competition, logistics costs, and trade policies implemented by exporting and importing countries (FAO, 2025a; World Bank, 2026). Consequently, studying price dynamics and forecasting export agricultural commodity prices is not only of academic significance but also of substantial practical value in supporting policy formulation, risk management, and the enhancement of international trade competitiveness (Sun et al., 2023; Hyndman and Athanasopoulos, 2021).

2.1.2. Rice as a Strategic Agricultural Commodity

Rice is one of the most important strategic agricultural commodities in the world, serving not only as a staple food source but also as a commodity of significant importance in international trade and global food security. According to the Food and Agriculture Organization of the United Nations (FAO), rice remains the primary staple food for more than half of the world's population and plays a crucial role in maintaining economic and social stability in many countries across Asia and Africa (FAO, 2025a). Due to its direct connection to social stability and food security, the rice market is highly influenced by export policies, stockpiling measures, and price control interventions implemented by major exporting countries such as India, Thailand, and Vietnam (Mottaleb et al., 2026).

One notable characteristic of the international rice market is its relatively low trade volume compared to total global rice production. According to Headey (2011), only a small proportion of the world's rice output is traded internationally, making rice prices particularly sensitive to shocks in supply and demand, inventory levels, and trade policies. Consequently, even minor changes in export regulations or supply conditions among major exporting countries can generate substantial fluctuations in global rice prices (Headey, 2011; Götz et al., 2013).

In international trade, rice prices are commonly quoted through standardized benchmark quotations, such as the FOB price of 5% broken rice. According to the FAO Rice Market Monitor and the World Bank Pink Sheet, these benchmark quotations are widely used to reflect international trading conditions and facilitate price comparisons across exporting countries (FAO, 2025a; World Bank, 2026). For Vietnam, the FOB price of 5% broken rice is considered a key indicator of the country's competitiveness and market position in the global rice trade (FAO, 2025b; World Bank, 2026).

Given its sensitivity to trade policies, supply-demand fluctuations, and external market shocks, rice export prices tend to exhibit considerable volatility over time. Events such as food crises, climate change, supply chain disruptions, and export restrictions imposed by major producing countries have significantly increased uncertainty in the global rice market (FAO, 2025a; Mottaleb et al., 2026). In this context, forecasting rice export prices has become an increasingly important research area in agricultural economics and international trade, particularly for major rice-exporting countries such as Vietnam, where price movements directly affect export revenues, farmers' incomes, and international competitiveness (Sun et al., 2023; FAO, 2025a).

2.1.3. Rice Export and International Rice Trade

In international trade, rice export refers to the commercial export of rice from producing countries to foreign markets for consumption, stockpiling, or processing purposes in importing countries. For agricultural economies such as Vietnam, Thailand, and India, rice exports not only generate foreign exchange earnings but also reflect a country's competitiveness within the global food supply chain (FAO, 2025a; Childs, 2025). According to the USDA Rice Outlook, Vietnam is currently among the world's leading rice-exporting countries, alongside Thailand, India, and Pakistan, with export markets spanning across Asia, Africa, and the Middle East (Childs, 2025).

In international trade research, rice export performance is commonly measured through three key indicators: export volume, export value, and export price. Each indicator captures a different dimension of trade activities. Export volume reflects the physical quantity of rice traded, export value represents the total monetary value of exports, while export price indicates the price at which a country sells its rice in international markets (FAO, 2024). In agricultural commodity forecasting studies, export price is often preferred because it can more promptly capture changes in market supply and demand, production costs, trade policies, and economic shocks affecting international markets (Sun et al., 2023; Hyndman and Athanasopoulos, 2021). Consequently, in studies forecasting agricultural export prices, particularly rice prices, the export price is commonly treated as the central research variable for evaluating market trends and international competitiveness over time.

Furthermore, it is important to distinguish between unit export value and benchmark export price. Unit export value is typically calculated as the ratio of total export value to total export volume. However, this indicator may be influenced by changes in rice product composition, destination markets, or delivery timing, and therefore may not fully reflect pure market price movements (FAO, 2025b; World Bank, 2026). In contrast, a benchmark export price refers to a standardized quotation for a specific rice variety, quality grade, and delivery condition. According to the World Bank Pink Sheet, benchmark quotations such as “Rice (Thailand), 5% broken, white rice, FOB Bangkok” and “Rice (Vietnam), 5% broken, white rice, FOB Hanoi” are widely used as representative indicators of the international rice export market (World Bank, 2026). Because these benchmark quotations are more highly standardized in terms of product quality and trading conditions, they are generally considered more suitable for time series forecasting and international price volatility analysis (Hyndman and Athanasopoulos, 2021; World Bank, 2026).

2.1.4. FOB Rice Export Price and Benchmark Rice Prices

In international trade, the FOB (Free on Board) price refers to the value of goods at the port of export, excluding international transportation and insurance costs. Under the FOB rule of Incoterms® 2020, the seller fulfills its delivery obligation once the goods are loaded onto the vessel at the agreed port of shipment. From that point onward, all risks and costs associated with the transportation of the goods are transferred to the buyer (International Chamber of Commerce, 2025). Because it reflects the value of goods at the point of export

while excluding post-shipment expenses, the FOB price is widely used as a benchmark indicator in trade analysis and export commodity price research (International Chamber of Commerce, 2025; World Bank, 2026).

In export price forecasting studies, FOB prices are generally preferred over CIF (Cost, Insurance and Freight) prices because FOB focuses on the commercial value of goods at the exporting country rather than costs associated with international transportation. If CIF prices are used, forecasting results may be influenced by fluctuations in freight rates, marine insurance costs, or global logistics expenses, which do not directly reflect the competitiveness of the exporting country. Consequently, most international rice quotations published by FAO, USDA, and the World Bank are reported on an FOB basis to ensure comparability across countries and over time (FAO, 2025a; Childs, 2025; World Bank, 2026).

Among international benchmark quotations, the FOB price of 5% broken white rice is considered one of the most important indicators of the Asian rice market. The term “5% broken” refers to rice containing no more than 5% broken kernels by weight, representing a highly standardized rice category in international trade (FAO, 2025b; Childs, 2025). Currently, the World Bank Commodity Price Data (Pink Sheet) publishes separate price series for “*Rice (Thailand), 5% broken, white rice, FOB Bangkok*” and “*Rice (Vietnam), 5% broken, white rice, FOB Hanoi*,” highlighting their importance as representative benchmark quotations for the regional rice export market (World Bank, 2026).

The selection of Vietnam’s FOB price for 5% broken white rice as the dependent variable in this study is supported by several economic considerations. First, it directly reflects Vietnam’s international offering price at the point of export. Second, this benchmark quotation is regularly monitored and published by major international organizations, ensuring data consistency and comparability over an extended period (FAO, 2025b; Childs, 2025; World Bank, 2026). Third, this price simultaneously captures the effects of various factors, including global supply and demand conditions, logistics costs, exchange rate movements, trade policies, and competitive interactions among major rice-exporting countries (Götz et al. 2013; FAO, 2025a).

In addition, Thailand’s FOB price for 5% broken white rice is selected as an exogenous variable in the ARIMAX model. Thailand has long been regarded as one of the key reference countries in the global rice market and possesses a continuously published benchmark price series, “*Rice (Thailand), 5% broken, white rice, FOB Bangkok*,” provided by the World

Bank over a long period (World Bank, 2026). Empirical studies have also shown that rice export prices in Vietnam and Thailand are closely related and exhibit long-run linkages in the price formation process of the international rice market. Chen and Saghaian (2016) argued that Thai rice prices play an important role in price transmission among major exporting countries, implying that price movements in Thailand may provide valuable information for forecasting Vietnam's rice export prices. Therefore, incorporating Thai rice prices as an exogenous variable is expected to improve the explanatory power and forecasting accuracy of the model (Chen and Saghaian, 2016).

Accordingly, within the scope of this study, the term *rice export price* refers to the FOB price of Vietnam's 5% broken white rice, a benchmark quotation that is highly representative of international rice trade. Meanwhile, Thailand's FOB price for 5% broken white rice is incorporated as an exogenous variable to capture the effects of regional market competition and international rice market dynamics on Vietnam's rice export prices.

2.1.5. Time Series and Forecasting

A time series is a sequence of observations of a variable recorded over time at regular intervals, such as monthly, quarterly, or annually. Time series data enable researchers to identify important characteristics of the variable under investigation, including long-term trends, seasonality, cyclical patterns, and random fluctuations. The analysis of these characteristics serves as a fundamental basis for developing forecasting models and supporting decision-making processes in economics and business (Makridakis et al., 2020).

Forecasting refers to the process of using historical data to estimate the future values of a variable through statistical methods or mathematical models. In environments characterized by uncertainty arising from factors such as fluctuations in supply and demand, changes in trade policies, exchange rate movements, and global economic shocks, forecasting helps identify the future trajectory of a variable over time, thereby supporting more informed and effective decision-making (Makridakis et al., 2020).

In international trade research, export forecasting represents an important application of time series forecasting methods, which are employed to estimate future values related to export activities based on historical observations. In the rice industry, export price is a particularly important indicator because it directly reflects the commercial value of rice in international markets while simultaneously being influenced by various economic and policy

related factors. Therefore, the application of time series forecasting models to predict Vietnam's rice export prices is considered appropriate from both theoretical and practical perspectives. By exploiting the dynamic characteristics embedded in historical data, forecasting models such as ARIMA, ARIMAX, and LSTM can help identify future price trends, thereby contributing to more effective production planning, business decision-making, and policy formulation within the rice sector.

2.2. Vietnam's Rice Export Market and Price Dynamics

2.2.1. Overview of Vietnam's Rice Export Industry

Rice plays a vital role in Vietnam's economy and food security. Following the *Doi Moi* reforms in 1986, Vietnam's rice sector experienced significant growth driven by agricultural reforms, production expansion, and integration into international markets. From being a food deficit country, Vietnam has transformed into one of the world's leading rice exporters. According to Nhung and Wang (2022), Vietnam accounted for approximately 10% of global rice exports during the 2017-2020 period and consistently maintained its position among the world's top three rice-exporting countries.

In recent years, Vietnam's rice export industry has continued to achieve remarkable performance. According to the General Department of Vietnam Customs, the country exported approximately 9 million tons of rice in 2024, generating around USD 5.7 billion in export revenue. Compared with 2023, export volume increased by approximately 11%, while export value rose by about 24%. This represents the highest level ever recorded in Vietnam's rice export history. Such achievements underscore the importance of the rice sector within the country's agricultural export structure and its substantial contribution to national economic growth.

The Mekong Delta serves as Vietnam's principal rice-producing region and plays a decisive role in the country's rice supply and export capacity (World Bank, 2024b). The region contributes approximately half of Vietnam's rice production and accounts for about 95% of the country's rice exports (World Bank, 2022). Benefiting from its significant agricultural production capacity, the Mekong Delta plays a central role in supporting Vietnam's position in the international rice market.

In addition to maintaining a large export volume, Vietnam's rice sector has gradually shifted from a quantity-based approach toward a quality- and value-oriented development

strategy. The country has increasingly emphasized high-quality rice varieties, including fragrant rice, Japonica rice, specialty rice, and other high-value-added rice products. This restructuring strategy focuses on improving product quality, increasing value added, promoting sustainable development, and strengthening the competitiveness of Vietnamese rice in international markets (Prime Minister of Vietnam, 2023).

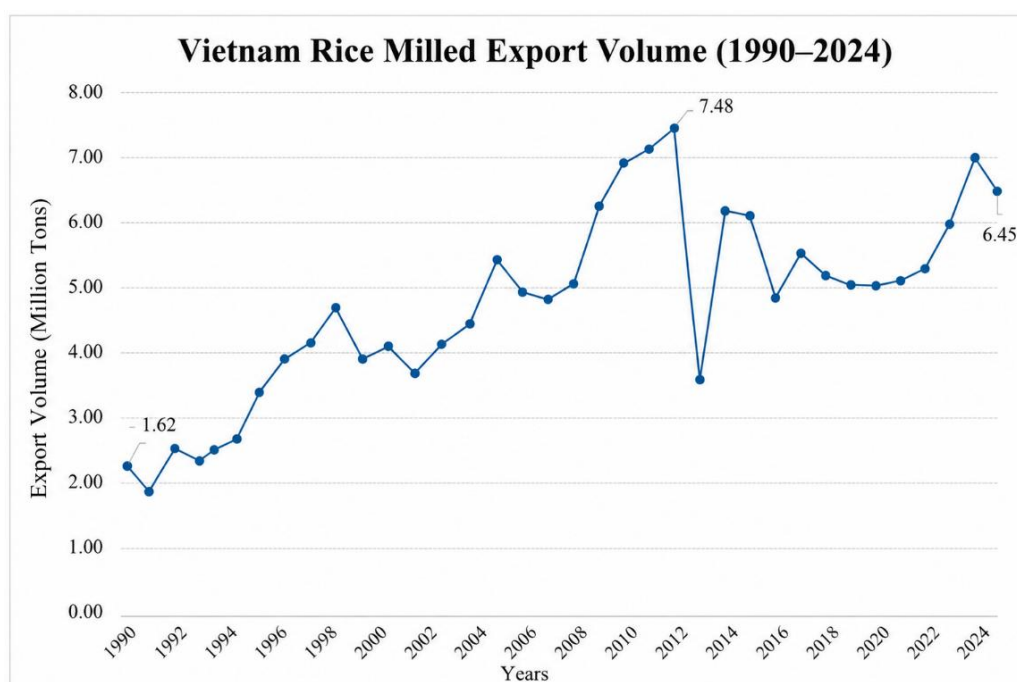


Figure 2.1. Vietnam Rice Milled Export Volume (1990-2024)

Source: Author's compilation based on FAOSTAT data.

Figure 2.1 illustrates the substantial growth of Vietnam's rice, milled export volume over the period 1990-2024. Based on the data presented in the figure, rice, milled exports increased from approximately 1.62 million tons in 1990 to a peak of around 7.48 million tons in 2012, before experiencing fluctuations in subsequent years and reaching approximately 6.45 million tons in 2024. These figures refer specifically to the "Rice Milled" series used in the figure and therefore should be distinguished from total rice export statistics reported by the General Department of Vietnam Customs, which may use different statistical classifications and reporting methodologies. FAOSTAT data may differ from customs statistics due to differences in product definitions, reporting periods, and data compilation methodologies. Overall, the figure reflects the remarkable expansion of Vietnam's milled rice export sector and the country's increasing integration into international trade. Although

rice, milled export volumes have fluctuated over time due to market conditions, trade policies, and changes in global supply and demand, the long-term trend remains positive. This trend highlights Vietnam's important role in the international rice trade and reinforces its position among the world's leading rice-exporting countries.

The figure also indicates that the growth of Vietnam's rice, milled exports has not followed a linear pattern but has experienced several periods of significant volatility. For instance, the years following 2012 witnessed a notable decline in export volume before a gradual recovery in recent years. These fluctuations reflect the influence of international competition, changing market demand, regional competitive pressures, shifts in trade policies, and global supply demand shocks on Vietnam's rice export sector. Such dynamics underscore the sensitivity of rice exports to both domestic and external factors and highlight the evolving nature of the global rice market.

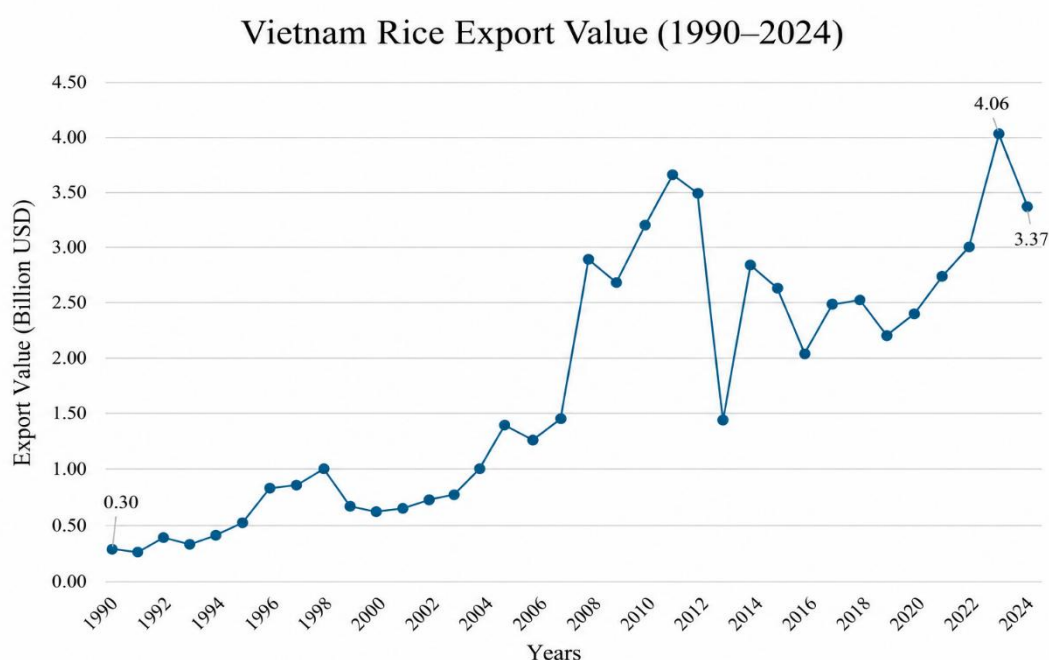


Figure 2.2. Vietnam Rice Export Value (1990-2024)

Source: Authors' compilation based on FAOSTAT data.

Figure 2.2 illustrates a substantial increase in the value of Vietnam's rice exports over the period 1990-2024. Compared with export volume, export value rose from approximately USD 0.30 billion in 1990 to around USD 3.37 billion in 2024 and reached a peak of approximately USD 4.06 billion during the study period. Significant increases were particularly evident during periods of global food price volatility, such as the 2007-2008 food crisis and the 2022-2024 period.

Notably, although export volumes did not increase substantially in certain years, export values continued to grow. This development coincided with higher international rice prices and Vietnam's ongoing transition toward higher-quality and higher-value rice varieties (World Bank, 2024a; Vietnam Institute of Industry and Trade, 2023). This trend suggests that price levels and price fluctuations have become increasingly important factors associated with Vietnam's rice export performance. It also highlights the growing importance of forecasting export price movements in an international market characterized by increasing uncertainty and volatility (World Bank, 2024a; Food and Agriculture Organization of the United Nations [FAO], 2025a).

The observed fluctuations in export volume and export value indicate that the performance of Vietnam's rice exports depends not only on the quantity of rice exported but also on changes in international rice prices (Nhung and Wang, 2022). As rice prices become increasingly volatile under the influence of economic, trade, and climate-related factors (World Bank, 2023, 2024a), analyzing rice price dynamics becomes essential for understanding the determinants of competitiveness within Vietnam's rice industry. Such analysis provides important insights into the factors shaping export performance and supports more effective decision-making in the context of a rapidly evolving global rice market (FAO, 2025a; Nhung and Wang, 2022).

2.2.2. Export Trends and Major Export Markets

Vietnam has maintained its position as one of the world's leading rice-exporting countries, with annual export volumes consistently exceeding 6 million tons (USDA FAS, 2023). In recent years, Vietnam's rice exports have experienced strong growth, driven by increasing food demand in many importing countries and fluctuations in global rice supply. According to the May 2023 Vietnam Rice Market Report, Vietnam exported approximately 3.6 million tons of rice during the first five months of 2023, with an export value of USD 1.92 billion, representing increases of 30.8% in volume and 41.6% in value compared with the same period of 2022 (Hoang, 2023).

In terms of market structure, the Philippines has remained Vietnam's largest rice import market, while Indonesia and Malaysia have recorded significant increases in rice imports in recent years (VFA, 2024). In contrast, rice exports to China have shown a declining trend due to the country's diversification of supply sources (Nhung and Wang, 2022). These

developments reflect changing import demand patterns and evolving food security policies among countries in the region.

Beyond its traditional Asian markets, Vietnam’s rice exports to Africa have also expanded considerably. Countries such as Côte d’Ivoire, Ghana, and Senegal have emerged as important import destinations, contributing to the diversification of Vietnam’s export markets and reducing dependence on a limited number of major trading partners (VFA, 2024). The expansion of export markets, together with the upward trend in international rice prices during the 2022-2024 period, has significantly enhanced the value of Vietnam’s rice exports. This development suggests that price factors are becoming increasingly important in determining the performance and competitiveness of the country’s rice export sector.

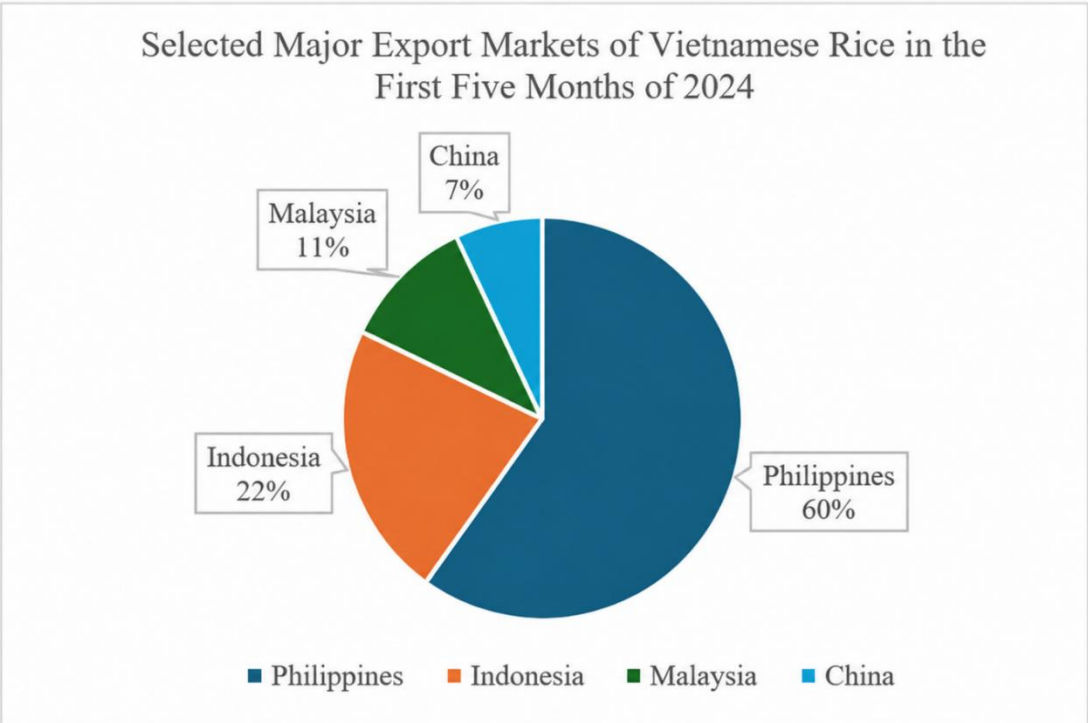


Figure 2.3. Vietnam’s major rice export markets in the first 5 months of 2024

Source: Authors’ compilation based on data from the General Department of Customs (2024).

Note: Percentages represent the share of the combined export volume among the four selected major destination markets.

Figure 2.3 shows that the Philippines was the largest destination market for Vietnamese rice during the first five months of 2024, accounting for approximately 60.1% of the combined export volume among the four selected major destination markets. It was followed

by Indonesia (22.2%), Malaysia (11.1%), and China (6.6%) (General Department of Customs, 2024). The Philippines was also reported as Vietnam's largest rice export market, accounting for 45.4% of Vietnam's total rice shipments during the same period (Reuters, 2024). These findings indicate the significant importance of the Southeast Asian region to Vietnam's rice exports and highlight the substantial roles of Indonesia and Malaysia within Vietnam's export market structure (Nhung and Wang, 2022).

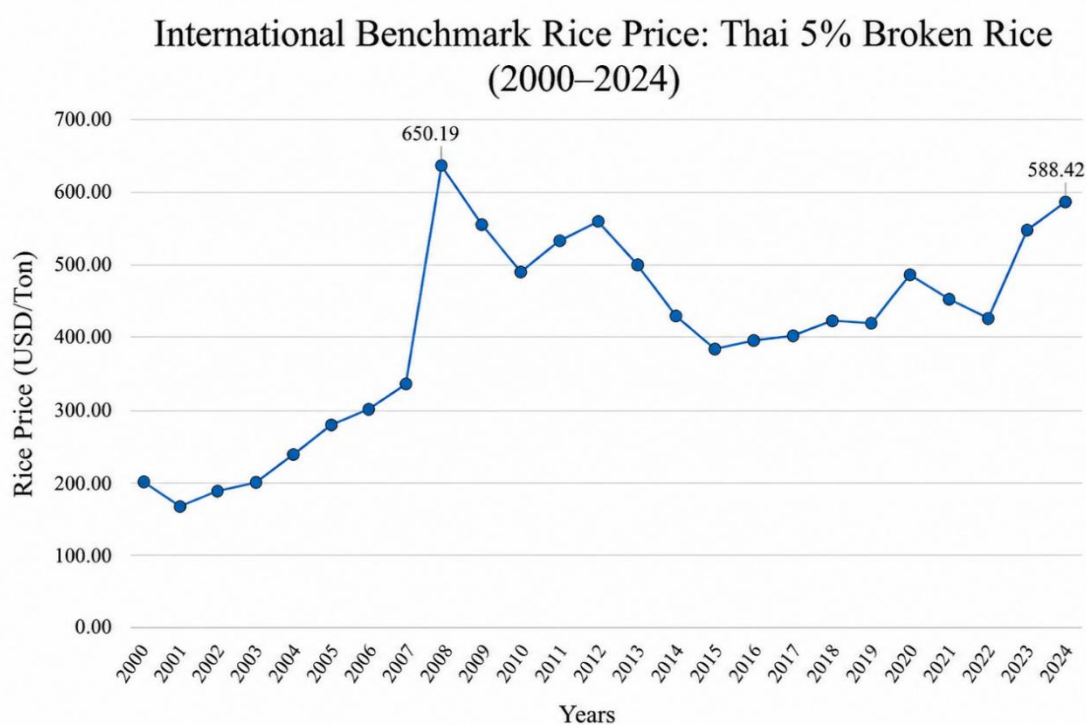


Figure 2.4. International Benchmark Rice Price: Thai 5% Broken Rice (2000-2024)

Source: Authors' compilation based on World Bank Commodity Price Data (Pink Sheet).

Figure 2.4 illustrates the fluctuations in international benchmark rice prices over the period 2000-2024. Rice prices increased sharply during the 2007-2008 global food crisis, reaching approximately USD 650 per ton before gradually declining in the following years. Since 2022, international rice prices have experienced a renewed upward trend, reaching around USD 588 per ton in 2024 (World Bank, 2026).

2.2.3. Price Dynamics in the Asian Rice Market

The Asian rice market is strongly influenced by major exporting countries such as Thailand, Vietnam, and India. Chen and Saghaian (2016) found that rice export prices in Vietnam, Thailand, and the United States exhibit a long-run cointegration relationship, indicating a high degree of integration among major rice-exporting markets. Among these countries, Thailand plays a particularly important role in the formation and transmission of rice

prices within the region, making Thai rice prices a widely recognized benchmark in the international rice market (Chen and Saghaian, 2016).

In addition to traditional supply and demand factors, rice prices in Asia are also affected by external shocks such as climate change, supply chain disruptions, and trade policy interventions. According to Götz et al. (2013), the 2007-2008 global food crisis led to a sharp increase in world rice prices as many exporting countries implemented trade restrictions to safeguard domestic food security. Similarly, India's rice export restrictions in recent years have reduced global market supply and contributed to increased rice price volatility across Asia.

These developments suggest that Vietnam's rice export prices are increasingly influenced by a range of complex and nonlinear factors, including supply-demand shocks, trade policies, climate-related disruptions, and geopolitical uncertainties (Chen and Saghaian, 2016; Götz et al., 2013). In such an environment, traditional forecasting models may face limitations in fully capturing modern price dynamics, particularly when markets experience unexpected shocks and rapid changes in the global economic landscape. This highlights the growing need for more advanced forecasting approaches, including machine learning and deep learning models, to improve the accuracy of rice export price predictions.

Note: The 2008 Vietnam rice price represents the average of available monthly observations from June to December 2008, as the World Bank Vietnamese 5% rice price series begins in June 2008.

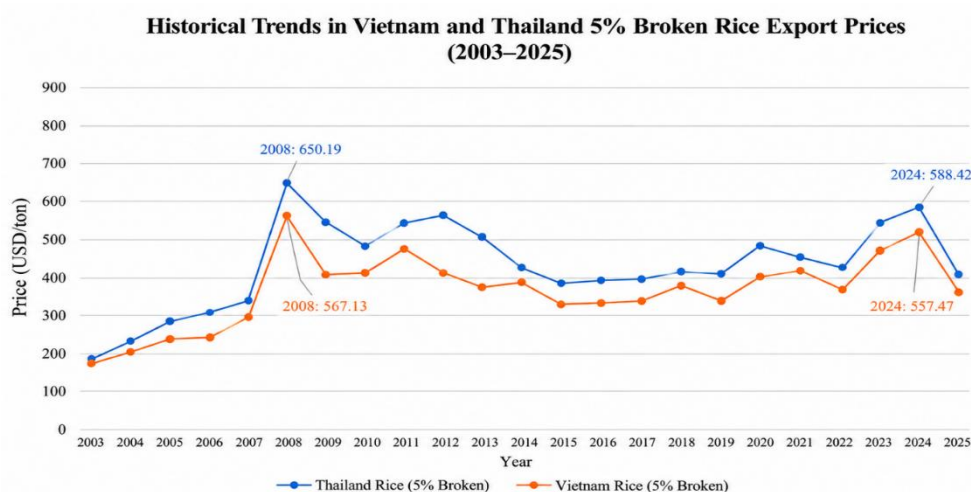


Figure 2.5. Historical Trends in Vietnam and Thailand 5% Broken Rice Export Prices (2003-2025).

Source: Authors' compilation based on World Bank Commodity Price Data (Pink Sheet, 2026).

Figure 2.5 presents the historical trends of Vietnam's and Thailand's 5% broken rice export prices over the period 2003-2025. It can be observed that the rice prices of the two countries generally moved in the same direction throughout most of the study period, reflecting a significant degree of integration between the two major rice-exporting markets in the region. Both price series increased sharply during the 2007-2008 global food price crisis, with Thailand's rice price reaching approximately USD 650.19 per ton, while Vietnam's rice price averaged approximately USD 567.13 per ton based on available observations from June to December 2008. Following this period, rice prices gradually declined and remained relatively stable between 2014 and 2022, with the two price series generally fluctuating within a range of approximately USD 330-500 per ton. A new upward price cycle emerged during 2023-2024, when Thailand's rice price increased to USD 588.42 per ton, and Vietnam's rice price reached USD 557.47 per ton. This surge reflects the impact of supply shocks and export restrictions imposed by several major rice-producing countries.

Furthermore, Thailand's rice prices generally remained higher than Vietnam's rice prices throughout most of the study period, highlighting Thailand's important benchmark role in the regional rice market. The similarity in the movement of the two price series provides descriptive support for considering Thailand's rice price as an exogenous variable in forecasting Vietnam's rice export prices (Chen and Saghaian, 2016). At the same time, the substantial price fluctuations observed during crisis periods indicate that the international rice market is highly sensitive to trade disruptions and supply shocks. These characteristics underscore the need for advanced forecasting approaches to improve the accuracy of rice export price predictions in an increasingly volatile market environment.

2.2.4. Challenges and Opportunities

Vietnam's rice export industry faces numerous challenges in an increasingly volatile global rice market. One of the most significant risks is the impact of climate change on the Mekong Delta, the country's primary rice-producing region. Droughts, saltwater intrusion, and declining freshwater resources may adversely affect productivity, output, and the long-term stability of rice export supply (National Key Laboratory of River and Coastal Dynamics, n.d.). In addition, international rice prices are becoming increasingly influenced by supply shocks, trade policy interventions, and geopolitical uncertainties, all of which contribute to greater market instability and higher risks for export activities (Götz et al., 2013).

Furthermore, competitive pressure from major rice-exporting countries such as India and Thailand, together with increasingly stringent requirements regarding product quality, food safety, and traceability, poses significant challenges for Vietnam's rice industry (Nhung and Wang, 2022). To maintain competitiveness, rice exporters must continue improving product quality and ensuring compliance with evolving international standards.

Despite these challenges, Vietnam also has substantial opportunities to strengthen its position in the global rice market. Demand for high-quality rice, fragrant rice varieties, and products that meet sustainable development standards is increasing across many importing countries, creating favorable conditions for the industry's value-added growth strategy (Center for WTO and International Trade, 2023). To capitalize on these opportunities, Vietnam's rice sector should continue enhancing product quality, strengthening brand development, and improving its competitiveness in international markets.

Overall, Vietnam's rice export industry faces both opportunities and challenges within an increasingly complex global business environment (FAO, 2024). Improving product quality, diversifying export markets, and enhancing resilience to external shocks will be critical factors in determining the future competitiveness of Vietnamese rice.

Against this increasingly complex market environment, Vietnam's rice export market is influenced by a wide range of factors, including trade policy shocks, climate change, fluctuations in global supply, and competitive pressures from major exporting countries (World Bank, 2023; FAO, 2024). These factors not only contribute to fluctuations in rice export prices but also make future price movements more difficult to predict (FAO, 2024). In such an environment, traditional descriptive analysis alone may be insufficient to effectively support forecasting and decision-making. Consequently, the application of forecasting models to generate more accurate predictions of rice export price movements has become increasingly important for policymakers, businesses, and market participants. Building on this context, the following section presents the theoretical foundations of forecasting and the major forecasting approaches used in time series research, which provide the basis for the development of the proposed forecasting model for Vietnam's rice export prices.

2.3. Theoretical Foundations of Time Series Forecasting

2.3.1. Forecasting Time Series

Time Series Forecasting is using available observations from a time series at time t to forecast its value at a future time $t+l$ can provide a basis for: (1) economic and business planning, (2) production planning, (3) inventory and production control, and (4) control and optimization of industrial processes. As originally described by Holt (1963) and Brown (1962), forecasts are often needed over a period called lead time, which varies depending on the problem. For example, in the inventory control problem, Harrison (1965) defined lead time as the period from when a restock order is placed with the factory until that order is received into inventory. We will assume that the available observations are at discrete and evenly spaced intervals. For example, in a sales forecasting problem, the sales z_t in the current month t and the sales z_{t-1} , z_{t-2} , ... in previous months can be used to forecast sales for the next 1, 2, ..., 12 months. The notation $\hat{z}_t(l)$ denotes the forecast made at the origin t for the value z_{t+l} at the future time $t + l$. The function $\hat{z}_t(l)$, which provides forecasts at the origin t for all future 12 months based on available information from the current and past values, will be called the forecast function at the origin t . Our goal is to obtain a forecasting function such that the mean square error of the deviations $z_{t+l} - \hat{z}_t(l)$ between the actual and predicted values is minimized for each waiting period l . In addition to calculating the best forecasts, determining their accuracy is also essential to calculating the risks associated with decisions based on those forecasts. Accuracy can be expressed by calculating probability limits on either side of each predicted value (e.g., 50% or 95% limits). These limits ensure that the actual value of the time series when it occurs will fall within this range with the given probability. Forecasting methods are developed based on the assumption that the time series z_t follows a known stochastic model. Therefore, the ARIMA (Integrated Moving Average Autocorrelation) model class is introduced to represent the behavior of the series z_t . The practical process for developing these models includes three stages: preliminary model identification, parameter estimation, and model diagnostic testing (Box et al., 2015).

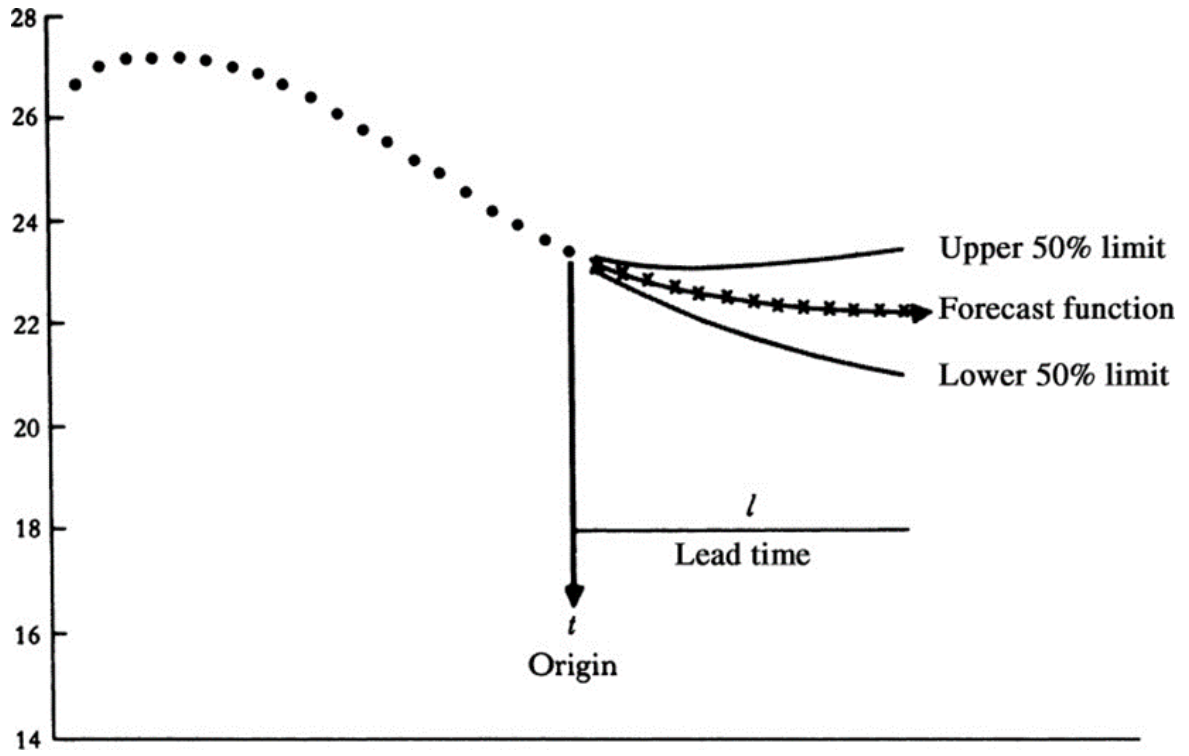


Figure 2.6. Values of a time series with a forecast function and 50% probability limits.

Source: Box et al., 2015.

2.3.2. Forecast Evaluation Framework

To evaluate the quality of a forecasting model, research typically relies on a loss function to measure the deviation between the forecasted and actual values. Since each loss function emphasizes a different type of risk, selecting the appropriate evaluation metric must align with the specific research objectives. In forecasting practice, Mean Absolute Error (MAE) and Mean Squared Error/Root Mean Squared Error (MSE/RMSE) are two fundamental and widely utilized metrics. MAE reflects the average absolute error, whereas MSE/RMSE heavily penalizes large errors; therefore, employing a combination of both indicators facilitates a more comprehensive evaluation (Hyndman and Athanasopoulos, 2021).

However, relying solely on in-sample evaluation is insufficiently reliable. Consequently, this study adopts an out-of-sample design adhering strictly to time series logic (employing a chronological train-test split or a rolling-origin approach) to assess the actual predictive performance. When selecting the ARIMA configuration, this study prioritizes the AIC/BIC criteria to strike a balance between model goodness-of-fit and parsimony. Subsequently, residual diagnostics are conducted to ensure that the residuals approximate white

noise (by examining the residual ACF/PACF and employing statistical checks such as the Ljung-Box test) before finalizing the model (Hyndman and Athanasopoulos, 2021).

In addition to point forecasts, modern research emphasizes the reporting of prediction intervals to reflect the considerable uncertainty surrounding future outcomes. This is consistent with Bergheim (2008), which highlights that forecasts are often accompanied by wide confidence bands (p. 110). In evaluating prediction intervals, this study explicitly addresses the trade-off between coverage (the proportion of actual observations falling within the interval) and width. Specifically, a wider interval generally yields higher coverage but offers limited utility for decision-making, whereas a narrower interval provides sharper insights but carries a higher risk of missing the actual values (Hyndman and Athanasopoulos, 2021).

To operationalize the evaluation of interval forecasts. These measures are used to assess not only the accuracy of point forecasts but also the reliability and usefulness of the uncertainty range around the forecasts.

For given confidence level $1 - \alpha$, the prediction interval of observation t is defined as:

$$PI_t^{(1-\alpha)} = [L_t, U_t] \quad (2.1)$$

Where L_t and U_t denote the lower and upper bounds of the prediction interval, respectively. In this study, two confidence levels are considered: 80% and 95%. Accordingly, $\alpha = 0.20$ for the 80% prediction interval and $\alpha = 0.05$ for the 95% prediction interval.

Coverage rate measures the proportion of actual observations that fall within the prediction interval during the evaluation period. It is calculated as:

$$Coverage = \frac{1}{n} \sum_{t=1}^n I(L_t \leq y_t \leq U_t) \quad (2.2)$$

Where y_t is the actual observed value, n is the number of observations in the evaluation period, and $I(.)$ is an indicator function that equals 1 if the actual value falls within the interval and 0 otherwise. A coverage rate close to the nominal confidence level indicates that the prediction interval is well calibrated.

Interval width reflects the sharpness of the forecast interval and is calculated as:

$$Width_t = U_t - L_t \quad (2.3)$$

A narrower interval indicates a sharper forecast and provides more specific information for decision-making. However, an excessively narrow interval may reduce coverage and increase the risk that actual observations fall outside the predicted range.

Finally, to closely reflect the practical export context, it is worth noting that the loss is sometimes asymmetric. Under-forecasting may result in inadequate preparation for procurement and logistics or a failure to fulfill contractual obligations. Conversely, overforecasting can lead to overcapacity and resource wastage. Thus, discussing asymmetric loss is highly relevant and logical when interpreting the forecast results.

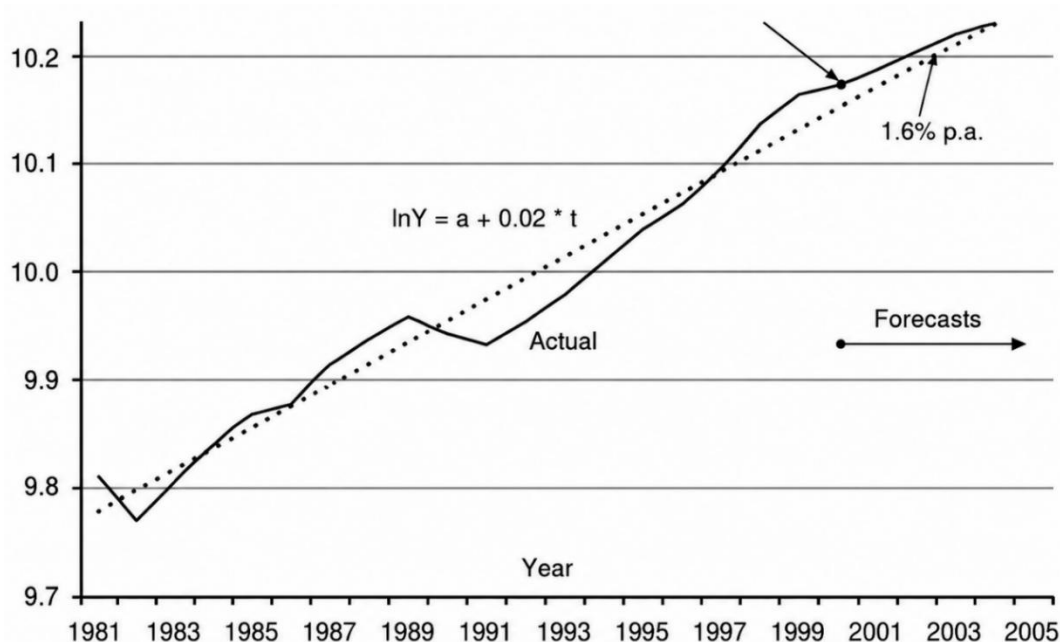


Figure 2.7. Intercept correction illustration for time-series forecasting to align forecasts with the last observation.

Source: Bergheim, 2008, p. 5.

Numerous studies in the field of commodity price dynamics assert the superiority of hybrid frameworks solely through the direct comparison of point-error metrics, without the necessity of conducting the Diebold-Mariano test. For instance, research on Thai rice export prices identified the ARIMAX-LSTM model as the optimal approach for both Hom Mali and white rice based on its achievement of the lowest RMSE and MAE values on the test dataset (Arya and Kasana 2026; Mahawan et al. 2022; Oktoviany, Knobloch, and Korn 2021). Similarly, the HySALS forecasting framework (2023) was proposed after observing

record-low MAPE values (4.31% - 7.83%) compared to standalone methods (Patil et al. 2023), while Rajana et al. (2026) demonstrated that hybrid architectures improved predictive accuracy by 5.51% to 11.96% across various commodities without resorting to formal statistical significance tests for error differentials (Rajana et al. 2026; Sudipa 2024). These findings suggest that distinct margins in empirical forecasting accuracy provide a sufficient basis for identifying superior performance in practical applications.

Evaluation methodologies centered on the magnitude of absolute or percentage error reduction are also widely employed to assert model superiority across the transport and industrial sectors. In forecasting Indonesian air passenger volumes, hybrid variants like ARIMAX-NN were confirmed to be superior due to their substantial reduction of MAPE compared to the original standalone ARIMAX model (Oktoviany et al. 2021; Sulistyowati et al. 2018). Within the steel industry, Jung et al. (2024) proved that the ARIMAX-LSTM model achieved an RMSE reduction of approximately 15.55% over a standalone ARIMAX specification (Jung and Kim 2024). Furthermore, Anggraeni et al. (2019) reported that hybrid structures enhanced accuracy—reducing MAPE by an average of 1.21%—compared to standalone artificial neural networks (ANN) in Indonesian rice price forecasting (Anggraeni et al. 2019). The consistent delivery of significant quantitative improvements by hybrid models allows researchers to validate their enhanced performance without the mandatory application of rigorous statistical hypothesis testing procedures such as the Diebold-Mariano test.

The Diebold-Mariano (DM) test is an important statistical tool used to assess whether the difference in forecasting accuracy between two competing models is statistically significant (Manogna et al., 2025; Ray et al., 2021). Rather than merely comparing point error metrics such as RMSE or MAPE, this test determines whether the superior performance of one model reflects genuine forecasting capability or is simply the result of random variation in the data. The null hypothesis (H_0) of the test assumes that both models have equal accuracy, while the alternative hypothesis (H_A) assumes that one model has a significantly lower forecast error than the other.

In terms of mechanism, the DM test is based on the loss differential series, which represents the difference in the loss function (typically absolute error or squared error) of the two models at each time step. This is an asymptotic z-test, computed based on the mean of the loss differential and an estimate of its long-run variance. In the practice of agricultural commodity price research, a low p-value (typically < 0.05) allows the null hypothesis to be

rejected, providing robust statistical evidence to confirm the superiority of hybrid models (such as ARIMA-LSTM) or machine learning techniques over traditional statistical models (Paul et al., 2022).

Random Walk without Drift

Random Walk without drift is a very simple baseline: it forecasts next month's price as exactly equal to the price of the most recent month. For example, if the rice price in June is USD 450/ton, this model forecasts July's price to also be USD 450/ton. It makes almost no attempt to be "smart," which is precisely why it serves as a useful benchmark — a more complex model is only convincing if it can outperform this simple baseline.

Forecast for month t = Actual price in month $t - 1$.

Random Walk with Drift

The term "drift" can simply be understood as the "average rate of change" or "average upward/downward trend." Random Walk with drift still starts from the previous month's price but adds the average monthly change observed in the historical data available at the time of forecasting.

For example, if last month's price was USD 450/ton and historical data show an average increase of USD 1.5/ton per month, Random Walk with drift would forecast approximately USD 451.5/ton. If the drift is negative, the average decrease is subtracted instead.

Forecast for month t = Price in month $t - 1$ + Historical average change.

2.3.3. The Role of Forecasting in Agricultural Markets

Forecasting plays a crucial role in agricultural markets, as prices often experience significant fluctuations due to factors such as weather conditions, seasonal production cycles, supply chain disruptions, and international trade policies (Sun et al., 2023). Therefore, accurate forecasting systems are essential for supporting production planning, export management, inventory control, and policy decision-making.

As markets become increasingly complex and difficult to predict, forecasting models capable of handling nonlinear and highly volatile data have attracted growing attention. Numerous studies have shown that machine learning, deep learning, and hybrid forecasting approaches can significantly improve forecasting accuracy for time-series data in the agricultural and commodity sectors (Dave et al., 2021; Mahawan et al., 2022).

For export-oriented economies such as Vietnam, fluctuations in rice prices can directly affect export revenues, farmers' incomes, and the overall performance of agricultural supply chains. Consequently, the development of accurate forecasting models not only supports risk management but also contributes to more effective strategic planning and enhances the competitiveness of the rice industry in international markets (Dave et al., 2021).

2.4. Forecasting Models and Methods

2.4.1. Traditional Econometric Forecasting Models

Before the widespread adoption of machine learning and deep learning techniques, traditional econometric forecasting models played a dominant role in time-series forecasting. These models were extensively applied to forecasting commodity prices, agricultural production, exchange rates, and various economic indicators due to their ability to extract statistical relationships from historical data (Sun et al., 2023; Sun et al., 2024). Traditional forecasting approaches are primarily based on the assumption that the past behavior of a time series can be utilized to predict future trends by modeling the autocorrelation structure and temporal dependencies within the data.

ARIMA Model

The Autoregressive Integrated Moving Average (ARIMA) model, developed by Box and Jenkins, is one of the most widely used approaches in time-series forecasting (Box et al., 2008). The model combines three components: Autoregressive (AR), Integrated (I), and Moving Average (MA) to capture the autocorrelation structure of time-series data.

ARIMA is denoted as $ARIMA(p, d, q)$, where p represents the order of the autoregressive component, d denotes the degree of differencing, and q indicates the order of the moving average component.

In practice, ARIMA is constructed by applying differencing of order d to the original time series in order to obtain a stationary series. The differenced series is then modeled using an $ARMA(p, q)$ structure. Therefore, the "Integrated" component of ARIMA reflects the differencing process performed prior to estimating the autoregressive and moving average components.

The general form of the ARIMA model can be expressed as follows:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \varepsilon_t \quad (2.5)$$

Where (y_t) is the value of the time series at time (t) , (c) is a constant term, (ϕ_i) represents the autoregressive coefficients, (θ_j) denotes the moving average coefficients, and (ε_t) is the random error term.

The AR component enables the model to capture the influence of past observations on the current value, while the MA component reflects the impact of previous forecasting errors. The Integrated (I) component is employed to address the non-stationarity of the data, allowing the time series to satisfy the assumptions required for model estimation.

Due to its simplicity, high interpretability, and strong theoretical foundation, ARIMA has been widely applied in forecasting commodity prices, exchange rates, and various economic and agricultural indicators (Sun et al., 2023). Furthermore, ARIMA continues to serve as a benchmark model in many contemporary forecasting studies (Dave et al., 2021).

However, ARIMA primarily relies on linear relationships and utilizes information solely from the target time series itself. As a result, its forecasting performance may be limited in situations where the variable of interest is significantly influenced by external factors.

ARIMAX Model

To overcome the limitations of ARIMA, the Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model was developed by incorporating exogenous variables into the ARIMA framework. As a result, the model not only utilizes historical information from the target time series but also allows relevant external information to be incorporated through one or more exogenous variables, thereby potentially improving forecasting performance when the selected variables contain useful predictive information (Wang et al., 2021). In the context of the international rice market, Thailand's rice price can serve as a market-based indicator that reflects information associated with broader market conditions, including supply and demand dynamics, climatic and production conditions, trade-policy developments, and other market shocks (FAO, 2024; World Bank, 2023; USDA FAS, 2024). This rationale is further supported by Thailand's prominent role in the global rice market and by evidence that Thai benchmark rice prices respond to changes in supply

conditions, exchange-rate movements, international demand, and other major market developments (World Bank, 2023; USDA FAS, 2024). Accordingly, this study employs Thailand's 5% broken white rice FOB price (THRICE) as the sole exogenous variable in the ARIMAX specification, rather than incorporating multiple external indicators separately.

The general form of the ARIMAX model can be expressed as follows:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \sum_{k=1}^m \beta_k X_{k,t} + \varepsilon_t \quad (2.6)$$

Where ($X_{k,t}$) represents the (k) exogenous variable at time (t), (β_k) denotes the coefficient measuring the impact of the exogenous variable, and the remaining notations are consistent with those used in the ARIMA model.

The inclusion of exogenous variables enables ARIMAX to incorporate additional predictive information beyond the historical dynamics of the target series, which may improve forecasting performance when the selected exogenous variable is relevant to the forecasting problem (Wang et al. 2021).

Accordingly, Thailand's 5% broken white rice FOB price is employed as the sole exogenous variable in the ARIMAX model to provide additional market information for forecasting Vietnam's 5% broken white rice FOB price.

Although ARIMAX can improve forecasting performance relative to a purely univariate ARIMA model when relevant exogenous information is available, the model still primarily represents linear relationships and may therefore be limited in capturing complex nonlinear patterns.

Overall, ARIMA and ARIMAX are important econometric forecasting models that remain widely used in time-series forecasting research. While ARIMAX extends ARIMA by incorporating relevant exogenous information, both models are primarily designed to capture linear time-series structures. Consequently, their ability to represent complex nonlinear patterns and abrupt market shocks may remain limited. This limitation provides the rationale for integrating ARIMAX with LSTM in the present study, where LSTM is subsequently applied to the ARIMAX residuals to capture nonlinear information that remains unexplained by the econometric component.

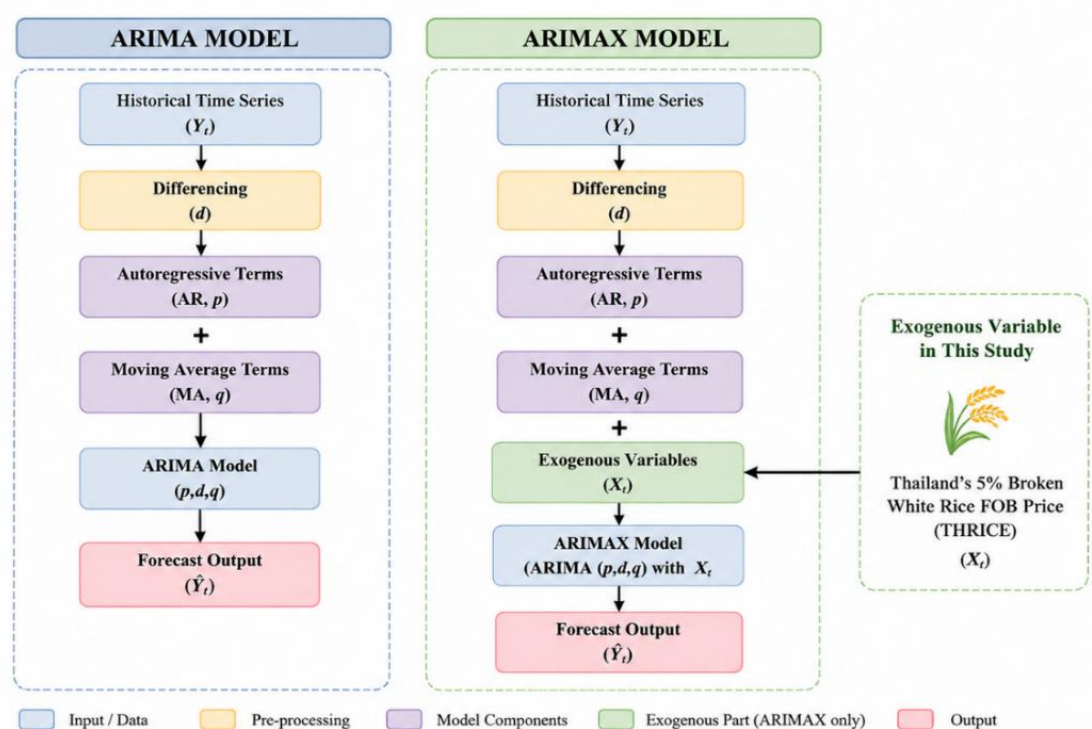


Figure 2.8. Conceptual Structure of ARIMA and ARIMAX Models

Source: Developed by the authors based on Box et al. (2008) and Wang et al. (2021).

Note: Y_t represents the dependent variable, Vietnam's 5% broken white rice FOB price; X_t represents the exogenous variable, Thailand's 5% broken white rice FOB price (THRICE); and $\hat{Y}_t$ represents the forecasted value of Vietnam's 5% broken white rice FOB price.

Figure 2.8 illustrates the basic structures of two widely used time-series forecasting models, namely ARIMA and ARIMAX. Both models utilize historical information through the Autoregressive (AR), Moving Average (MA), and Differencing (I) components to address the non-stationarity of time-series data. The primary distinction between the two models is that ARIMAX allows relevant exogenous information to be incorporated in addition to the historical dynamics of the target series. In the present study, Thailand's 5% broken white rice FOB price is incorporated as the sole exogenous variable, providing additional market information for forecasting Vietnam's 5% broken white rice FOB price.

2.4.2. Machine Learning and Deep Learning in Agricultural Forecasting

The limitations of traditional econometric models have driven the development of forecasting approaches based on machine learning and deep learning techniques. Unlike conventional statistical models, which often rely on assumptions of linearity and data stability, machine learning methods are capable of learning directly from data and identifying complex or nonlinear relationships without requiring the functional form of the model to be specified in advance (Sun et al., 2023). This capability is particularly important in agricultural markets, where prices are simultaneously influenced by multiple factors, including weather conditions, supply and demand dynamics, trade policies, and global economic shocks.

In recent years, various machine learning algorithms, such as Support Vector Machine (SVM), Random Forest (RF), and Artificial Neural Networks (ANNs), have been applied to forecasting agricultural prices and market indicators. These approaches have demonstrated the ability to improve forecasting accuracy compared with many traditional econometric models, particularly when dealing with nonlinear and highly volatile datasets (Dave et al., 2021; Sun et al., 2023).

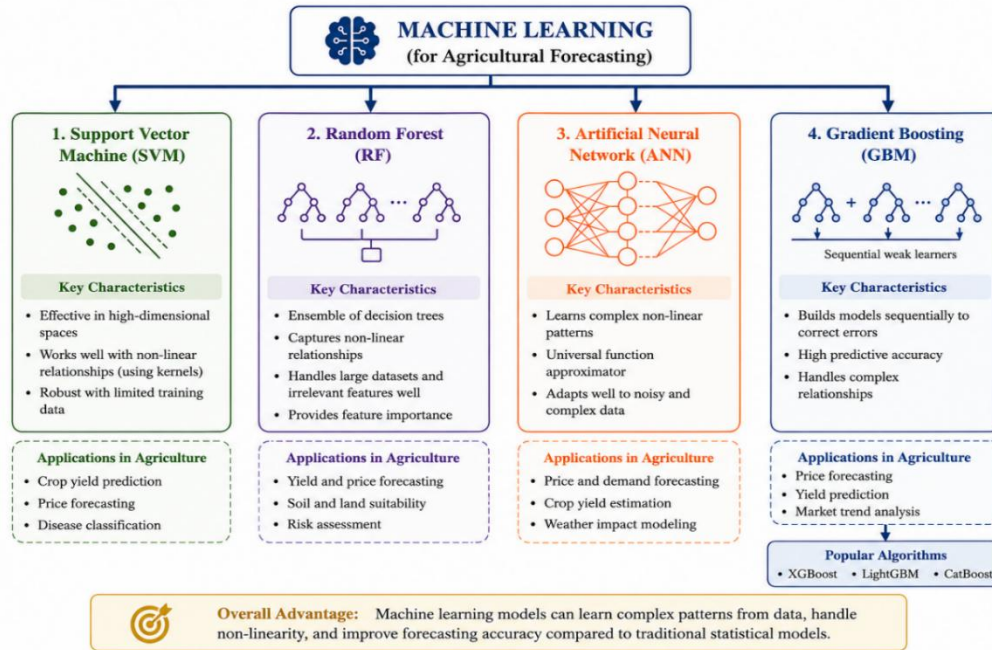


Figure 2.9. Machine Learning Approaches Used in Agricultural Forecasting

Source: Developed by the authors based on Sun et al. (2023) and Dave et al. (2021).

Figure 2.9 presents several popular machine learning algorithms that have been widely applied in agricultural forecasting. Among them, Support Vector Machine (SVM) is particularly effective in handling nonlinear data with small to medium-sized datasets; Random

Forest (RF) is recognized for its ability to process high-dimensional data and evaluate variable importance; Artificial Neural Networks (ANNs) are capable of learning complex nonlinear relationships; while Gradient Boosting Machines (GBMs) focus on improving forecasting accuracy through the combination of multiple weak learners. These algorithms have been extensively applied to forecasting agricultural prices, crop yields, and market trends, demonstrating superior performance over many traditional econometric models in dealing with nonlinear and volatile data (Dave et al., 2021; Sun et al., 2023).

However, most traditional machine learning algorithms still face limitations in effectively capturing long-term dependencies within time-series data. As agricultural prices become increasingly influenced by complex factors and continuous market fluctuations over time, there is a growing need for models capable of learning deeper sequential patterns from data. This demand has driven the development of deep learning approaches, particularly architectures such as Recurrent Neural Networks (RNNs), Long Short-Term Memory (LSTM), and Gated Recurrent Units (GRUs), to enhance forecasting performance for nonlinear and highly dynamic time-series datasets.

Among these deep learning architectures, LSTM is considered one of the most widely adopted models for time-series forecasting due to its ability to capture long-term dependencies and mitigate information loss during the learning process. A detailed discussion of the LSTM architecture and its operational mechanism is presented in the following section. To further illustrate the evolution of forecasting approaches from traditional econometric models to modern artificial intelligence techniques, Figure 2.9 presents the progression of forecasting methodologies in time-series research.

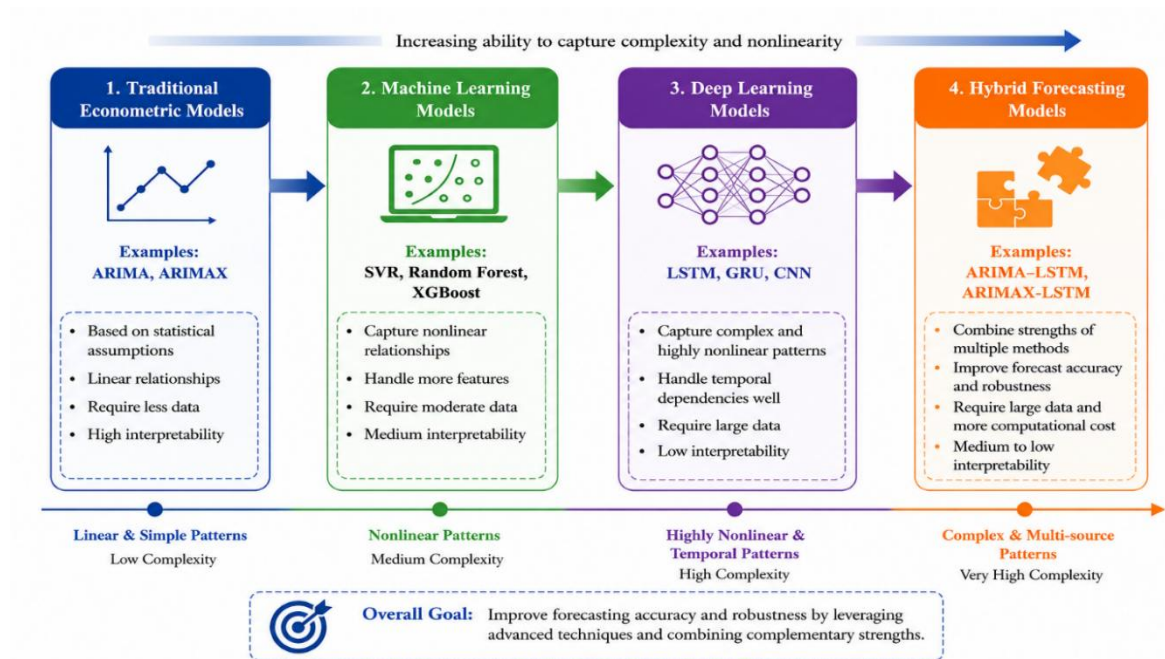


Figure 2.10. Evolution of Forecasting Methods

Source: Developed by the authors based on Sun et al. (2023), Dave et al. (2021), and Sun et al. (2024).

Figure 2.10 illustrates the evolution of forecasting approaches, from traditional econometric models such as ARIMA and ARIMAX to machine learning, deep learning, and hybrid forecasting methods. This development trend reflects the growing need to handle nonlinear data, multi-source information, and complex relationships within time-series datasets. Modern forecasting approaches not only improve forecasting accuracy but also enable more effective extraction of dynamic market characteristics, particularly in agricultural markets where volatility has become increasingly significant.

2.4.3. Deep Learning and the Long Short-Term Memory (LSTM) model

Among the deep learning approaches applied to time-series forecasting, Long Short-Term Memory (LSTM) is regarded as one of the most widely used and effective architectures. Unlike traditional machine learning methods, which often rely heavily on feature engineering, deep learning models are capable of automatically extracting relevant features from data and identifying complex nonlinear relationships within time series (Sun et al., 2023).

Among various deep learning architectures, Long Short-Term Memory (LSTM) is considered one of the most popular and effective models for time-series forecasting. LSTM

is a specialized form of Recurrent Neural Network (RNN) developed by Hochreiter and Schmidhuber to address the vanishing gradient problem commonly encountered in traditional recurrent networks. Through its memory cell structure and information control mechanisms known as gates, LSTM can retain and utilize information over extended periods, thereby improving forecasting performance for time-series data characterized by temporal dependencies (Gu et al., 2022; Wang et al., 2024).

One of the most important advantages of LSTM is its ability to capture both short-term and long-term relationships within time-series data. This characteristic makes LSTM particularly suitable for forecasting applications in finance, economics, and agriculture, where data are often influenced by multiple factors with persistent effects over time. Numerous studies have shown that LSTM can improve forecasting accuracy for nonlinear and highly volatile datasets compared with many traditional statistical approaches (Dave et al., 2021; Sun et al., 2023).

In the agricultural sector, LSTM has been increasingly applied to forecasting agricultural commodity prices, crop yields, product demand, weather conditions, and various supply-chain-related factors. Its ability to handle nonlinear patterns, high volatility, and temporal dependencies makes LSTM a suitable tool for agricultural market forecasting in an environment characterized by increasing climatic, economic, and trade-related shocks (Gu et al., 2022; Wang et al., 2024).

Despite its notable advantages, LSTM also has certain limitations. Deep learning models generally require large amounts of training data and involve substantially higher computational costs than traditional econometric models. Furthermore, model interpretability is often lower due to the complexity of neural network structures, making it more difficult to explain the underlying economic relationships captured by the model. These limitations have encouraged researchers to develop hybrid forecasting models that combine the interpretability of statistical methods with the nonlinear learning capabilities of deep learning techniques to enhance forecasting performance.

To provide a clearer understanding of the LSTM mechanism, Figure 2.11 illustrates the basic structure of an LSTM memory cell and the information-control gates involved in processing time-series data.

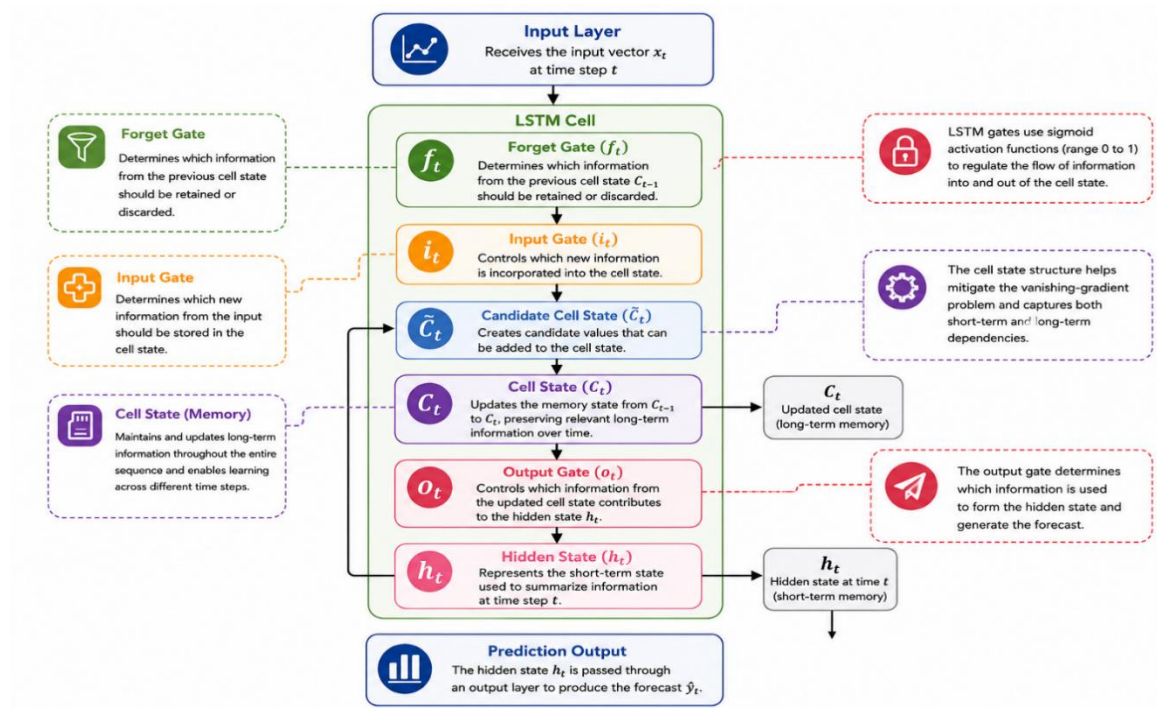


Figure 2.11. Architecture of an LSTM Network

Sources: Developed by the authors based on Gu et al. (2022) and Wang et al. (2024).

Figure 2.11 illustrates the basic architecture of the Long Short-Term Memory (LSTM) network. The model receives time-series input data and processes information through three main gates: the Forget Gate, Input Gate, and Output Gate in conjunction with the Cell State. The Forget Gate determines which information from previous time steps should be retained or discarded; the Input Gate decides which new information should be incorporated into the memory state; and the Output Gate selects the relevant information to generate the forecasting output. Through this mechanism, LSTM is able to maintain and update important information over extended periods while mitigating the vanishing gradient problem commonly encountered in traditional Recurrent Neural Networks (RNNs) (Gu et al., 2022; Wang et al., 2024).

The ability to capture both short-term and long-term dependencies makes LSTM well suited to modeling complex nonlinear patterns in time-series data, including agricultural commodity prices and economic indicators. Numerous studies have demonstrated that LSTM often achieves higher forecasting accuracy than traditional statistical models in complex and dynamic data environments (Dave et al., 2021; Sun et al., 2023).

Despite these advantages, LSTM also has several limitations, including the requirement for relatively large training datasets, high computational costs, and limited interpretability. As a result, recent studies have increasingly developed hybrid forecasting models that combine the linear modeling capabilities of statistical approaches with the nonlinear learning strengths of deep learning techniques. Among these approaches, ARIMA-LSTM and ARIMAX-LSTM models have demonstrated promising results in time-series forecasting. Rather than necessarily using LSTM as an independent forecasting model, residual-based hybrid frameworks can employ LSTM to capture nonlinear information remaining after the linear component has been modeled by ARIMA or ARIMAX. This structure provides the methodological basis for the Hybrid ARIMAX-LSTM framework adopted in the present study.

2.4.4. Hybrid Forecasting Models

In recent years, forecasting research has increasingly focused on the development of hybrid forecasting models that combine the strengths of traditional econometric approaches with machine learning and deep learning techniques. This approach stems from the fact that each forecasting method possesses its own advantages and limitations. While econometric models are effective in capturing linear relationships and offer high interpretability, machine learning and deep learning methods excel at identifying nonlinear patterns and complex interactions within data. Consequently, hybrid models are designed to leverage the strengths of multiple approaches while mitigating the individual limitations of each model (Mahawan et al., 2022; Allassak et al., 2025).

One of the most widely adopted hybrid approaches is the integration of ARIMA or ARIMAX models with Long Short-Term Memory (LSTM) networks. In this framework, ARIMA or ARIMAX is typically employed to model the linear components of the target series, while ARIMAX additionally incorporates information from the specified exogenous variable(s). LSTM can then be applied to the residuals generated by the econometric model to learn nonlinear patterns and temporal dependencies that remain unexplained by the linear component. This combination enables researchers to simultaneously benefit from the interpretability of statistical models and the powerful nonlinear learning capability of deep learning techniques (Dave et al., 2021; Mahawan et al., 2022).

Prior studies have reported that hybrid ARIMA-LSTM and ARIMAX-LSTM models can achieve improved forecasting performance relative to their corresponding baseline models, particularly in environments characterized by high volatility and pronounced nonlinear behavior (Dave et al., 2021; Allassak et al., 2025).

In agricultural markets, hybrid forecasting approaches are considered particularly suitable because agricultural price series often contain both linear economic relationships and nonlinear behaviors arising from weather fluctuations, changes in supply and demand, trade policies, and international market shocks. Recent review studies on agricultural price forecasting indicate a growing trend toward the adoption of hybrid approaches to improve forecasting performance in complex and highly volatile data environments (Sun et al., 2023).

Although the number of studies on hybrid forecasting has increased substantially, most existing research has focused on areas such as stock markets, exchange rates, energy markets, and production demand forecasting (Dave et al., 2021; Mahawan et al., 2022; Allassak et al., 2025). In contrast, the application of ARIMAX-LSTM models in agricultural trade forecasting remains relatively limited compared with their use in finance, energy, and industrial production. This gap suggests significant potential for extending hybrid forecasting approaches to agricultural price forecasting in increasingly volatile market conditions. Meanwhile, Vietnam's rice export prices are simultaneously influenced by multiple factors, including fluctuations in global supply, climatic conditions, and international trade policies (Sun et al., 2023). These characteristics highlight the need for forecasting models capable of integrating linear information, nonlinear patterns, and exogenous variables within a unified framework (Mahawan et al., 2022; Allassak et al., 2025).

Furthermore, the selection of the ARIMAX-LSTM model in this study is motivated by the characteristics of Vietnam's rice export price data, which may contain linear dynamics, information associated with the exogenous variable Thailand's rice export price and nonlinear patterns that may remain unexplained by the linear component. Therefore, the Hybrid ARIMAX-LSTM framework is designed to combine ARIMAX-based linear and exogenous information with LSTM-based residual learning, with the objective of improving forecasting performance relative to the standalone ARIMAX baseline.

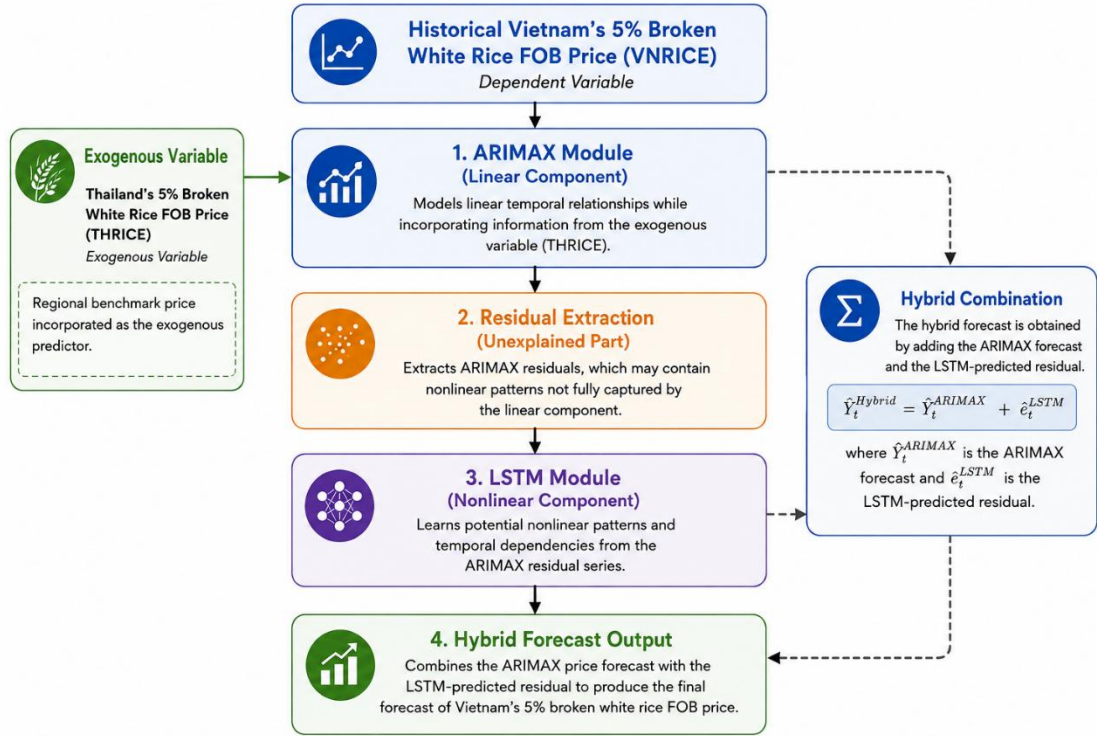


Figure 2.12. Proposed Hybrid ARIMAX-LSTM Forecasting Framework

Source: Developed by the authors based on Dave et al. (2021), Mahawan et al. (2022), and Allassak et al. (2025).

Figure 2.12 illustrates the proposed Hybrid ARIMAX-LSTM framework employed in this study. Within this framework, the ARIMAX model is utilized to capture the linear dynamics of Vietnam's rice export price series while incorporating Thailand's 5% broken white rice FOB price (THRICE) as the sole exogenous variable. The residuals generated by the ARIMAX model are subsequently used as the input to the LSTM network, which is designed to learn the remaining nonlinear patterns and temporal dependencies not captured by the econometric component. Finally, the ARIMAX forecast and the LSTM-based residual forecast are combined to generate the final Hybrid ARIMAX-LSTM forecast.

2.5. Empirical Evidence on Agricultural Price Forecasting

Research on agricultural price forecasting has expanded significantly in recent years due to the increasing need to enhance risk management and support decision-making in production, trade, and policy formulation. Along with advances in data availability and compu-

tational technologies, forecasting approaches have gradually evolved from traditional econometric models to hybrid models in order to improve forecasting accuracy for highly volatile time series data.

Empirical Studies Using Econometric Forecasting Models

Traditional econometric models, particularly ARIMA and its extensions, have been widely applied in agricultural price forecasting due to their strong theoretical foundation and clear interpretability. According to Wang et al. (2021), incorporating exogenous variables into the ARIMAX model can significantly improve forecasting performance when the target variable is influenced by external economic and market factors. Their findings indicate that ARIMAX generally achieves higher forecasting accuracy than ARIMA when relevant exogenous variables are included in the model.

In the context of the rice market, Chen and Saghaian (2016) examined the relationships among rice export prices in Vietnam, Thailand, and the United States using time-series econometric techniques. Their results revealed the existence of a long-run cointegration relationship among major rice-exporting markets worldwide. The study also found that Thailand's rice price plays a significant role in regional price transmission, suggesting that Thai rice prices can serve as an important exogenous variable in forecasting models for Vietnam's rice export prices.

Although traditional econometric models continue to be widely used as benchmark models, many studies have shown that their forecasting performance often deteriorates when the data exhibit nonlinear characteristics or are affected by unusual market shocks (Sun et al., 2023). This limitation has encouraged the development of more advanced forecasting approaches based on machine learning, deep learning, and, in particular, hybrid forecasting models.

Empirical Studies Using Hybrid Forecasting Models

In recent years, hybrid forecasting models have attracted considerable attention from researchers due to their ability to combine the strengths of traditional statistical models with deep learning techniques. Numerous studies have demonstrated that integrating linear and nonlinear components can significantly improve forecasting performance for highly volatile time series.

Dave et al. (2021) developed an ARIMA-LSTM model to forecast Indonesia's export value. The results showed that the hybrid model achieved substantially lower RMSE and MAPE values than the standalone ARIMA and LSTM models. The authors concluded that combining statistical methods with deep learning techniques can significantly enhance forecasting accuracy for highly volatile economic data.

Similarly, Mahawan et al. (2022) proposed an ARIMA-LSTM model for handling nonlinear time series. Their empirical findings indicated that the hybrid model was capable of capturing both the linear and nonlinear characteristics of the data, thereby improving forecasting performance compared with individual models.

More recently, Allassak et al. (2025) applied an ARIMAX-LSTM model to time series forecasting with the inclusion of exogenous variables. The results demonstrated that combining the linear modeling capability of ARIMAX with the nonlinear learning capacity of LSTM significantly improved forecasting accuracy. This study provides important empirical evidence regarding the potential of the ARIMAX-LSTM framework for forecasting problems characterized by complex dynamics and the influence of multiple external factors.

Overall, recent empirical studies indicate a growing shift from traditional econometric models toward hybrid forecasting approaches. Among these, the ARIMAX-LSTM model has emerged as a promising framework due to its ability to simultaneously incorporate linear information, nonlinear patterns, and exogenous variables within a unified forecasting structure. Empirical evidence suggests that this approach has considerable potential to improve forecasting accuracy for complex time series influenced by multiple external factors.

Although previous studies have provided substantial evidence regarding the effectiveness of modern forecasting methods, several limitations remain concerning research contexts, methodological applications, and their applicability to export rice markets. These issues are discussed in the following Research Gap section.

2.6. Research Gap

Although significant advancements have been made in time series forecasting in recent years, studies specifically focusing on forecasting Vietnam's rice export prices remain relatively limited. Most existing research has concentrated on commonly traded agricultural commodities or sectors such as finance, energy, and industrial production. In addition, many

studies continue to rely on single forecasting models, which may limit their ability to fully capture the complex characteristics of the rice market.

In practice, Vietnam's rice export prices are influenced by a combination of linear components, nonlinear relationships, and external factors such as Thailand's rice export price. However, studies that simultaneously incorporate these characteristics within a unified forecasting framework are still scarce. This highlights the need for forecasting models capable of leveraging historical price information, exogenous variables, and nonlinear patterns within the data.

To address this research gap, this study proposes the development of a hybrid ARIMAX-LSTM model for forecasting Vietnam's rice export prices. The proposed model is expected to combine the strengths of ARIMAX in capturing linear relationships and the effects of exogenous variables with the capability of LSTM to learn nonlinear patterns. As a result, the model is anticipated to improve forecasting accuracy and provide additional empirical evidence to the field of agricultural price forecasting.

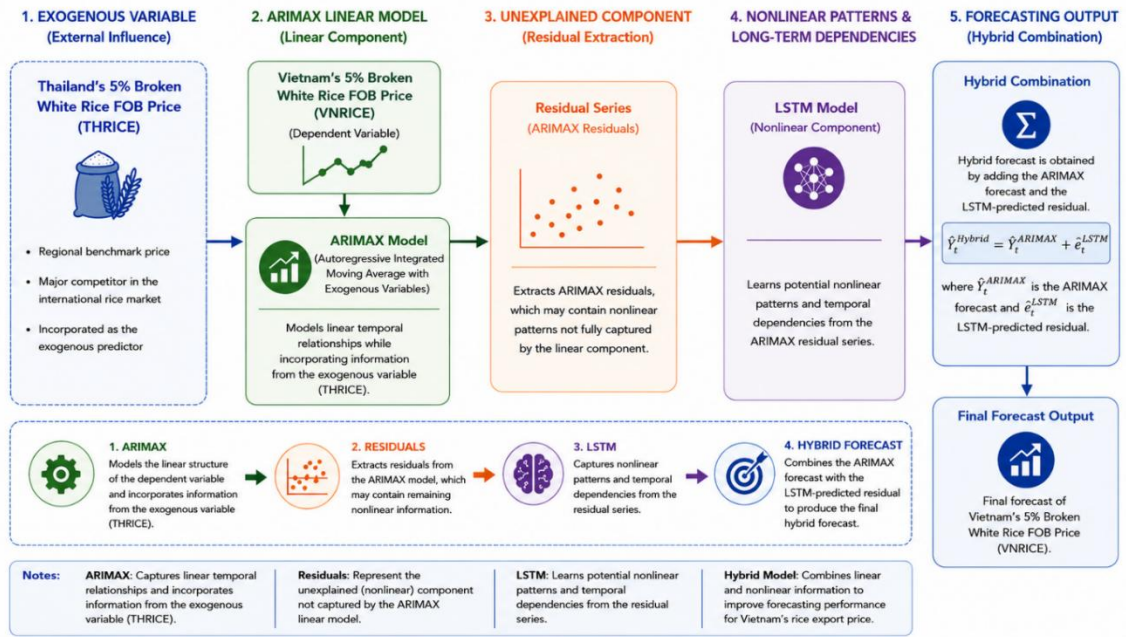
2.7. Conceptual Framework

Based on the research objectives and the selected variables, this study proposes a conceptual framework to forecast Vietnam's rice export price using a hybrid ARIMAX-LSTM model.

In this framework, Vietnam's rice export price serves as the target variable. One exogenous variable, namely Thailand's rice export price, is incorporated into the model as an input variable. Historical data on these variables are utilized to develop the forecasting model.

The research process consists of two stages. In the first stage, the ARIMAX model is employed to capture the linear components of Vietnam's rice export price series and the effects of the exogenous variable. Subsequently, the residuals generated by the ARIMAX model are fed into the LSTM model to learn nonlinear patterns and long-term dependencies that remain unexplained in the previous stage. The outputs from these two stages are then combined to generate the final forecast of Vietnam's rice export price.

The proposed research framework is presented in Figure 2.13, illustrating the relationships among the input variables, the processing mechanism of the ARIMAX-LSTM model, and the forecasting output of the study.



Source: Developed by the authors based on Chen and Saghaian (2016), Wang et al. (2021), Dave et al. (2021), Mahawan et al. (2022), and Sun et al. (2023).

Figure 2.13. Conceptual Framework of the Proposed Hybrid ARIMAX-LSTM Model for Forecasting Vietnam's Rice Export Price Volatility

Source: Developed by the authors based on Chen and Saghaian (2016), Wang et al. (2021), Dave et al. (2021), Mahawan et al. (2022), and Sun et al. (2023).

Figure 2.13 illustrates the research framework proposed in this study. Vietnam's rice export price is identified as the target variable to be forecast. One exogenous variable, namely Thailand's rice export price, is incorporated as an input variable in the forecasting model.

In the first stage, the ARIMAX model is employed to capture the dynamics of Vietnam's rice export price series and the effect of the exogenous variable. The residuals generated by the ARIMAX model are then passed to the LSTM model to capture nonlinear patterns and temporal dependencies that remain unexplained in the previous stage.

The forecasting results from these two stages are subsequently combined to generate the final forecast of Vietnam's rice export price. Based on this research framework, the next chapter will present the research data, research methodology, and the process of developing the ARIMAX-LSTM model in detail.

CHAPTER 3. METHODOLOGY

This chapter presents the methodological framework employed to develop and evaluate the forecasting models for Vietnam's 5% broken white rice export prices. It describes the research design, data collection process, and preprocessing procedures applied to the monthly time-series data of Vietnam and Thailand rice prices. The chapter then details the analytical procedures for constructing the ARIMAX model, including stationarity testing, model identification, estimation, and diagnostic checking, followed by the development of the Hybrid ARIMAX-LSTM framework in which LSTM is used to capture nonlinear patterns remaining in the ARIMAX residuals. Finally, the forecasting performance of the models is evaluated using MAE, RMSE, MAPE, and SMAPE, providing a systematic methodological basis for the empirical analysis and model comparison presented in the following chapter.

3.1. Research Design

This study adopts a quantitative research design to analyze and forecast the volatility of Vietnam's rice export prices over a 12-month horizon. The research design integrates traditional time-series econometrics with modern deep learning, forming a comprehensive hybrid analytical framework. According to Box and Jenkins (1976), a quantitative approach is most appropriate when the research objective is to model the underlying temporal patterns in numerical data and generate statistically grounded forecasts.

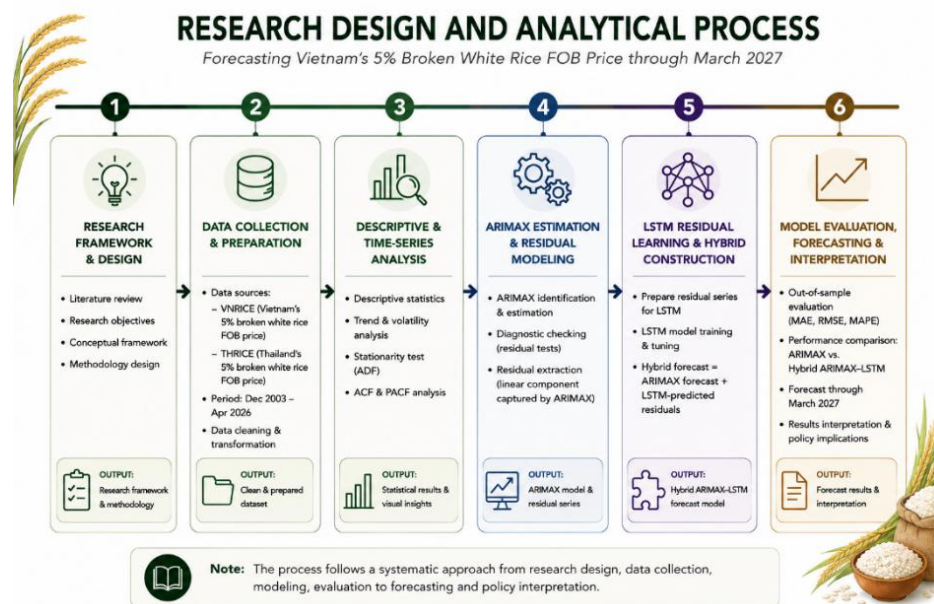


Figure 3.1. Research design and analytical process for forecasting Vietnam's Rice Export Prices to March 2027.

Source: Author's compilation.

In terms of analytical methods, the study employs both descriptive analysis and forecasting methods. First, descriptive analysis is used to examine the fluctuation trends of Vietnam's 5% broken white rice export price over the period from December 2003 to March 2026. This stage addresses Research Objective 1 by identifying long-term trends, periods of strong volatility, and the characteristics of the rice export price series over time.

Subsequently, the study applies forecasting methods to construct and evaluate the ARIMAX model and the hybrid ARIMAX-LSTM model for forecasting Vietnam's rice export prices. This content addresses Research Objective 2 by comparing the forecasting performance of the traditional econometric model against the hybrid deep-learning-integrated model, thereby identifying the model with higher accuracy.

Regarding the research strategy, the study uses secondary monthly time-series data collected from the World Bank Commodity Price Data (Pink Sheet) database. Two data series are used: the FOB price of Vietnam's 5% broken white rice and the FOB price of Thailand's 5% broken white rice. Vietnam's FOB rice price is selected as the dependent variable to be forecast, while Thailand's FOB rice price is used as an exogenous variable in the ARIMAX model.

The ARIMAX model was selected for its ability to analyze time-series data while simultaneously accounting for the effects of exogenous variables on the target variable. The ARIMAX model-building process follows the four main steps of the Box-Jenkins methodology: model identification, parameter estimation, diagnostic checking, and forecasting. By incorporating Thailand's rice price into the model, the study expects to better capture the competitive relationship and price linkages within the international rice market.

However, rice export prices are typically influenced by numerous complex factors, such as fluctuations in global supply and demand, export policies, logistics costs, weather conditions, and international economic shocks. As a result, the data series may contain nonlinear components that the ARIMAX model cannot fully capture. To address this limitation, the study further employs a Long Short-Term Memory (LSTM) network. Specifically, the residuals obtained from the ARIMAX model are used as input data for the LSTM network to learn the remaining nonlinear patterns in the data series. The forecast results from ARIMAX and LSTM are then combined to form the hybrid ARIMAX-LSTM model.

To evaluate the forecasting performance of the models, the study uses common error metrics, including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and

Mean Absolute Percentage Error (MAPE). These metrics are widely used in time-series forecasting studies and enable an objective comparison across models. In addition, the study applies out-of-sample testing by splitting the dataset into training and testing sets to accurately assess the actual forecasting capability of each model.

The hybrid ARIMAX-LSTM model is considered well-suited to this study because rice export prices are influenced by both linear relationships and nonlinear characteristics as well as unpredictable market shocks. While ARIMAX effectively explains economic relationships and the effects of exogenous variables, LSTM is capable of learning nonlinear structures and long-term dependencies within the data. Combining the two models is expected to improve forecasting accuracy compared with using either model individually.

Regarding the study's time scope, monthly data were collected from December 2003 to March 2026, resulting in a final analytical dataset of 268 monthly observations. Based on this historical dataset, the study forecasts Vietnam's rice export prices for the period from April 2026 to March 2027. The use of monthly data allows the model to better capture short-term fluctuations and market changes compared with annual data.

The unit of analysis in this study is Vietnam's rice export market at the national level. The study does not conduct separate forecasts for individual localities, enterprises, or specific import markets. Instead, the study focuses on the FOB price of Vietnam's 5% broken white rice as a representative indicator of the competitiveness and price fluctuation trends of Vietnam's rice export industry in the international market.

Overall, the research design ensures consistency between the theoretical foundation presented in Chapter 2, the empirical analytical strategy based on the ARIMAX model and the hybrid ARIMAX-LSTM model, and the research objective of forecasting the volatility of Vietnam's rice export prices through March 2027. By integrating econometric modeling with deep learning, the study aims to improve forecasting accuracy while providing a scientific basis for policymaking and decision-making in the rice export sector.

3.2. Data Collection

3.2.1. Secondary Data

To support the analysis and forecasting of Vietnam's rice export price volatility, the study uses secondary time-series data collected from the World Bank Commodity Price Data (Pink Sheet) database over the period from December 2003 to March 2026. The data collection process was carried out systematically to ensure completeness, consistency, and reliability for the subsequent analysis and forecasting model development.

In this study, two main variables are used: Vietnam's 5% broken white rice export price and Thailand's 5% broken white rice export price. Both data series are published periodically by the World Bank and are regarded as benchmark prices reflecting developments in the international rice market.

Specifically, Vietnam's 5% broken white rice export price (Vietnam Rice Price - VNRICE) is selected as the dependent variable in the study. This is the FOB (Free on Board) price of Vietnam's 5% broken white rice, measured in USD/ton. This variable reflects the export price level representative of Vietnam's rice industry in the international market and is the main object to be forecast in this study.

For the exogenous variable, the study uses Thailand's 5% broken white rice export price (Thailand Rice Price - THRICE). Thailand is one of the world's largest rice exporters and a direct competitor of Vietnam in the international market. Consequently, fluctuations in Thailand's rice export price often have a significant influence on Vietnam's rice export price. Incorporating Thailand's rice price into the ARIMAX model is expected to improve the explanatory power and forecasting accuracy of the model.

Overall, the simultaneous use of Vietnam's and Thailand's rice prices allows the study to examine both the fluctuation trends of Vietnam's export prices and the impact of competitive factors from the regional market. The variables used in the study are summarized in Table 3.1 below.

Table 3.1. Description of Variables Used in the Study

Variable	Abbreviation	Definition	Unit of Measurement	Data Source
Vietnam Rice Price	VNRICE	FOB export price of Vietnam's 5% broken white rice	USD/Ton	World Bank
Thailand Rice Price	THRICE	FOB export price of Thailand's 5% broken white rice	USD/Ton	World Bank

Regarding the World Bank Commodity Price Data (Pink Sheet) database, the data were extracted from the Agricultural Commodities group, selecting two benchmark price series: Rice (Vietnam), 5% broken, white rice, FOB Ha Noi, and Rice (Thailand), 5% broken, white rice, FOB Bangkok. These price series are regularly published by the World Bank and widely used in studies on the international rice market. The World Bank data extraction interface is illustrated in Figure 3.2.

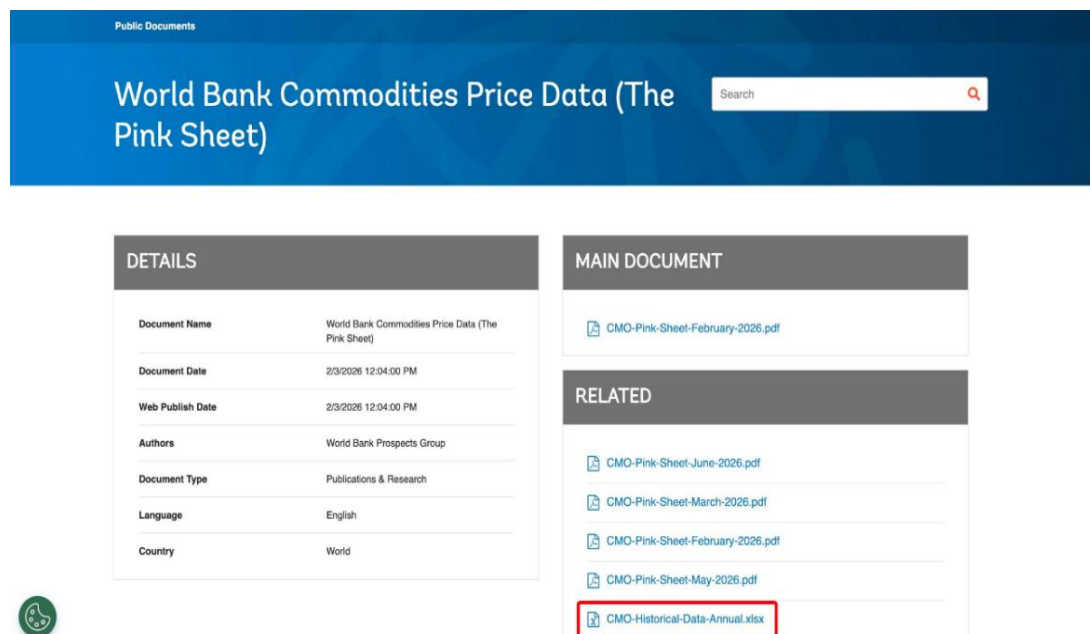


Figure 3.2. World Bank interface used to extract Vietnam’s Rice and Thailand’s rice data.

Source: Screenshot from World Bank

After collection, the data from the World Bank were checked for completeness, time-format consistency, and consistency across the data series. The subsequent data-processing steps included checking for missing values, testing the stationarity of the time series, and standardizing the data before proceeding to the construction of the ARIMAX and Hybrid ARIMAX-LSTM models.

In addition, the price levels published on the World Bank's data system were cross-checked against the downloaded dataset to enhance the reliability and accuracy of the input data.

In summary, the study uses monthly rice export price data collected from the World Bank Commodity Price Data (Pink Sheet). This dataset serves as the foundation for the data preprocessing, descriptive analysis, stationarity testing, model construction, and forecast evaluation steps presented in the following sections.

3.2.2. Data Preprocessing

The data collected from the World Bank Commodity Price Data (Pink Sheet) were processed prior to model construction to ensure accuracy, consistency, and compliance with the requirements of time-series forecasting methods. The data preprocessing was carried out in several steps to transform the raw data into a complete dataset suitable for building the ARIMAX and Hybrid ARIMAX-LSTM models.

First, the data were downloaded from the World Bank database in spreadsheet format. Descriptive rows, technical notes, and information unrelated to the price series were removed to retain only the observations necessary for the study. The dataset was then filtered to include only the two price series used in the study -the FOB price of Vietnam's 5% broken white rice and the FOB price of Thailand's 5% broken white rice-over the period from December 2003 to March 2026.

Following this step, the data were standardized to ensure consistency across variables. The time variable was arranged in ascending chronological order from December 2003 to March 2026. All observations were rechecked for numerical format, unit of measurement, and consistency across the data series. Both study variables were recorded in USD per ton (USD/Ton), consistent with international price-quotation standards for exported rice.

After completing the data-cleaning step, the dataset was reorganized into a monthly time series, in which each observation corresponds to a given month and the corresponding

values of the variables at that time. The time variable was also declared in the statistical software to establish the time-series structure of the data. This is an important step to ensure that subsequent procedures-such as stationarity testing, determination of the differencing order, ACF/PACF plot analysis, and construction of the ARIMAX model -are carried out in accordance with the principles of time-series analysis.

For the ARIMAX model, the data were tested for stationarity using the Augmented Dickey-Fuller (ADF) test. Where the data series was found to be non-stationary at the level, the study applied differencing until statistical stationarity was achieved. Stationarity testing is a necessary condition to ensure the validity of the ARIMAX model and to avoid the phenomenon of spurious regression.

In addition, the study examined the completeness of the data to identify the presence of missing values and outliers in the time series. The review revealed that certain time points in the World Bank Commodity Price Data dataset had no recorded observations. To ensure the continuity of the data series and to meet the requirements of time-series forecasting models, missing values were addressed using linear interpolation. This method allows missing observations to be estimated based on the trend of neighboring data points, thereby maintaining the consistency of the time series while minimizing disruption to the analysis and model-building process.

After the data-cleaning and processing procedures, the final dataset comprises the standardized Vietnam and Thailand rice export price series, checked for completeness and with missing values addressed. This dataset serves as the foundation for the stationarity testing, ARIMAX model construction, LSTM model training, and forecast evaluation steps presented in the following sections.

3.3. Research Analysis

3.3.1. Descriptive Statistics

Descriptive statistics is a method used to summarize, describe, and present the basic characteristics of the research data through statistical indicators. According to Brooks (2019), descriptive statistics provide an overview of the data by reflecting its central characteristics and dispersion, thereby helping researchers understand the nature of the data before conducting more in-depth analyses.

In this study, descriptive statistics are used to examine the characteristics of the 5% broken rice export price series of Vietnam and Thailand over the period from December 2003 to March 2026. The indicators used include the Mean, Standard Deviation, Minimum, and Maximum values. These indicators help assess the average price level, the degree of volatility, and the range of fluctuation of the data throughout the study period.

In addition, descriptive statistics help detect outliers and provide a preliminary assessment of the fluctuation trends of the time series. This is a necessary step to understand the characteristics of the data before conducting statistical tests and building forecasting models. According to Cooksey (2020), descriptive statistical analysis plays an important role in quantitative research because it ensures that the data are fully understood and evaluated before applying more advanced analytical methods.

The results of the descriptive statistics step will serve as the basis for subsequent analyses, including testing the stationarity of the data series and constructing the Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model to forecast the volatility of Vietnam's rice export prices.

3.3.2. ARIMAX Model Development and Diagnostics

a. ARIMAX Model Specification

To forecast Vietnam's rice export prices, the study employs the Autoregressive Integrated Moving Average with Exogenous Variables (ARIMAX) model. This is an extension of the traditional ARIMA model that allows exogenous variables to be incorporated into the forecasting process. Compared with ARIMA, the ARIMAX model not only exploits information from the past values of the dependent variable but also considers the effects of external factors related to the target variable (Hyndman and Athanasopoulos, 2021).

In this study, the FOB price of Vietnam's 5% broken white rice (VNRICE) is used as the dependent variable, while the FOB price of Thailand's 5% broken white rice (THAIRICE) is used as the exogenous variable. The selection of Thailand's rice price is based on Thailand's role as one of the world's largest rice exporters and a direct competitor of Vietnam in the international market. Previous studies have shown that rice prices between the two countries exhibit a positive co-movement and that a long-run price transmission mechanism exists between them (Chen and Saghaian, 2016).

Since the study data are monthly time series, the ARIMAX model is constructed following the Box-Jenkins methodology, comprising four main steps: model identification, parameter estimation, diagnostic checking, and forecasting. This process allows for the selection of the ARIMAX(p,d,q) structure that best fits the characteristics of the study's data series (Box et al., 2015).

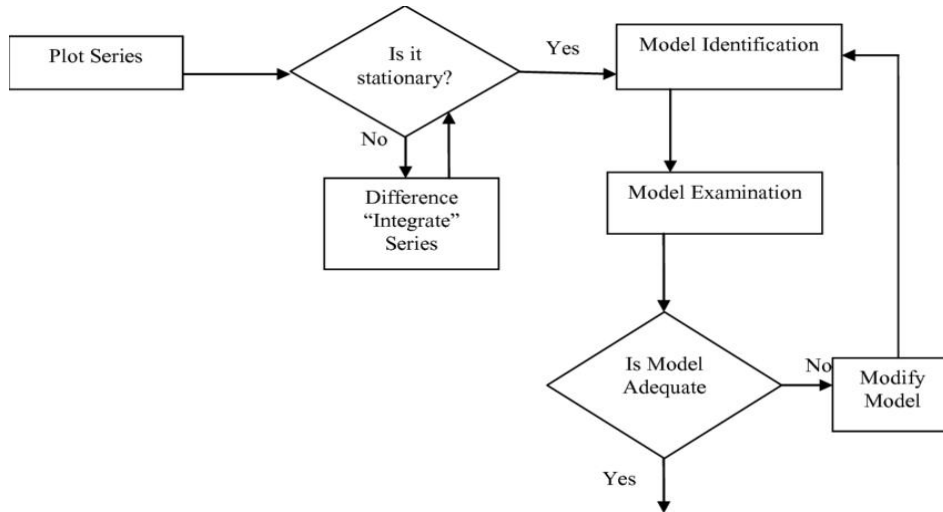


Figure 3.3. Steps in forecasting using the Box-Jenkins methodology

Source: Adapted from Box et al. (2015); Brooks (2019).

First, the time series must be tested for stationarity. A series is considered stationary when its basic statistical properties such as the mean, variance, and covariance do not change over time. If the series is non-stationary, differencing must be applied until stationarity is achieved before proceeding with model identification.

According to Box et al. (2015), the general ARIMAX model is expressed as follows:

$$y_t = c + \sum_{i=1}^p \phi_i y_{t-i} + \sum_{j=1}^q \theta_j \varepsilon_{t-j} + \sum_{k=1}^m \beta_k X_{k,t} + \varepsilon_t \quad (3.1)$$

Where (y_t) is the value of the series at time (t) , (c) is a constant, (ϕ_i) represents the autoregressive coefficients, (θ_j) represents the moving average coefficients, and (ε_t) is the random error term; $(X_{k,t})$ is the (k) exogenous variable at time (t) , (β_k) is the coefficient representing the effect of the exogenous variable, and the remaining notations are consistent with the ARIMA model.

For stationary time series, the Autoregressive Moving Average (ARMA) model serves as the fundamental analytical framework. The general form of this model is defined as follows:

$$Y_t = \phi_1 Y_{t-1} + \dots + \phi_p Y_{t-p} + a_t - \theta_1 a_{t-1} - \dots - \theta_q a_{t-q} \quad (3.2)$$

Where ϕ_i represents the autoregressive coefficients, θ_j represents the moving average coefficients, and a_t is the random error term (white noise). When the data series is non-stationary, the Box-Jenkins methodology applies differencing techniques to transform the series into a stationary state. On this basis, the ARIMA(p,d,q) model is defined as:

$$\phi_p(B)(1-B)^d Y_t = \theta_q(B) a_t \quad (3.3)$$

In this equation, d represents the order of differencing applied to the data series. In practice, the optimal values of p , d , and q are selected based on the specific characteristics of the data, combined with the analysis of the Autocorrelation Function (ACF) and Partial Autocorrelation Function (PACF) plots.

However, in real-world economic problems, the target variable y_t depends not only on its own past values but is also influenced by external macroeconomic factors. Therefore, the ARIMA model is extended into the ARIMAX(p,d,q) model by incorporating exogenous variables. The mathematical form of the ARIMAX model is restructured as follows:

$$\phi_p(B)(1-B)^d Y_t = \sum_{k=1}^K \beta_k X_{k,t} + \theta_q(B) a_t \quad (3.4)$$

Where $X_{k,t}$ is the k exogenous variable at time t (for example, prices of related commodities, exchange rates, or macroeconomic indicators), and β_k is the coefficient representing the effect of that exogenous variable on the target variable. The procedure for determining the internal parameters (p, d, q) of the ARIMA component within this hybrid model continues to follow the principles of ACF and PACF analysis, after the data series has been made stationary and the effects of the exogenous variables have been accounted for. In addition, the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) are used to select the most appropriate model.

The final model is selected based on the principle of parsimony, meaning a model with a reasonable number of parameters that still ensures good explanatory and forecasting ability.

At the same time, the model's residuals must satisfy the property of white noise, that is, no significant autocorrelation should remain after estimation.

b. Stationarity Testing

Before constructing the ARIMAX model for Vietnam's rice export price series, it is necessary to test the stationarity of the time series. According to Brooks (2019), before conducting formal tests, stationarity can be preliminarily assessed through visual inspection of time-series plots. The illustrations provided by Brooks clearly distinguish between a stationary series, a non-stationary series with a stochastic trend, and a series with a deterministic trend.

Specifically, a white noise process fluctuates around a constant meaning and frequently crosses this mean level. In contrast, a random walk and a random walk with drift typically exhibit prolonged fluctuations, deviating from the mean for extended periods and rarely returning to that level. For a deterministic trend process, the series does not have a fixed meaning but instead fluctuates randomly around a trend that increases over time.

Brooks also demonstrates that in a first-order autoregressive process, AR (1), the value of the autoregressive coefficient ϕ determines the nature of the series:

When $\phi = 0$, the series is white noise. When $\phi = 0.8$, the series is a stationary AR (1) process. When $\phi = 1$, the series becomes a random walk, indicating a non-stationary process.

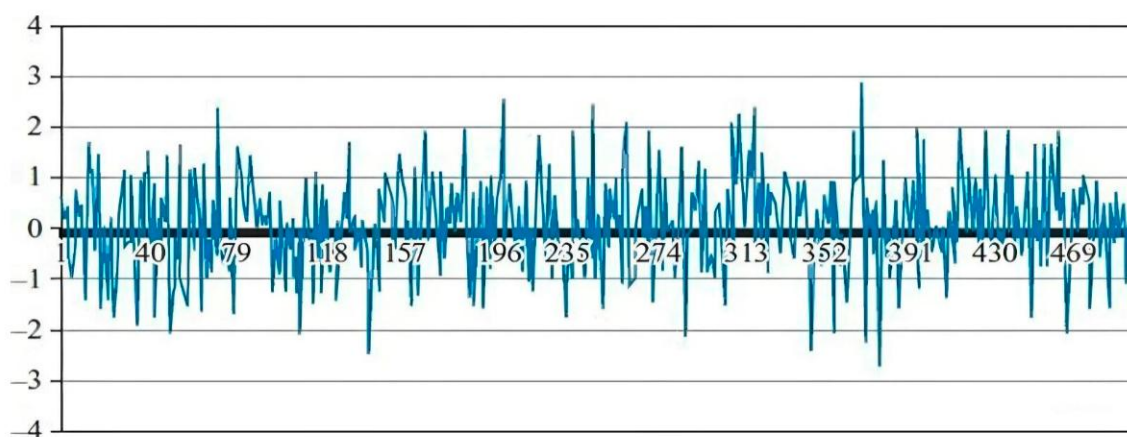


Figure 3.4. Example of a white noise process.

Source: Brooks, C. (2019). Introductory econometrics for finance (4th ed., p. 444). Cambridge University Press.

Figure 3.4 illustrates a white noise series, in which the observations fluctuate randomly around a constant mean of zero and display no discernible trend. White noise represents a specific form of stationary process and is used here as a visual reference for illustrating stable behavior over time. In the context of this study, if the differenced series of Vietnam's rice export price (ΔVNRICE_t) exhibits a broadly stable pattern around a constant mean after differencing, this provides preliminary visual evidence that the series may have achieved stationarity. However, visual inspection alone is not sufficient to establish stationarity; therefore, the Augmented Dickey-Fuller (ADF) test is subsequently employed for formal verification (Brooks, 2019).

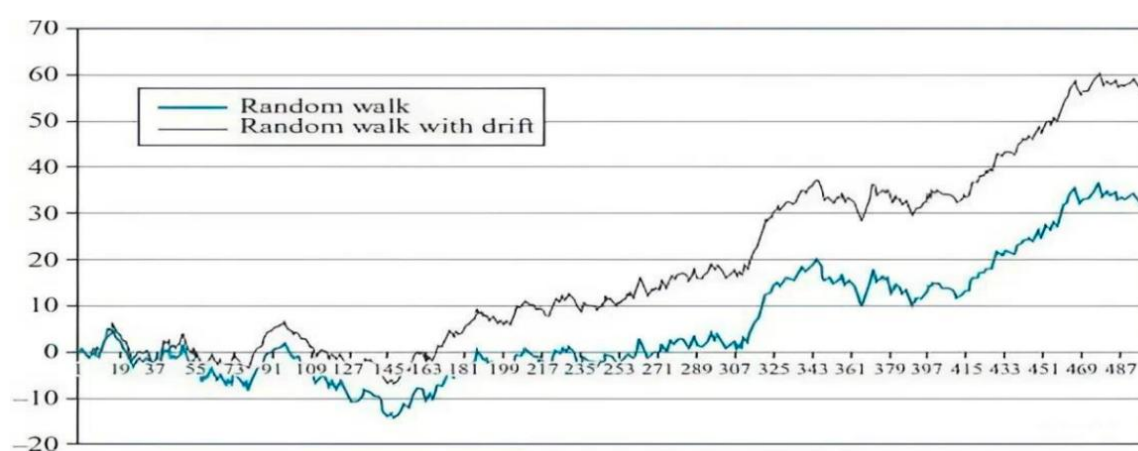


Figure 3.5. Time-series plot of a random walk versus a random walk with drift.

Source: Brooks, C. (2019). Introductory econometrics for finance (4th ed., p. 444). Cambridge University Press.

In contrast, Figure 3.5 demonstrates that a random walk and a random walk with drift exhibit prolonged fluctuations around the mean, meaning the series can deviate significantly from the mean for extended periods without quickly reverting. In particular, a positive drift component causes the series to display an upward trend over time. These are typical manifestations of a non-stationary series characterized by a stochastic trend. This characteristic is directly relevant to Vietnam's rice export price series in its price-level form, as the period from December 2003 to March 2026 witnessed prolonged price surges (for example, the 2007-2008 food crisis period and the 2023 price shock) followed by extended adjustment phases - a classic feature of a random walk with drift.

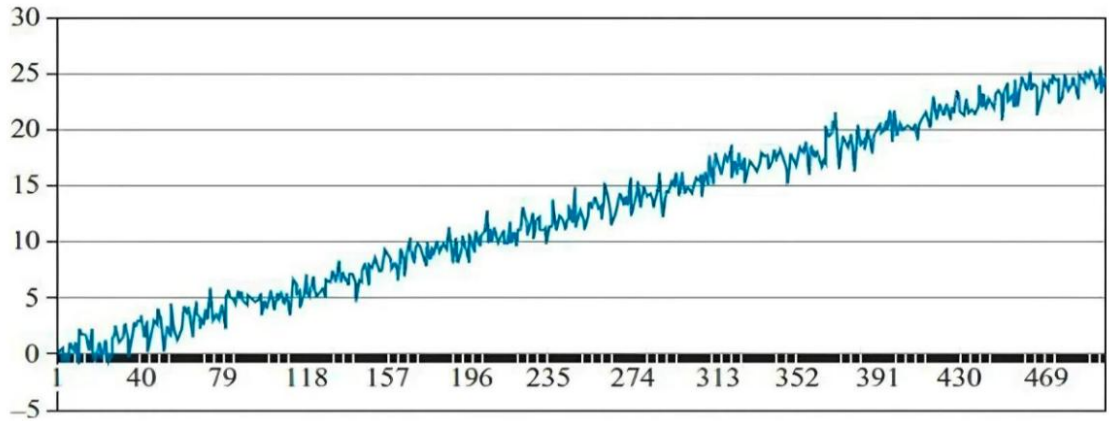


Figure 3.6. Time-series plot of a deterministic trend process.

Source: Brooks, C. (2019). Introductory econometrics for finance (4th ed., p. 445). Cambridge University Press.

Figure 3.6 illustrates a deterministic trend process. In this case, the series has no fixed meaning but instead fluctuates around a trend that increases gradually over time. If this trend component is removed, the resulting series may become stationary and exhibit characteristics similar to a white noise process.

Brooks (2019) also shows that in a first-order autoregressive model, AR (1), the nature of the series depends on the value of the autoregressive coefficient ϕ :

When $\phi = 0$, the series is white noise.

When $\phi = 0.8$, the series is a stationary AR (1) process.

When $\phi = 1$, the series becomes a random walk and is non-stationary.

Brooks (2019) emphasizes that graphical identification is only a preliminary visual tool. To formally determine whether a data series contains a unit root, a unit root test must be conducted. If a series has a unit root, it is non-stationary; conversely, if it lacks a unit root, the series is considered stationary and more suitable for time-series analysis.

In the ARIMAX model, ensuring stationarity applies not only to the dependent variable series (y_t) but also to the exogenous explanatory variables (x_t) before they are incorporated into the model, in order to avoid spurious regression results.

Brooks begins with a first-order autoregressive process:

$$y_t = \phi y_{t-1} + u_t \quad (3.5)$$

Where u_t is the random error term. The purpose of the test is to evaluate the null hypothesis $H_0: \phi = 1$ (the series has a unit root) against the alternative hypothesis $H_1: \phi < 1$ (the series is stationary).

In practice, for ease of computation and interpretation, Brooks uses the following regression form:

$$\Delta y_t = \psi y_{t-1} + u_t \quad (3.6)$$

Where $\psi = \phi - 1$. Based on this specification, testing $H_0: \phi = 1$ is equivalent to testing $H_0: \psi = 0$. If the coefficient ψ is negative and statistically significant, this provides evidence to reject the null hypothesis of a unit root.

In cases where the test accounts for a drift term and a deterministic time trend, the test equation is written as:

$$\Delta y_t = \psi y_{t-1} + \mu + \lambda t + u_t \quad (3.7)$$

Where μ is a constant and λt is the deterministic trend component. This form allows stationarity to be tested when the data have a non-zero mean or a trend that changes over time.

According to Brooks, the Dickey-Fuller test statistic is calculated as the ratio between the estimated coefficient and its standard error:

$$t\text{-statistic} = \frac{\hat{\psi}}{SE(\hat{\psi})} \quad (3.8)$$

This test statistic does not follow the standard t-distribution under the null hypothesis but instead follows the Dickey-Fuller distribution, which has its own set of critical values. The decision rule is to reject the null hypothesis if the test statistic is more negative than the corresponding critical value.

Furthermore, Brooks notes that the standard Dickey-Fuller test is valid only when the residual u_t is white noise. If the residuals exhibit autocorrelation, the Augmented Dickey-Fuller (ADF) test must be used instead, by adding lagged difference terms of the dependent variable to the model:

$$\Delta y_t = \psi y_{t-1} + \mu + \lambda t + \sum_{i=1}^p \alpha_i \Delta y_{t-i} + u_t \quad (3.9)$$

Where the lagged terms Δy_{t-i} are included to eliminate autocorrelation in the residuals.

Accordingly, in this study, the ADF test is used to assess the stationarity of all input data series before proceeding with the differencing steps and determining the structure of the ARIMAX model.

c. Differencing and Transformation

In the Box-Jenkins methodology, a time series must be transformed into a stationary state before model identification and estimation can proceed. In practice, many financial and economic data series exhibit non-stationarity due to trends, time-varying means, or seasonal fluctuations. Therefore, the differencing step is used to remove these non-stationary components.

For the ARIMAX model, ensuring stationarity is required for both the dependent variable (y_t) and all exogenous explanatory variables (x_t), in order to avoid spurious regression and ensure that statistical inferences are reliable.

For series exhibiting a trend, the standard procedure is to apply first-order differencing. The first difference of a series y_t is defined as follows:

$$W_t = Y_t - Y_{t-1} \quad (3.10)$$

Using the backshift operator B (or L), the expression above can be rewritten as:

$$W_t = (1 - B)Y_t \quad (3.11)$$

Where B is the backshift operator. This differencing operation removes the trend and stabilizes the data series. If the series remains non-stationary after the first difference, higher-order differencing can be applied until the autocorrelation structure becomes appropriate.

In its general form, the ARIMAX (p, d, q) model is represented by the following equation:

$$\phi_p(B)(1 - B)^d y_t = \sum_{j=1}^k \beta_j (1 - B)^{d_x} x_{jt} + \theta_q(B)e_t \quad (3.12)$$

In this specification, d is the order of integration (the number of times differencing is applied) to render the series stationary, $\phi_p(B)$ represents the autoregressive polynomial of order p , $\theta_q(B)$ represents the moving average polynomial of order q , e_t is the random error

term (white noise), and $\sum \beta_j x_{jt}$ is the exogenous component added to the model to enhance its explanatory and forecasting power. Thus, the differencing process plays an important role in transforming the data series from a non-stationary to a stationary state, laying the foundation for determining the parameters of the ARIMAX model in the next step.

d. Model Identification (p, d, q)

Once the data series have been appropriately different, the next stage is model identification to determine the optimal values of p, d, and q for the ARIMAX (p, d, q) model. According to the Box-Jenkins methodology, this step initially relies primarily on analyzing the plots of the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF). The objective is to select a parsimonious model -that is, the simplest model that still adequately captures the dynamic characteristics of the data series.

First, the order of differencing d is assessed based on the rate at which the autocorrelation coefficients decay. When the ACF coefficients decline rapidly toward zero, the current level of differencing can be considered appropriate. Conversely, if the ACF remains large and persists across many lags, the series may still be non-stationary and require further differencing. For the ARIMAX model, this requirement for stationarity and determination of the order d must be applied simultaneously to both the dependent variable (y_t) and all exogenous explanatory variables (x_t) to ensure the validity of statistical inferences.

In addition to graphical analysis, in practice, for complex financial data series, the identification of the orders p and q is often supported by information criteria such as the Akaike Information Criterion (AIC) or the Bayesian Information Criterion (BIC). These criteria help the balance model fit against the number of parameters, thereby avoiding overfitting. Within the ARIMAX structure, exogenous variables are incorporated into the model based on economic theory or causality tests, after which the autoregressive (p) and moving average (q) orders of the residuals are determined to complete the forecasting structure.

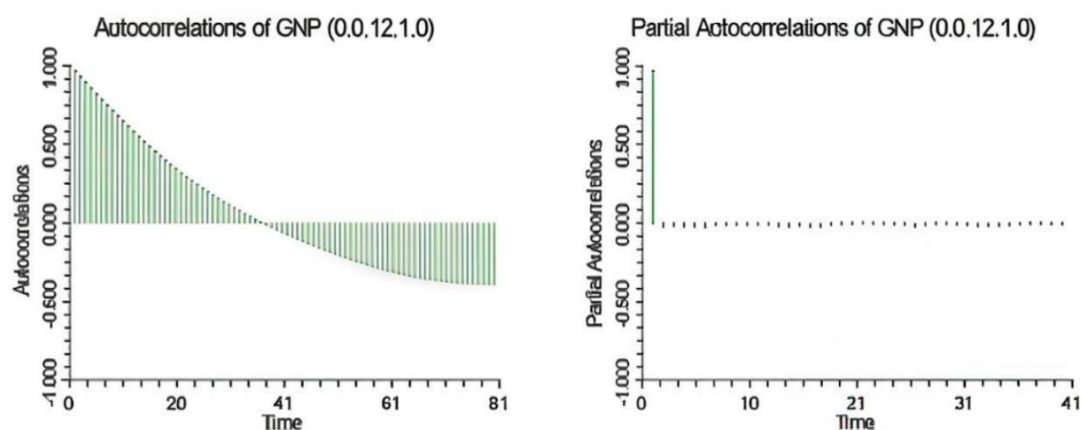


Figure 3.7. Correlogram of GNP before differencing.

Source: NCSS, The Box-Jenkins Method, p. 470-476.

Note: The original source does not specify the country corresponding to the GNP series. Therefore, this figure is presented only to illustrate the correlogram pattern before differencing.

Figure 3.7 illustrates the correlogram of a series prior to differencing. It can be observed that the ACF coefficients decay very slowly and remain large across many lags, while the PACF exhibits several notable spikes at the initial lags. This is a common indication of a non-stationary series that still contains a trend or a unit root. In this case, directly determining the parameters p and q would not be appropriate, since a model built on non-stationary data may produce unreliable estimation results.

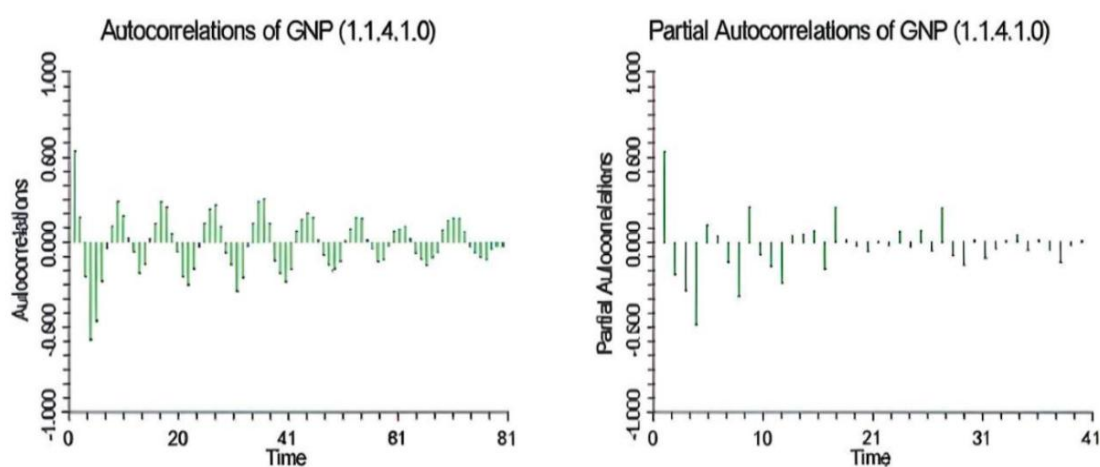


Figure 3.8. Correlogram of GNP after differencing.

Source: NCSS, The Box-Jenkins Method, p. 470-476.

Note: The original source does not specify the country corresponding to the GNP series. Therefore, this figure is presented only to illustrate the correlogram pattern after differencing

In contrast, Figure 3.8 shows the correlogram of the series after differencing. Compared with the original series, the ACF and PACF coefficients fluctuate more clearly around zero and no longer exhibit prolonged slow decay. This indicates that the differencing process has removed most of the trend in the series and helped the data achieve stationarity. Only at this point do the ACF and PACF plots become meaningful for determining the model's parameters.

After determining the order of differencing d , the autoregressive parameter p is identified based on the PACF plot of the differenced series. If the PACF cuts off after a certain number of lags, the last statistically significant lag is typically taken as the appropriate value of p . Conversely, if the PACF tails off gradually rather than cutting off sharply, the series may not follow a pure AR model but instead belong to a mixed model form.

Similarly, the moving average parameter q is determined through the ACF plot. If the ACF cuts off after a certain number of lags, the last statistically significant lag is selected as the appropriate value of q . If the ACF tails off gradually instead of cutting off sharply, the model may contain an AR component or take the form of a mixed ARIMAX model rather than a pure MA model.

To help determine which autocorrelation coefficients are statistically significant, Box and Jenkins propose the following approximate threshold:

$$|r_k| > \frac{2}{\sqrt{N}} \quad (3.13)$$

Where r_k is the autocorrelation coefficient at lag k , and N is the sample size. According to the Box-Jenkins method, an autocorrelation coefficient is considered statistically significant only when its absolute value is at least $2/\sqrt{N}$. However, it should be noted that a statistically significant coefficient is not always large enough to be practically meaningful for model selection. Therefore, the model identification process should rely not only on the ACF and PACF but also be combined with model selection criteria such as the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC).

In this study, after determining the feasible values of p , d , and q , multiple candidate ARIMAX models will be estimated and compared. The selected model will be the one with

the lowest AIC and BIC values, while also satisfying the conditions for residual behavior and statistical adequacy. The results of this stage will serve as the basis for constructing the optimal ARIMAX model before proceeding to develop the hybrid ARIMAX-LSTM model to further improve the accuracy of forecasting Vietnam's rice export price volatility.

e. Model Estimation

After determining the appropriate values of p , d , and q through stationarity testing and ACF/PACF plot analysis, the next step is to estimate the parameters of the ARIMAX model. The ARIMAX model is an extension of the traditional ARIMA model, incorporating exogenous variables to improve the model's explanatory and forecasting capability (Box et al., 2015).

In this study, Vietnam's FOB 5% broken white rice export price is used as the dependent variable, while Thailand's FOB 5% broken white rice export price is used as the exogenous variable. The general ARIMAX model is expressed as follows:

$$Y_t = c + \sum_{i=1}^p \phi_i Y_{t-i} + \sum_{j=1}^q \theta_j e_{t-j} + \beta X_t + \varepsilon_t \quad (3.14)$$

Where:

Y_t is the export price of Vietnam's rice at time t ;

X_t is the export price of Thailand's rice at time t ;

c is the constant term;

ϕ_i are the autoregressive (AR) coefficients;

θ_j are the moving average (MA) coefficients;

β is the coefficient of the exogenous variable;

ε_t is the random error term;

The model parameters are estimated using Maximum Likelihood Estimation (MLE), a method commonly used in ARIMA-family models due to its ability to produce efficient and consistent estimates (Box et al., 2015). The estimation process is carried out using Stata 17 software.

After the model is estimated, the statistical significance of the parameters is assessed based on the corresponding z-statistics and p-values. Coefficients with a p-value below 0.05 are considered statistically significant at the 5% level. In addition, the residuals of the ARIMAX model are extracted for use in the subsequent LSTM model-building stage. Using the residuals as input allows the LSTM model to examine whether temporal or nonlinear predictive information remains after the linear component has been modeled (Zhang, 2003).

f. Model Selection Criteria (AIC and BIC)

To select the optimal ARIMAX model, the study estimates multiple models with different combinations of the autoregressive order (p) and moving average order (q). Model selection is based on two widely used information criteria -the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) -which help assess model fit while controlling for overfitting (Hyndman and Athanasopoulos, 2021).

The AIC is calculated as follows:

$$AIC = -2 \ln(L) + 2k \quad (3.15)$$

Where:

L is the maximum likelihood value of the model;

k is the number of parameters to be estimated

The BIC is calculated as follows:

$$BIC = -2 \ln(L) + k \ln(n) \quad (3.16)$$

Where:

n is the sample size

k is the number of parameters in the model

Both AIC and BIC are used to compare competing models. Smaller AIC and BIC values indicate a better model fit to the data while maintaining parsimony and avoiding overfitting. However, BIC applies a larger penalty for models with more parameters, and therefore tends to favor simpler models compared with AIC (Burnham and Anderson, 2004).

In this study, multiple candidate ARIMAX models are constructed and compared based on their AIC and BIC values. The model with the lowest AIC and BIC values -whose

parameters are statistically significant and which satisfies the residual diagnostic requirements -is selected as the optimal ARIMAX model. The selected model is then used to forecast the linear component of Vietnam's rice export price series before being combined with the LSTM model within the hybrid ARIMAX-LSTM framework.

g. Diagnostic Checking

After the ARIMAX model is estimated, the next step is to perform diagnostic checking to assess the model's adequacy for the study data. According to Box et al. (2015), the purpose of this stage is to determine whether the model has fully captured the information contained in the time series. A model is considered adequate when its residuals reduce to white noise -that is, the residuals have a mean of zero, constant variance, and no autocorrelation across lags.

For this study, diagnostic checking is performed after the ARIMAX model has been constructed to forecast Vietnam's rice export price volatility. Although the model incorporates Thailand's rice export price as an exogenous variable, including this explanatory variable does not guarantee that the entire dynamic structure of the series has been fully modeled. Therefore, checking the residuals is a mandatory requirement before using the model for forecasting purposes or combining it with the LSTM neural network within the hybrid ARIMAX-LSTM model.

Within the Box-Jenkins framework, diagnostic checking is typically performed by analyzing the ACF and PACF plots of the residuals. If the residual autocorrelation coefficients fluctuate randomly around zero and fall within the 95% confidence interval, it can be concluded that the residuals no longer contain systematic information and that the model adequately describes the characteristics of the original data (Box et al., 2015).

Conversely, if the ACF or PACF plots of the residuals still exhibit statistically significant coefficients at certain lags, this indicates that the model has not fully captured the structure of the data. In this case, the choice of parameters p , d , and q should be reconsidered, and the adequacy of the exogenous variable included in the model should be reassessed.

In addition to visual inspection of the plots, the study employs the Ljung-Box test to formally assess residual autocorrelation. This is a portmanteau test used to evaluate the null hypothesis that all autocorrelation coefficients of the residuals are jointly equal to zero. If

the Q-statistic is significant (small p-value), this indicates that the model has not yet fully captured the dynamic structure of the data and needs to be re-specified.

Thus, diagnostic checking plays an important role in confirming the adequacy of the ARIMAX model. Only when the residuals no longer contain significant autocorrelation structure is the model considered qualified for forecasting and for extracting residuals for the subsequent deep learning stage.

h. White Noise Checking

After the ARIMAX model has been selected and has passed the diagnostic checking steps, the final step in the Box-Jenkins procedure is to verify whether the model's residuals satisfy the property of white noise. This is an important step to confirm that the model has fully captured the linear information in the data series and that no systematic structure remains unaccounted for (Box et al., 2015).

According to Box et al. (2015), a residual series is considered white noise when it satisfies three basic conditions:

The expected value of the residuals is zero:

$$E(e_t) = 0 \quad (3.17)$$

The variance of the residuals is constant over time:

$$Var(e_t) = \sigma^2 \quad (3.18)$$

There is no autocorrelation between residuals at different lags:

$$Cov(e_t, e_{t-k}) = 0, \quad k \neq 0 \quad (3.19)$$

Where:

e_t is the residual at time t ;

σ^2 is the variance of the residual series;

k is the lag order.

These three conditions indicate that if the residuals are reduced to white noise, all the predictable information in the data series has been captured by the ARIMAX model. Conversely, if the residuals still exhibit autocorrelation, this demonstrates that the model has not fully described the dynamic structure of the time series, and reconsideration of the parameter selection or a more suitable model specification is required.

In this study, testing the white noise property of the residuals is conducted using two methods. First, the ACF and PACF plots of the residuals are used to visually assess the remaining degree of autocorrelation. If most of the autocorrelation coefficients fall within the 95% confidence interval and are randomly distributed around zero, the residuals can be considered white noise.

In addition to visual assessment, the study employs the Ljung-Box test to statistically evaluate whether the residual series still exhibits autocorrelation. This is a test widely used in time-series analysis and recommended within the Box-Jenkins procedure (Ljung and Box, 1978).

The test statistics are defined as follows:

$$Q = n(n + 2) \sum_{k=1}^m \frac{\widehat{\rho}_k^2}{n - k} \quad (3.20)$$

Where:

Q : the Ljung-Box test statistic;

n : the number of observations;

m : the number of lags tested;

$\widehat{\rho}_k$: the sample autocorrelation coefficient at lag k .

For this study, the ARIMAX model is selected only when it simultaneously satisfies the following conditions:

The model's parameters are statistically significant;

The AIC and BIC values are optimal;

The residuals show no autocorrelation according to the Ljung-Box test;

The residual series exhibits the characteristics of white noise.

Ensuring that the residuals are white noise is important because it demonstrates that the ARIMAX model has fully captured the linear component as well as the effect of the exogenous variable -Thailand's rice export price. This is also a necessary condition before using the ARIMAX model's results to construct the hybrid ARIMAX-LSTM model in the next stage.

3.3.3. Hybrid ARIMAX-LSTM Forecast Construction

a. Long Short-Term Memory (LSTM)

Long Short-Term Memory (LSTM) is a variant of the Recurrent Neural Network (RNN), developed by Hochreiter and Schmidhuber (1997) to overcome the vanishing gradient problem when processing time-series data with long-term dependencies. Through its long-term memory mechanism together with control gates -the forget gate, the input gate, and the output gate -LSTM is capable of effectively selecting, retaining, and discarding information during the learning process.

Compared with traditional statistical models such as ARIMA or ARIMAX, LSTM has an advantage in modeling nonlinear relationships and complex data patterns that linear models struggle to capture (Goodfellow et al., 2016; Géron, 2022). As a result, LSTM has been widely applied in time-series forecasting studies, particularly for problems characterized by strong volatility and influenced by numerous difficult-to-observe factors.

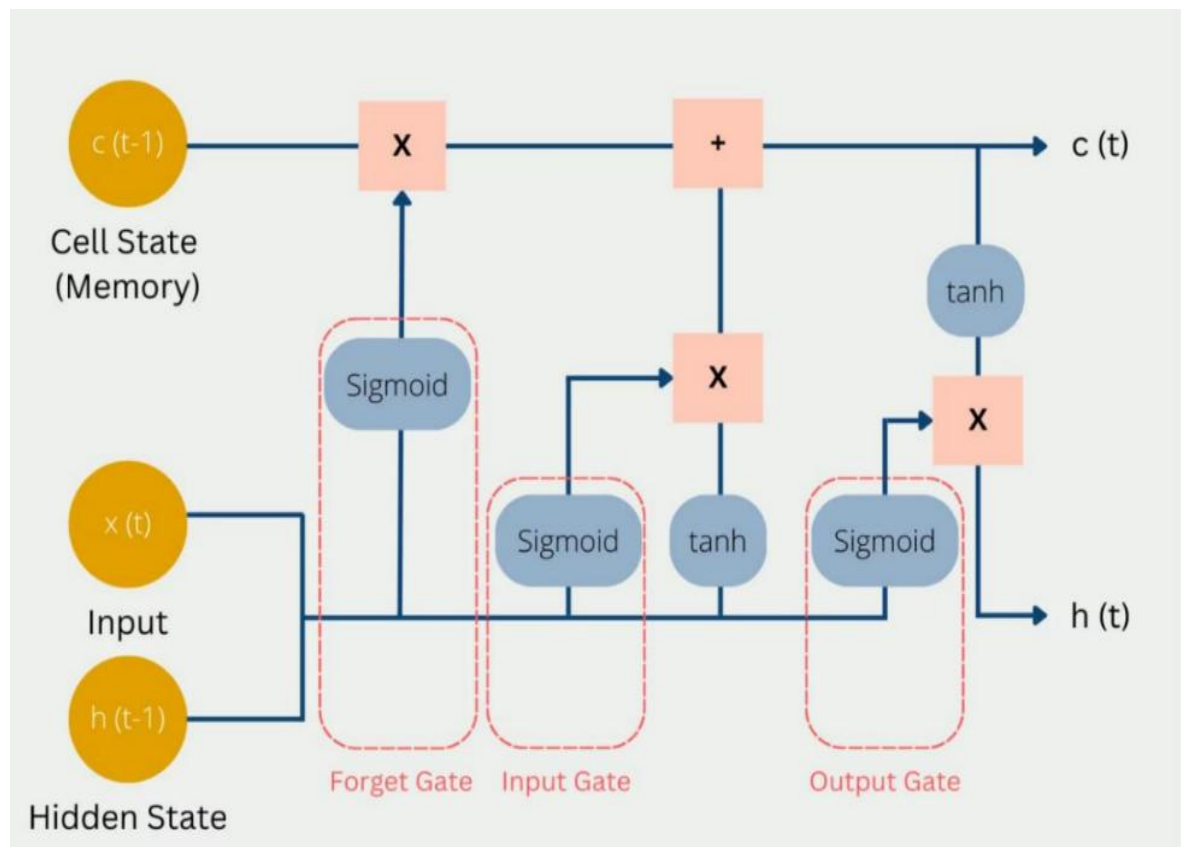


Figure 3.9. LSTM Architecture

Source: Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow (3rd ed.)*. O'Reilly Media.

b. Residual Data Preparation

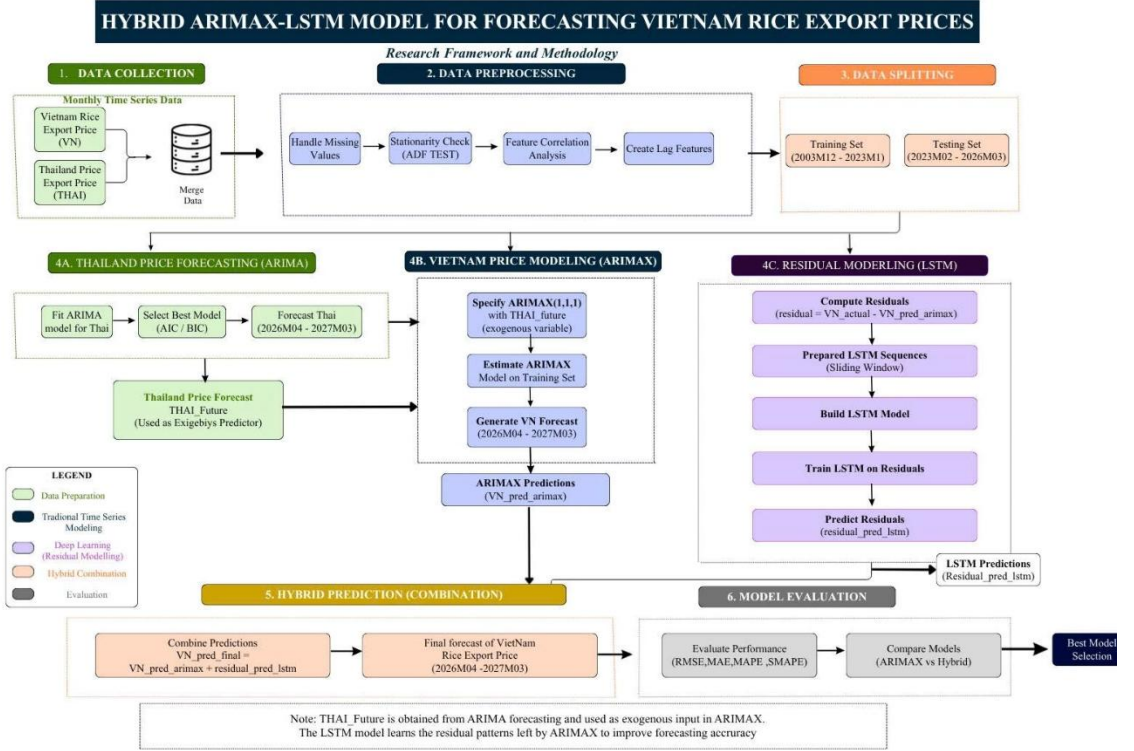


Figure 3.10. Research Framework for Hybrid ARIMAX-LSTM Forecasting

Source: Author's compilation.

After the ARIMAX model passes the diagnostic checks and is confirmed to be adequate, the next step is residual extraction. This is a particularly important step in this study, as the model is built following a hybrid approach combining ARIMAX and Long Short-Term Memory (LSTM).

Within the structure of the hybrid ARIMAX-LSTM model, the actual time series (y_t) is assumed to be a combination of two components: a linear component (L_t) and a nonlinear component (N_t), which reflects irregular, non-additive behaviors and complex structures that a linear model cannot represent. This relationship is expressed as:

$$y_t = L_t + N_t \quad (3.21)$$

In addition, the model's forecast value is regarded as an estimate of the linear component:

$$\hat{L}_t = y_t^{\widehat{\text{ARIMAX}}} \quad (3.22)$$

After the ARIMAX model is estimated, the residual series is calculated as the difference between the actual observed price and the fitted value from the ARIMAX model:

$$e_t = Y_t - \widehat{Y_t^{\text{ARIMAX}}} \quad (3.23)$$

Where:

e_t : the residual at time t ;

Y_t : the actual value;

$\widehat{Y_t^{\text{ARIMAX}}}$: the forecast value from the ARIMAX model.

The residual series obtained from this step represents the portion of the observed price variation that is not explained by the ARIMAX specification. This remaining information may contain nonlinear or other temporal patterns that are not captured by the linear model. Therefore, the residual series is subsequently used as input for the LSTM model to examine whether it provides additional predictive information.

c. Data Splitting

After completing the data preprocessing process, the study proceeds to split the data into training and testing sets. Because time-series data exhibit temporal dependence, random splitting is not applied, in order to avoid data leakage. The data are divided chronologically into a training set of 230 observations and a testing set of 38 observations. This approach helps preserve the temporal structure of the data and accurately reflects the model's forecasting capability (Hyndman and Athanasopoulos, 2021).

d. Hyperparameter Tuning

The performance of the LSTM model depends significantly on the selection of appropriate hyperparameters. Therefore, the study conducts hyperparameter tuning using the Grid Search method to determine the optimal configuration for the model. The hyperparameters considered include the number of neurons, the number of LSTM layers, batch size, learning rate, and the number of training epochs. The configuration yielding the lowest error value on the validation set is selected for the final training process, in order to improve forecasting accuracy and mitigate overfitting (Goodfellow et al., 2016).

e. Model Training and Validation

After determining the optimal hyperparameter set, the LSTM model is trained on the training dataset. During training, the Adam algorithm is used to optimize the network's weights, while Root Mean Square Error (RMSE) and Mean Absolute Percentage Error (MAPE) are used to evaluate forecasting error.

At the same time, the validation set is used to monitor the model's performance during training. The Early Stopping technique is applied to prevent overfitting by halting training when the validation error no longer improves after a certain number of iterations (Goodfellow et al., 2016).

f. Hybrid ARIMAX-LSTM Modeling

The construction of the hybrid forecast represents the central methodological contribution of this study, as it constitutes the mechanism that combines the linear modeling strength of ARIMAX with the nonlinear learning capability of LSTM into a unified forecasting framework with higher accuracy. This combination strategy is grounded in the additive decomposition principle first proposed by Zhang (2003), which has since become the most widely applied ARIMA(X) neural network combination framework in the field of time-series forecasting (Dave et al., 2021; Mahawan et al., 2022; Allassak et al., 2025).

According to this principle, an observed time series y_t is assumed to result from the additive interaction of two distinct, theoretically separable components: a linear component L_t , reflecting the autocorrelation structure and systematic effects of exogenous variables; and a nonlinear component N_t , reflecting irregular, non-additive behaviors and complex structures that a linear model cannot represent. This relationship is expressed as:

$$y_t = L_t + N_t \quad (3.24)$$

The theoretical basis for this decomposition is that no single model is well suited to simultaneously capture both types of structure. ARIMAX has a strong theoretical foundation and high economic interpretability for linear dynamics but is constrained by nonlinearity; meanwhile, LSTM is highly flexible in approximating nonlinear functions but lacks comparable interpretability and economic structure (Sun et al., 2023; Mahawan et al., 2022). By assigning each model the component it is best suited to estimate, the hybrid approach is expected to capture a more complete representation of the actual data-generating process than either model operating independently.

Stage 1 - Estimating the linear component. As presented in Section 3.3.2, the ARIMAX (p,d,q) model is first estimated on the original rice export price series y_t , using Thailand's 5% broken rice price as the exogenous regressor. The fitted values (in-sample) and forecast values (out-of-sample) produced by this model are denoted $\hat{Y}_t^{ARIMAX}$ and are used as the estimate of the linear component:

$$\hat{L}_t = y_t^{\widehat{ARIMAX}} \quad (3.25)$$

Stage 2 - Extracting the residual series. After the ARIMAX model is estimated, the residual series is calculated as the difference between the actual observed price and the fitted value from the ARIMAX model:

$$e_t = Y_t - \hat{Y}_t^{\widehat{ARIMAX}} \quad (3.26)$$

Under Zhang's (2003) decomposition assumption, this residual series e_t is regarded as the empirical representation of the nonlinear component N_t , since the linear, autocorrelated structure of the series has already been absorbed by the ARIMAX model. If the ARIMAX model is correctly specified, the residual series will exhibit little to no remaining linear autocorrelation (as verified by the Ljung-Box test), meaning that any predictable structure remaining in e_t would, by nature, belong to nonlinear dynamics, asymmetric shocks, or higher-order temporal dependencies that the linear framework cannot represent (Dave et al., 2021).

Stage 3 - Modeling the nonlinear residuals using LSTM. The residual series e_t is treated as a new, independent time series; it is itself normalized, divided into sliding windows, and used to train the LSTM network. Rather than directly forecasting the original price level, the LSTM is trained separately to learn the remaining temporal dependencies within the residuals:

$$\hat{N}_t = f_{LSTM}(e_{t-1}, e_{t-2}, \dots, e_{t-\tau}; W, b) \quad (3.27)$$

Where $f_{LSTM}(\dots)$ denotes the trained LSTM mapping function, τ is the length of the lookback window, and W, b represents the full set of weight matrices and bias vectors learned through the gating mechanism described in Section a. This step constitutes the conceptual core of the hybrid method: rather than requiring the LSTM to learn the entire price-generating process from scratch - a task made more complex by the strong linear trend and

dependence on the exogenous variable already present in the original series - the neural network is assigned a "cleaner" and considerably more tractable learning task, since the dominant linear signal has already been removed beforehand (Zhang, 2003; Dave et al., 2021).

Stage 4 - Additive recombination. The final hybrid forecast at time t is computed by adding the linear forecast from ARIMAX to the nonlinear forecast from LSTM on the residual series:

$$\widehat{y}_t^{\text{Hybrid}} = \widehat{y}_t^{\text{ARIMAX}} + \widehat{N}_t \quad (3.28)$$

This additive recombination is the exact inverse operation of the decomposition step performed in Stage 2, ensuring complete theoretical consistency between how the components are separated and how they are recombined. It should be noted that this combination is purely additive and unweighted - meaning the framework does not estimate separate weighting coefficients for the ARIMAX component and the LSTM component, unlike other hybrid strategies such as weighted ensemble averaging or the functional combination approach proposed by Khashei and Bijari (2010), in which $y_t = f(L_t, N_t)$ rather than a simple sum. The additive formula in Equation (3.28) was chosen for this study because it preserves the clearest interpretability for the contribution of each component: the ARIMAX term can always be understood as "the price level explained by historical autocorrelation and Thailand's competitive price," while the LSTM term can always be understood as "the additional adjustment needed to reflect the nonlinear behavior of the market."

g. Final Forecast Generation

After the combination mechanism presented in Equation (3.28) is verified on the training data, it is applied to generate forecasts over two separate time horizons: the out-of-sample test period, and the true future forecast period lying beyond the range of observed data.

Forecasts on the out-of-sample test set. For the test period (February 2023 - March 2026), the trained ARIMAX model is used to generate one-step or multi-step linear forecasts $\widehat{Y}_t^{\text{ARIMAX}}$, given that the observed values of the exogenous variable (Thailand's rice price) over the same period are known. In parallel, the trained LSTM model generates nonlinear residual forecasts $\widehat{N}_t$ using the lagged residual series as input. These two forecast series are then summed according to Equation (3.28) to produce the hybrid forecast series for the entire

test period. This out-of-sample hybrid forecast series forms the basis for the forecast evaluation procedure and allows for a like-for-like comparison with the standalone ARIMAX benchmark model.

Generating future forecasts (through March 2027). To address Research Objective 2 (RO2), the final hybrid model - re-estimated on the full dataset from December 2003 to March 2026 - is used to generate genuine out-of-period forecasts, extending twelve months beyond the last observation, through March 2027. Because Thailand's future rice price values are not observed beyond March 2026, this study applies a recursive forecasting approach consistent with the procedures used in related ARIMAX-based agricultural commodity forecasting studies (Mahawan et al., 2022; Hyndman and Athanasopoulos, 2021), in which the exogenous variable itself is forecast independently using its own separate univariate ARIMA specification, and these forecast exogenous values are then fed into the ARIMAX equation to generate the linear forecast component for the future horizon. The LSTM component is applied in a similarly recursive manner: the residual forecast at time t is fed back into the input window to generate the residual forecast at time $t + 1$, and this process continues until the entire forecast horizon is filled.

This multi-step recursive procedure inevitably accumulates greater uncertainty as the forecast horizon extends, since the forecast error at each step propagates into subsequent steps (Hyndman and Athanasopoulos, 2021). For this reason, prediction intervals - constructed at the 80% and 95% confidence levels according to the framework - are reported alongside the point forecasts for the future horizon, and the width of these prediction intervals is expected to widen progressively from April 2026 to March 2027. This treatment of forecast uncertainty is consistent with the considerations regarding asymmetric loss functions discussed in Chapter 2, since both under-forecasting and over-forecasting carry distinct practical consequences for rice export planning.

The final output of this stage is a complete forecast series comprising: (i) historical fitted values, (ii) out-of-sample forecasts compared with the realized actual prices, and (iii) genuine future point forecasts from April 2026 to March 2027. Because multi-step recursive forecasting may accumulate uncertainty as the forecast horizon extends, these projections should be interpreted as model-based estimates rather than deterministic predictions.

3.3.4. Forecast Performance Evaluation

To evaluate and compare the forecasting performance of the Hybrid ARIMAX-LSTM model against the standalone ARIMAX model, four out-of-sample forecast error measures are used: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Symmetric Mean Absolute Percentage Error (SMAPE) (Hyndman and Athanasopoulos, 2021).

Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{t=1}^n |Y_t - \hat{Y}_t| \quad (3.29)$$

MAE indicates the average absolute value of the forecast error, measuring the average magnitude of deviation between the forecast and the actual value. The advantage of MAE is that it is easy to interpret and robust to outliers: all errors contribute linearly, regardless of whether they are large or small (Hyndman and Athanasopoulos, 2021). MAE is less sensitive to outliers than RMSE and is more easily interpreted in economic terms (the average error expressed in USD/ton).

Root Mean Squared Error (RMSE):

$$RMSE = \sqrt{\frac{1}{n} \sum_{t=1}^n (Y_t - \hat{Y}_t)^2} \quad (3.30)$$

RMSE measures the dispersion of forecast errors in the same unit as the dependent variable (USD/ton). Because errors are squared, RMSE is more sensitive to large errors than MAE, which is particularly useful in this study: RMSE is interpreted as a measure of "resilience to extreme volatility" -a model with an RMSE substantially higher than its MAE indicates that, while the model may perform well under normal conditions, it struggles to control large errors during periods of sharp price surges or declines. In the field of international agricultural commodities, this characteristic is often considered more important than average accuracy alone (Manogna et al., 2025).

Mean Absolute Percentage Error (MAPE):

$$MAPE = \frac{100}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \quad (3.31)$$

MAPE converts the error into a percentage of the actual value, removing the dependence on the unit and scale of the data. This allows for (i) comparison of results with other studies across commodity markets with different absolute price levels, and (ii) intuitive interpretation for non-technical stakeholders such as exporting enterprises, regulatory agencies, or funding organizations (Dave et al., 2021; Makridakis et al., 2020). According to common practice in agricultural commodity forecasting, a MAPE below 10% is considered highly accurate forecasting, 10% to 20% is considered acceptable, and above 20% reflects low accuracy (Manogna et al., 2025).

Symmetric Mean Absolute Percentage Error (SMAPE):

$$SMAPE = \frac{100\%}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t + \hat{Y}_t} \right| \quad (3.32)$$

SMAPE measures the average percentage difference between the actual and forecast values while using the average of their absolute values as the denominator. Unlike MAPE, SMAPE is symmetric because it considers the relative difference between the actual and forecast values. In this study, SMAPE is used as an additional percentage-based measure to evaluate forecasting accuracy and to provide a more balanced assessment of forecast errors across the evaluation period.

Interpretive relationship among the four measures. Reporting all four measures simultaneously is not redundant but necessary, since each measure answers a different question. If RMSE is substantially larger than MAE ($RMSE > MAE$), this indicates the presence of certain time points with particularly large errors, reflecting that the model has difficulty forecasting extreme price shocks. Conversely, if $RMSE \approx MAE$, the error distribution is relatively even and the model performs stably across different periods. MAPE provides a third layer of information by normalizing the error relative to price scale, which is particularly useful given that the rice price series ranges from approximately USD 200/ton to over USD 600/ton over the course of the study period. SMAPE provides an additional symmetric percentage-based measure that complements MAPE when assessing forecast accuracy. (Hyndman and Athanasopoulos, 2021).

CHAPTER 4. ANALYSIS AND FINDINGS

This chapter presents empirical analysis and forecasting results of the study. It begins with a descriptive analysis of the FOB prices of Vietnam's and Thailand's 5% broken white rice to examine their historical movements and volatility over the 2003-2026 period. The chapter then reports the estimation and diagnostic results of the ARIMAX model, including stationarity testing, model identification, parameter estimation, and residual diagnostics. Building on the selected ARIMAX specification, the Hybrid ARIMAX-LSTM model is subsequently developed by applying LSTM to the ARIMAX residuals to capture the remaining nonlinear patterns. The forecasting performance of the ARIMAX and Hybrid ARIMAX-LSTM models is then compared using MAE, RMSE, MAPE, and SMAPE, followed by the presentation and interpretation of the final forecasts for Vietnam's rice export prices through 2027. Overall, the chapter provides the empirical evidence required to evaluate the effectiveness of the proposed hybrid forecasting approach and address the research objectives of the study.

4.1. Descriptive Analysis of Vietnam's and Thailand's 5% Broken White Rice FOB Prices (2003-2026)

The scope of the study covers the period from December 2003 to March 2026, determined based on the availability, continuity, and suitability of the FOB price data for 5% broken white rice, in order to reflect the movement of Vietnam's rice export prices within the context of international economic integration and the development of the global rice market. This time span encompasses the modern era of Vietnam's rice industry, during which export activities have increasingly played an important role in agricultural trade growth and international competitiveness. In addition, the monthly frequency of the data series allows for a comprehensive record of both short-term and long-term market fluctuations, thereby providing a suitable empirical foundation for analyzing and forecasting rice export prices.

From a methodological standpoint, focusing on the period after 2003 helps limit the influence of historical changes from the early stages of integration, while producing a dataset that is relatively homogeneous in terms of market conditions. This is particularly important for time-series forecasting methods such as ARIMAX and Hybrid ARIMAX-LSTM, since data consistency contributes to the reliability of the estimation and model evaluation process.

The study dataset comprises 268 monthly observations for the two main study variables collected from the World Bank Commodity Price Data (Pink Sheet), namely the FOB price of Vietnam's 5% broken white rice (VNRICE) and the FOB price of Thailand's 5% broken white rice (THRICE). VNRICE is used as the dependent variable to be forecast, while THRICE is used as the exogenous variable in the ARIMAX model to reflect the price linkage between the two leading rice-exporting countries in the region. The descriptive statistics presented in Table 4.1 include the number of observations (Obs), the mean, the standard deviation (Std. Dev.), the minimum (Min), and the maximum (Max), thereby providing an overview of the distributional characteristics and volatility of the data before constructing the forecasting models.

Table 4.1. Descriptive Statistics of the FOB Prices of Vietnam's and Thailand's 5% Broken White Rice (2003-2026)

Variable	Obs	Mean	Std. dev.	Min	Max
VietNam	268	399.079	97.530	197.000	780.000
ThaiLand	268	446.053	112.892	198.250	907.000

Source: Author's calculations based on World Bank Commodity Price Data (2003-2026).

According to Table 4.1, the FOB price of Vietnam's 5% broken white rice averaged approximately USD 399.08/ton over the entire study period. The standard deviation of USD 97.53/ton indicates a relatively large dispersion of the price series around its mean. Statistically, this result suggests that Vietnam's rice export price did not remain stable throughout the study period but instead exhibited periods of considerable volatility. The range from USD 197/ton to USD 780/ton reflects the presence of pronounced upward and downward phases within the data series, indicating a significant degree of volatility in Vietnam's rice export price over time.

For the FOB price of Thailand's 5% broken white rice, the average price reached USD 446.05/ton, considerably higher than that of Vietnam. At the same time, the standard deviation of Thailand's price series reached USD 112.89/ton, approximately USD 15.7/ton higher than that of Vietnam, indicating a stronger degree of volatility in Thailand's rice export market. The range from USD 198.25/ton (2003-2004) to USD 907/ton (2008) reflects the occurrence of severe price shocks, particularly during the 2008 global food crisis and the renewed

price surge in 2023-2024. Statistically, the greater volatility of the THRICE series indicates that Thailand's rice price exhibited a wider fluctuation band than Vietnam's over the same study period. This finding is consistent with Thailand's role as one of the leading countries shaping the international rice market, where shifts in stockpiling, export, and supply policies tend to have spillover effects across the entire region.

A comparison between the two data series shows that Thailand's rice price maintained a higher average level while also exhibiting greater volatility compared with Vietnam. This reflects differences in market positioning, export strategy, and each country's role in leading international price movements. At the same time, the similarity in the timing of fluctuations and the degree of volatility between the two price series suggests the existence of a close relationship between the two major rice-exporting markets in the region. This characteristic provides an important theoretical basis for incorporating Thailand's rice price into the ARI-MAX model as an exogenous variable, in order to improve the model's explanatory and forecasting power.

In addition to the descriptive statistics, visual inspection of the raw time-series plots plays an important role in identifying long-term trends, the degree of volatility, and structural break points in the data. These plots complement the observations drawn from Table 4.1, while also providing an initial basis for the stationarity testing and forecasting model construction steps presented in the following sections.

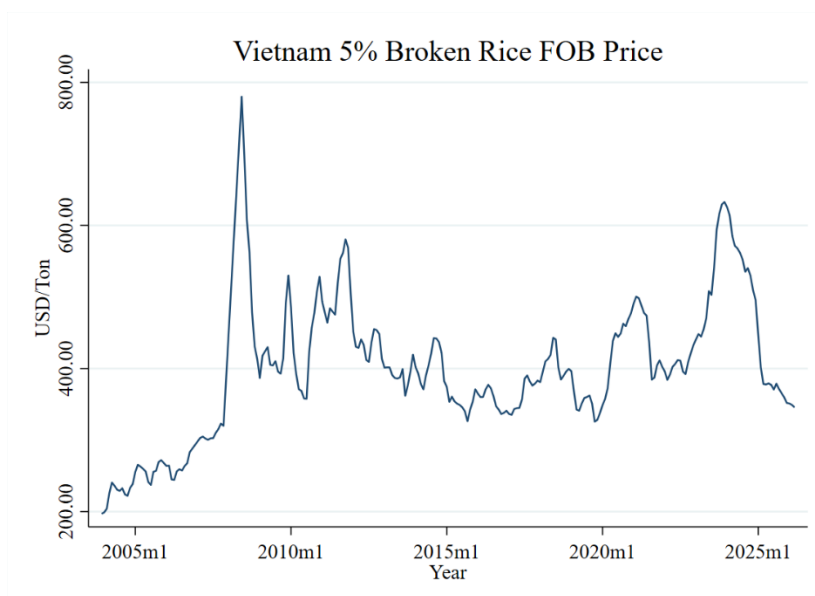


Figure 4.1. Time series of the FOB price of Vietnam's 5% broken white rice (2003-2026)

Source: World Bank Commodity Price Data (2003-2026).

Observation of Figure 4.1 shows that the FOB price of Vietnam's 5% broken white rice does not follow a stable linear growth trend but instead undergoes several distinct phases of volatility. The period 2004-2007 recorded a relatively steady upward trend, before a severe price shock occurred in 2008, when rice prices reached their historical peak due to the impact of the global food crisis. The price subsequently declined sharply and entered a period of fluctuation within the range of USD 350-500/ton for most of the 2010-2022 period.

The final years of the data series recorded a renewed price surge before declining again in early 2026. The presence of multiple turning points, together with time-varying volatility, suggests that the data series may contain both linear and nonlinear components. Consequently, the combination of ARIMAX and LSTM is expected to leverage the strengths of both approaches in modeling the dynamics of rice export prices.

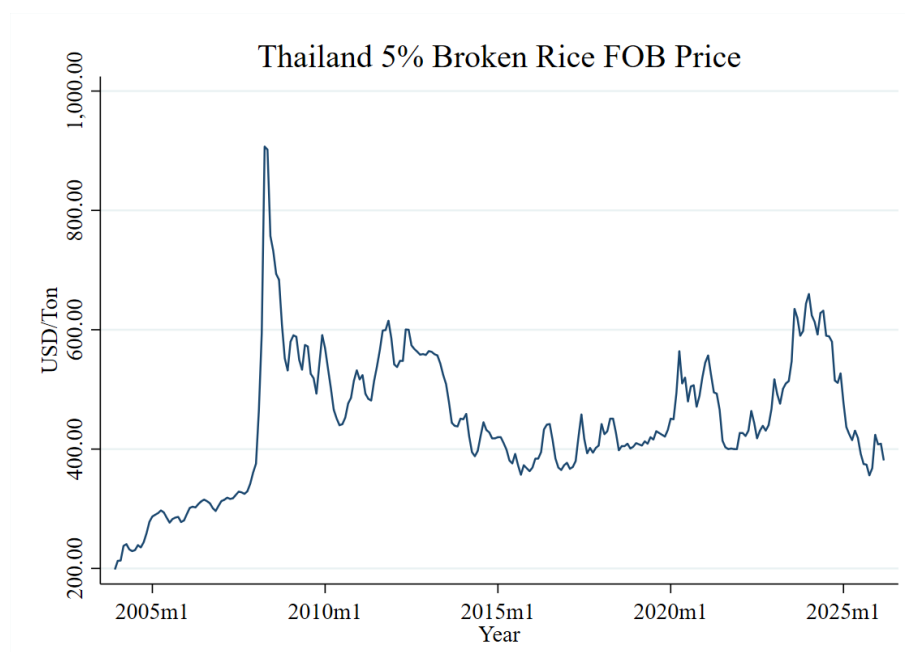


Figure 4.2. Time series of the FOB price of Thailand's 5% broken white rice (2003-2026)

Source: World Bank Commodity Price Data (2003-2026).

Similar to Vietnam, Thailand's rice price series also exhibits strong fluctuations throughout the study period. However, Thailand's fluctuation range is considerably wider, as reflected in a peak price of over USD 900/ton in 2008. Following this shock, the price gradually declined and remained within the range of USD 400-600/ton for several subsequent years, before entering a new upward cycle during the 2021-2024 period.

A notable feature is that the turning points and the upward/downward phases of Thailand's rice price tend to move in sync with Vietnam's price series. This similarity in the timing of price movements between the two data series suggests a certain degree of linkage between the Vietnamese and Thai rice markets. From a modeling perspective, this characteristic reinforces the suitability of using THRICE as an exogenous variable in the ARIMAX model to help explain the fluctuations in VNRICE.



Figure 4.3. Normalized Price Index of Vietnam's and Thailand's FOB 5% Broken White Rice, 2003-2026

Source: World Bank Commodity Price Data (2003-2026).

Figure 4.3 presents the movements of VNRICE and THRICE after normalization, in order to remove the difference in absolute scale between the two data series and highlight the relative movement patterns of each market. Normalization allows the analysis to focus on the upward/downward trends, the degree of volatility, and the co-movement of the two price series over time, rather than on differences in absolute average price levels.

Observation of the graph shows that VNRICE and THRICE tend to move relatively in sync throughout most of the study period. Although Thailand's rice price maintains a higher level and a wider fluctuation range, the major turning points of the two series appear fairly similar, particularly during periods of strong volatility in the international food market. This

suggests that both markets are jointly affected by external factors such as global supply-demand conditions, trade policies, and supply shocks.

This co-movement is clearly illustrated through two pivotal price surges during the study period. The first surge occurred in 2008, associated with the global food crisis, when many major rice-exporting countries such as India, Egypt, and Vietnam imposed restrictions or temporary export bans to safeguard domestic food security. The resulting contraction in international supply, combined with stockpiling behavior and commodity speculation, pushed world rice prices to record highs not seen in decades (Headey and Fan, 2008; FAO, 2009). Against this backdrop, both VNRICE and THRICE recorded sharp increases, reflecting the sensitivity of both markets to global supply shocks.

The second price surge occurred during 2023-2024 and was primarily driven by a tightening of rice supply at the international level. The El Niño phenomenon heightened risks for agricultural production in many Asian countries, while India's decision to restrict exports of non-basmati rice from mid-2023 significantly reduced the volume of supply circulating in the world market. According to the World Bank (2024), these factors sustained upward price pressure on rice throughout 2024, pushing the price level to its highest point since the 2008 crisis. Similarly, the FAO (2024) also recorded a significant increase in global rice prices due to disruptions in international trade and supply-related uncertainties.

From an economic perspective, the relatively synchronized response of VNRICE and THRICE to external shocks indicates that the price dynamics of the two countries are shaped not only by domestic factors but also strongly influenced by global supply-demand conditions, trade policies, and climate-related risks. Although this co-movement is not, in itself, sufficient to confirm the existence of a price transmission mechanism in the strict econometric sense, it suggests the possible existence of a significant linkage between the two markets and provides further empirical grounds for using THRICE as an exogenous variable in the ARIMAX model to enhance the explanatory and forecasting power for Vietnam's rice export prices.

Overall, the results from the descriptive statistics and visual analysis show that both VNRICE and THRICE exhibit characteristics typical of agricultural commodity time-series data, including long-term trends, time-varying volatility, and the occurrence of major price shocks. At the same time, the relatively high degree of co-movement between the two series

reinforces the hypothesis that fluctuations in Thailand's rice price may help explain and improve the forecasting performance for Vietnam's rice export price. On this basis, the following section proceeds to test the stationarity of the time series before constructing and evaluating the ARIMAX and Hybrid ARIMAX-LSTM models.

4.2. Forecasting Results for Vietnam's Broken White Rice FOB Price through March 2027

4.2.1. ARIMAX Model Results Based on World Bank Data (2003-2026)

a. Stationarity Test

Before estimating the ARIMAX model, the Augmented Dickey-Fuller (ADF) test was conducted to examine the stationarity of the time series. As shown in Table 4.2, both Vietnam's rice price series (VN) and Thailand's rice price series (Thai) are non-stationary at the level, with p-values of 0.1244 and 0.0531, respectively, both exceeding the 5% significance level. After first difference, both series become stationary, with a p-value of 0.0000. Therefore, both variables are integrated of order one, $I(1)$, and the first-differenced series are suitable for use in estimating the ARIMAX model.

Table 4.2. ADF Unit Root Test Results for Variables (2003-2026).

Variable	At Level		1st Difference		Conclusion
	T-Statistic	P-Value	T-Statistic	P-Value	
VietNam Price	-2.464	0.124	-8.473	0.000	$I(1)$
ThaiLand Price	-2.837	0.053	-11.698	0.000	$I(1)$

To eliminate non-stationarity and satisfy the assumptions of the ARIMAX model, first-order differencing was applied to both variables. After differencing, the ADF test statistics for Vietnam's rice price (VN) and Thailand's rice price (Thai) reached -8.473 and -11.698, respectively, with a p-value of 0.0000 for both series. These results are statistically significant at the 1% level, indicating that both series became stationary after first differencing.

Therefore, both Vietnam's and Thailand's rice export price series are integrated of order one, $I(1)$. Accordingly, the first-differenced data are suitable for use in the model identification and model estimation steps of the ARIMAX model.

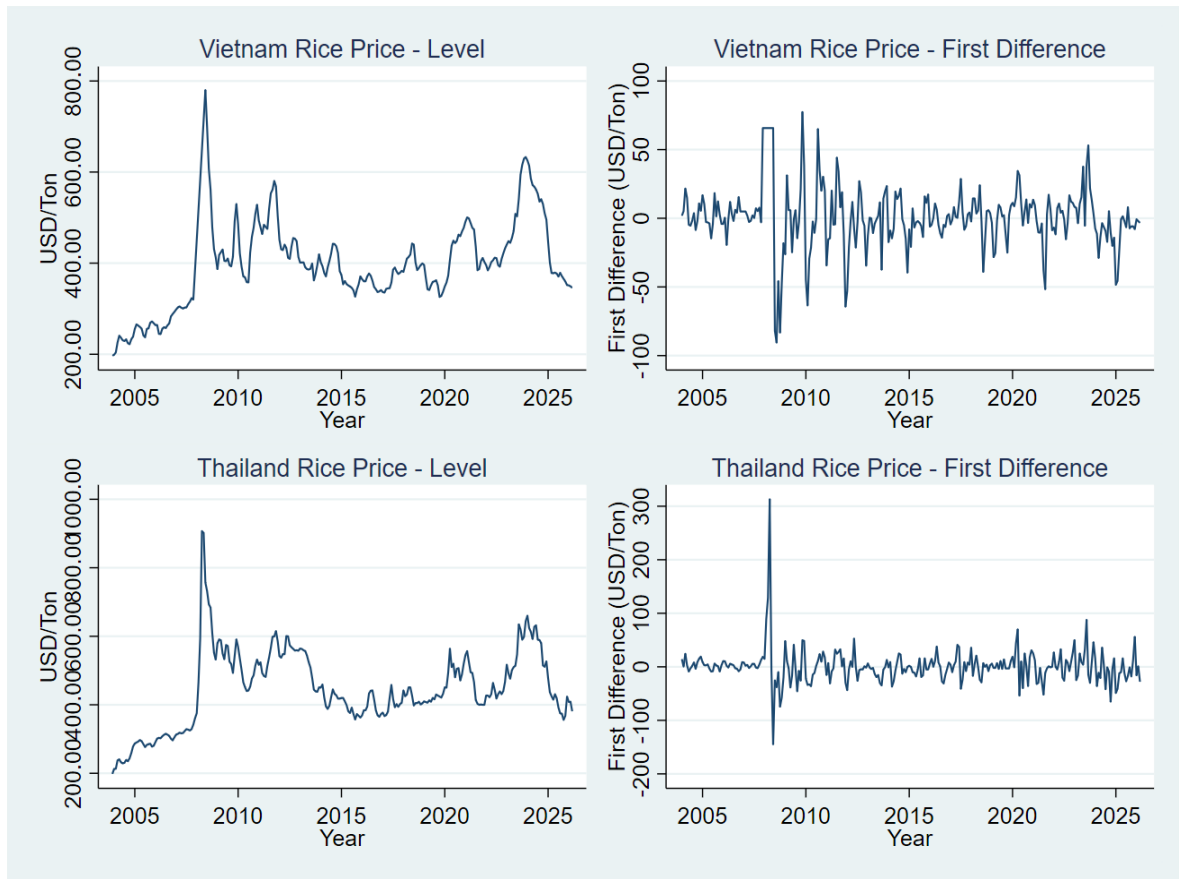


Figure 4.4. Time Series Plots of Vietnam and Thailand Rice Export Prices at Level and First Difference.

Figure 4.4 presents a visual comparison between the rice export price series before and after first differencing. In their original form, both Vietnam's and Thailand's rice export price series exhibit clear trends together with prolonged fluctuations over time, indicating the presence of non-stationarity. After applying first differencing, the series fluctuate around a relatively stable mean and no longer display a clear long-term trend. This visual evidence is consistent with the results of the ADF test, confirming that first differencing removed the unit root and produced stationary series, satisfying the conditions required for use in the construction and estimation of the ARIMAX model.

b. Model Identification and Selection

After confirming that both Vietnam's rice export price (VN) and Thailand's rice export price (Thai) are integrated of order one, $I(1)$, the next step is to determine the appropriate ARIMAX specification for forecasting purposes. The model identification process begins with an analysis of the Autocorrelation Function (ACF) and the Partial Autocorrelation

Function (PACF) of Vietnam's rice export price series after first differencing. Since Thailand's rice export price is incorporated into the ARIMAX model as an exogenous variable, the ACF and PACF analysis is conducted only for the dependent variable, in order to determine the appropriate autoregressive (AR) and moving average (MA) orders of the model.

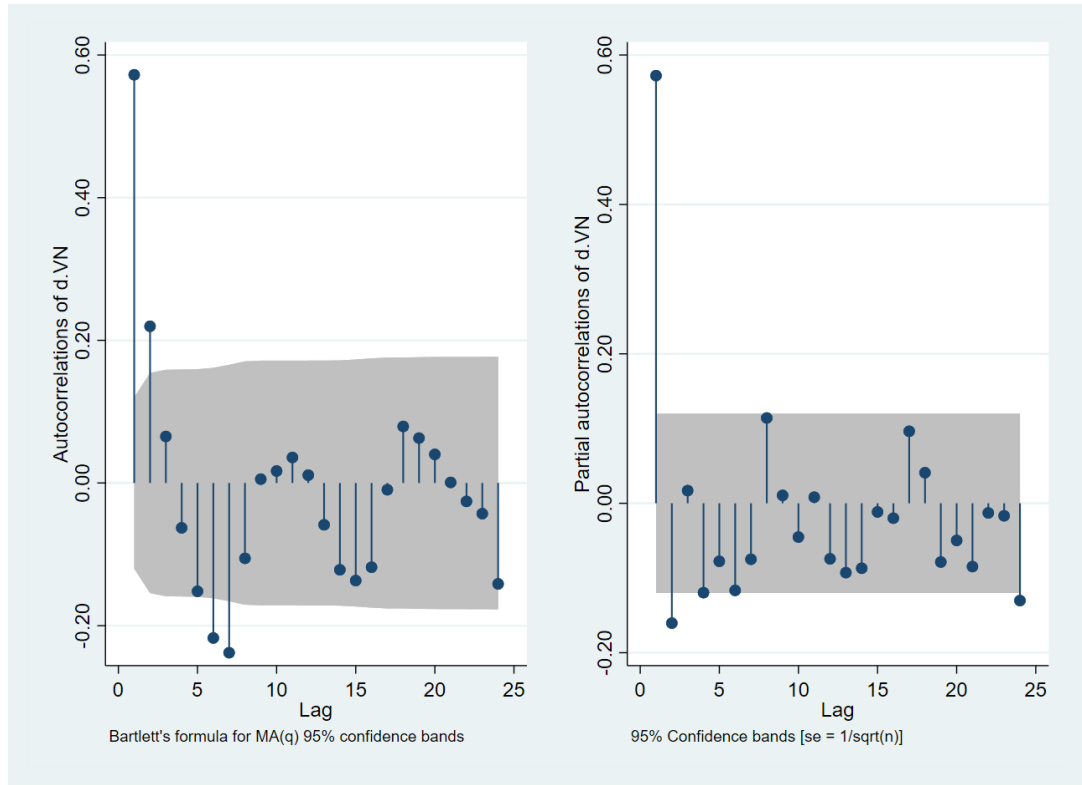


Figure 4.5. ACF and PACF plots of the Vietnam Rice Price variable at first difference.

Figure 4.5 presents the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF) of Vietnam's rice export price series after first differencing. The results show that the ACF plot displays a statistically significant spike at the first lag, while the autocorrelation coefficients at the remaining lags generally fall within the 95% confidence interval. Similarly, the PACF plot also shows a prominent spike at lag 1, after which the partial autocorrelation coefficients at higher lags are not statistically significant.

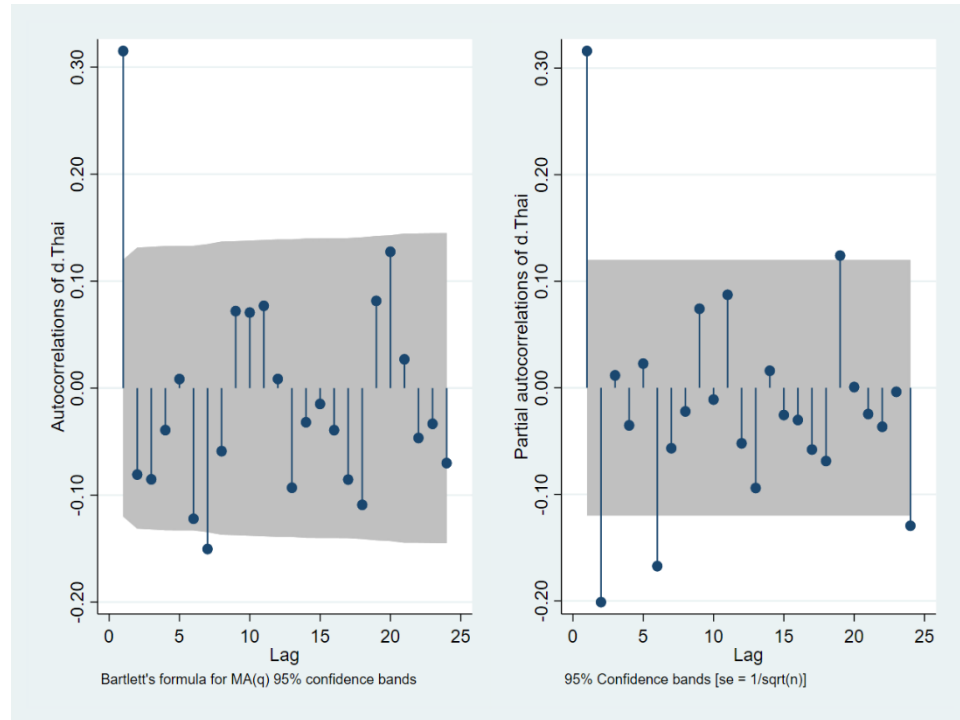


Figure 4.6. ACF and PACF plots of the Thailand Rice Price variable at first difference.

Figure 4.6 presents the Autocorrelation Function (ACF) and the Partial Autocorrelation Function (PACF) of Thailand's rice export price series after first differencing. Similar to Vietnam's price series, the autocorrelation coefficients are mainly concentrated at the initial lags, while most of the autocorrelation coefficients at higher lags fall within the 95% confidence interval. This indicates that Thailand's rice price series, after differencing, also exhibits short-memory characteristics and is suitable for inclusion in the ARIMAX model as an exogenous variable.

Based on preliminary evidence from the ACF and PACF analysis, five candidate ARIMAX models were considered, namely ARIMAX (1,1,0), ARIMAX (0,1,1), ARIMAX (1,1,1), ARIMAX (2,1,0), and ARIMAX (2,1,1). These models were subsequently estimated and compared based on the statistical significance of the estimated coefficients, together with two model selection criteria -the Akaike Information Criterion (AIC) and the Bayesian Information Criterion (BIC) -in order to determine the most appropriate specification for forecasting Vietnam's rice export price. The comparison results across the models are presented in Table 4.3.

Table 4.3. Selected Candidate ARIMAX Models

Candidate Model	Log Likelihood	AR p value	MA p value	Thai p value	AIC	BIC	Decision
ARIMAX (1,1,0)	-1159.999	0.000	-	0.000	2327.997	2342.346	Rejected
ARIMAX (0,1,1)	-1156.722	-	0.000	0.000	2321.444	2335.793	Alternative
ARIMAX (1,1,1)	-1155.354	0.008	0.000	0.000	2320.708	2338.644	Approved
ARIMAX (2,1,0)	-1156.461	0.000/ 0.000	-	0.000	2322.922	2340.858	Rejected
ARIMAX (2,1,1)	-1155.122	0.640/ 0.389	0.004	0.000	2322.244	2343.767	Rejected

The estimation results show that the coefficient of the exogenous variable -Thailand's rice export price -is statistically significant across all candidate models ($p < 0.001$), confirming that fluctuations in Thailand's rice export price have a significant effect on Vietnam's rice export price.

As shown in Table 4.3, both the ARIMAX (0,1,1) and ARIMAX (1,1,1) models demonstrate good empirical performance. Specifically, the ARIMAX (1,1,1) model achieves the lowest Akaike Information Criterion value ($AIC = 2320.708$), while the ARIMAX (0,1,1) model achieves the lowest Bayesian Information Criterion value ($BIC = 2335.793$). In addition, the AR(1) coefficient ($p = 0.008$) and the MA(1) coefficient ($p < 0.001$) of the ARIMAX (1,1,1) model are both statistically significant, indicating that the model is able to effectively capture both the time-dependence (persistence) and the short-term shock effects on Vietnam's rice export price.

Although the ARIMAX (0,1,1) model achieves a slightly lower BIC value, the difference in BIC between the two competing models is relatively small (approximately 2.85). Since the primary objective of this study is forecasting, the AIC criterion is given priority in the model selection process. According to Akaike (1974) and Hyndman and Athanasopoulos (2021), AIC is generally recommended in forecasting studies because it achieves a better balance between model fit and forecasting ability compared with more conservative criteria such as BIC. Therefore, considering the lowest AIC value together with the statistical significance of all estimated parameters, the ARIMAX (1,1,1) model is selected as the optimal model for proceeding with the subsequent diagnostic checking and forecasting steps.

In contrast, the ARIMAX (2,1,1) model was not selected, as both autoregressive coefficients are statistically insignificant (AR (1): $p = 0.640$; AR (2): $p = 0.389$), indicating that adding these autoregressive components does not meaningfully improve the model's explanatory power. Similarly, although the ARIMAX (2,1,0) model has statistically significant autoregressive coefficients, its AIC and BIC values remain higher than those of ARIMAX (1,1,1), indicating that adding an additional autoregressive order does not yield a sufficient improvement to offset the increased complexity of the model. Accordingly, ARIMAX (1,1,1) is retained as the final model for the diagnostic checking and forecasting steps in the following sections.

Table 4.4. ARIMAX (1,1,1) Model Estimation Results

Variable	Coefficient	Std. Error	z-Statistic	p-value
Thai.D1	0.1162	0.0232	5.00	0.000
Constant	0.4837	2.2106	0.22	0.827
AR (1)	0.2262	0.0853	2.65	0.008
MA (1)	0.3900	0.0899	4.34	0.000

The coefficient of the exogenous variable Thai.D1 is 0.1162, with a standard error of 0.0232, a z-statistic of 5.00, and a p-value < 0.001 , indicating a very high level of statistical significance (1%). This confirms that fluctuations in Thailand's rice export price have a positive and statistically significant effect on fluctuations in Vietnam's rice export price. Specifically, when the first-differenced Thai rice price increases by 1 USD/ton, the first-differenced Vietnamese rice price tends to increase by approximately 0.116 USD/ton, holding other factors constant. This result is consistent with the findings from the earlier descriptive and visual data analysis, which indicated a close linkage between the Vietnamese and Thai rice markets, and further reinforces the empirical basis for including THRICE in the model as an exogenous variable.

The AR (1) coefficient is 0.2262, statistically significant at the 1% level ($z = 2.65$; $p = 0.008$). This positive and moderately sized coefficient indicates that Vietnam's rice price in the current period is influenced by its own fluctuations in the previous period, reflecting a certain degree of persistence in the differenced price series; however, this dependence is not particularly strong (coefficient below 0.3).

The MA (1) coefficient is 0.3900, showing the highest level of statistical significance in the model ($z = 4.34$; $p < 0.001$). This positive and substantial MA coefficient indicates

that shocks (forecast errors) from the previous period have a lasting effect and exert a positive influence on the current value of the series, reflecting the fact that Vietnam's rice market requires a certain amount of time to fully absorb information shocks or supply-demand shocks that occurred in the past.

Overall, the estimation results show that the ARIMAX (1,1,1) model adequately captures three important dimensions of the dynamics of Vietnam's rice price: (i) the linkage with the Thai rice market through the exogenous variable, (ii) short-term persistence through the AR component, and (iii) the persistence of shocks through the MA component. This combination allows the model to better capture the linear structure of the data series before being used as an input (linear residual) for the LSTM model within the hybrid framework.

c. Diagnostic Checking

After selecting the optimal ARIMAX model based on model selection criteria, the study proceeds to perform diagnostic checking to assess the model's goodness of fit. An important part of this step is to examine whether the model residuals still exhibit serial autocorrelation. To this end, the Ljung-Box test is applied to the model residuals.

Table 4.5. Ljung-Box Test Results for the Residuals of the ARIMAX Model

Statistic	Value
Lags	24
Ljung-Box Q Statistic	21.884
P-value	0.586
Diagnosis	White noise

To evaluate the adequacy of the selected ARIMAX (1,1,1) model, the Ljung-Box test is applied to the model residuals to test the hypothesis of no serial autocorrelation in the residuals -a necessary condition to confirm that the model has fully exploited the linear information structure present in the data.

The results in Table 4.5 show that, at 24 lags, the Ljung-Box Q statistic is 21.884, corresponding to a p-value of 0.586. Since this p-value is considerably larger than conventional significance levels (1%, 5%, 10%), the null hypothesis H_0 (that the residuals exhibit no serial autocorrelation, i.e., the residuals are white noise) cannot be rejected.

This result implies that the residuals of the ARIMAX (1,1,1) model no longer contain exploitable autocorrelation structure, indicating that the model has absorbed most of the lin-

ear information present in Vietnam's rice export price series. This constitutes important statistical evidence confirming the adequacy of the selected model specification, while also reinforcing the reliability of the estimated coefficients presented in Table 4.4.

In the context of this study, the conclusion that "the residuals are white noise" is of particular importance for the Hybrid ARIMAX-LSTM modeling strategy. Specifically, because the linear residuals no longer contain autocorrelation structure, the LSTM component in the hybrid model is not used to "patch" remaining errors of the linear component in the sense of autocorrelation; rather, its role is to capture nonlinear, asymmetric, or complex pattern that the linear ARIMAX model cannot capture. This provides a valid rationale for combining the two methods within a hybrid analytical framework to improve the forecasting performance of Vietnam's rice export price.

LAG	AC	PAC	Q	Prob>Q	-1 [Autocorrelation]	0 [Partial autocor]	1 -1	0 1
1	-0.0016	-0.0016	.0007	0.9788				
2	0.0100	0.0100	.02762	0.9863				
3	0.0473	0.0474	.63646	0.8880				
4	-0.0412	-0.0413	1.1006	0.8942				
5	-0.0544	-0.0558	1.9104	0.8614				
6	-0.0533	-0.0553	2.691	0.8465				
7	-0.1976	-0.1960	13.481	0.0612				
8	-0.0069	-0.0059	13.494	0.0959				
9	0.0376	0.0424	13.887	0.1264				
10	-0.0304	-0.0193	14.146	0.1664				
11	0.0546	0.0364	14.982	0.1833				
12	-0.0020	-0.0304	14.983	0.2424				
13	-0.0077	-0.0275	15	0.3074				
14	-0.0760	-0.1256	16.639	0.2759				
15	-0.0716	-0.0816	18.1	0.2574				
16	-0.0603	-0.0520	19.141	0.2614				
17	0.0037	-0.0007	19.145	0.3203				
18	0.0909	0.1144	21.528	0.2536				
19	-0.0187	-0.0281	21.629	0.3031				
20	0.0170	-0.0206	21.713	0.3563				
21	-0.0231	-0.0981	21.869	0.4071				
22	0.0052	-0.0436	21.877	0.4673				
23	0.0038	-0.0141	21.881	0.5274				
24	-0.0032	0.0103	21.884	0.5862				

Figure 4.7. Residual Correlogram (ACF and PACF)

Figure 4.7 presents the correlogram of the residuals following estimation of the ARIMAX model, including the Autocorrelation Function (ACF), the Partial Autocorrelation Function (PACF), and the Ljung-Box Q test statistic.

The results show that the ACF and PACF coefficients at nearly all lags from lag 1 to lag 24 are very small and fluctuate around zero. There is no evidence of prolonged autocorrelation coefficients or any clear decay pattern across lags. This indicates that the model residuals no longer contain significant autocorrelation structure, meaning the ARIMAX model has explained most of the linear information in Vietnam's rice export price series.

In addition, the Prob > Q values from the Ljung-Box test at all lags exceed the 5% significance level (0.05). Specifically, the p-values range from 0.0612 to 0.9863, with the

smallest value occurring at lag 7, which is still above 0.05. Therefore, there is insufficient evidence to reject the null hypothesis (H_0) that the residuals exhibit no serial autocorrelation. This result confirms that the model residuals possess white-noise properties.

Furthermore, the ACF and PACF plots show that all correlation bars fall within the confidence bounds and are randomly distributed around the horizontal axis. This reflects that the remaining errors are largely random, containing no unmodeled linear patterns.

Overall, the correlogram results combined with the Ljung-Box test indicate that the ARIMAX model satisfies the requirement of residual independence and meets the standard of the Diagnostic Checking stage in the Box-Jenkins methodology. Since the residuals exhibit white-noise characteristics, the model is considered suitable for use in forecasting and as the basis for constructing the Hybrid ARIMAX-LSTM model in the subsequent stage.

Alongside estimating the ARIMAX model for Vietnam's rice export price, the study also separately estimates an ARIMA (0,1,1) model for the Thai rice price variable. Since the ARIMAX (1,1,1) model uses Thailand's rice export price (THRICE) as an exogenous variable, forecasting Vietnam's future rice price first requires constructing a forecast for the Thai price series itself. In other words, because future values of the exogenous variable cannot be observed, a separate ARIMA model for the THRICE series is necessary to generate a 12-month forecast for Thailand, which is then used as the exogenous input for the ARIMAX model when forecasting Vietnam's rice price.

Table 4.6. Estimation Results of the ARIMA (0,1,1) Model for Thailand

Variable	Coefficient	Std. Error	z-Statistic	p-value
Constant	0.6598	2.9038	0.23	0.820
MA (1)	0.4121	0.0217	19.02	0.000
Sigma (σ)	29.1492	0.3694	78.92	0.000

The results show that the MA (1) coefficient is 0.4121 and is statistically significant at the 1% level ($p < 0.001$), indicating that shocks from the previous period continue to influence current fluctuations in the Thai rice price. In addition, the sigma (σ) coefficient is highly significant ($p < 0.001$), reflecting that the model residuals still retain a certain level of volatility, consistent with the characteristics of international rice price series. Overall, the results indicate that the ARIMA (0,1,1) model is suitable for generating forecast values of the exogenous variable to be used in constructing the ARIMAX model for forecasting Vietnam's rice export price in subsequent steps.

d. Forecasting

The forecast results in Table 4.7, generated from the ARIMA (0,1,1) model estimated in Table 4.6, show that Thailand's rice export price is expected to increase gradually along a relatively stable linear trend from March 2026 to March 2027.

Table 4.7. Forecasting Results for Thailand Rice Prices (April 2026-March 2027)

Month	Thailand Rice Price
Apr-26	381.658
May-26	382.315
Jun-26	382.973
Jul-26	383.630
Aug-26	384.288
Sep-26	384.945
Oct-26	385.603
Nov-26	386.261
Dec-26	386.918
Jan-27	387.576
Feb-27	388.233
Mar-27	388.891

Specifically, Thailand's rice price is forecast to increase from 381.66 USD/ton in April 2026 to 388.89 USD/ton in March 2027, corresponding to an increase of about 7.23 USD/ton over 12 months. Notably, the month-to-month increase remains almost constant, stable at around 0.6576 USD/ton per month. This feature reflects the underlying nature of the ARIMA(0,1,1) model: since there is no autoregressive (AR) component, the influence of the MA(1) component operates only in the short run (mainly at the first forecast horizon) and decays rapidly thereafter; the forecast values then converge toward a linear trend line determined by the model's constant (drift) coefficient.

It should be noted that the steady monthly increase (~ 0.66 USD/ton) in the forecast results is quite close in magnitude to the estimated constant (0.6598) in Table 4.6. This is theoretically consistent, since once the MA(1) component no longer exerts influence after the first forecast period, the expected value of the differenced series converges to the estimated constant, causing the forecast price series (in levels) to increase linearly with a slope equal to this coefficient.

From an economic standpoint, the forecast results suggest that Thailand's rice price is expected to continue a mild and stable upward trend in the short-to-medium term (the next 12 months), with no abrupt fluctuations or trend reversals. However, it should be noted that this "linearization" characteristic is an inherent limitation of linear forecasting models such as ARIMA when forecasting over longer horizons: due to the absence of an autoregressive component, the model cannot capture nonlinear fluctuations, unusual shocks, or complex cyclical patterns that may occur in reality (such as shocks related to export policy or weather conditions). Therefore, the forecast results in Table 4.7 should be viewed as a baseline reference scenario reflecting the market's expected central tendency, rather than an absolutely precise forecast of future price movements.

Table 4.8. Vietnam and Thailand Rice Prices (April 2026-March 2027)

Month	Thailand Rice Price	Vietnam Rice Price
Apr-26	381.658	346.340
May-26	382.315	346.900
Jun-26	382.973	347.460
Jul-26	383.630	348.020
Aug-26	384.288	348.580
Sep-26	384.945	349.140
Oct-26	385.603	349.700
Nov-26	386.261	350.260
Dec-26	386.918	350.820
Jan-27	387.576	351.380
Feb-27	388.233	351.940
Mar-27	388.891	352.500

Table 4.8 presents the forecast results for Vietnam's and Thailand's rice export prices from April 2026 to March 2027. The results show that both price series exhibit a mild and stable upward trend throughout the forecast period, reflecting a positive recovery trend in the rice export market in the short term.

For Thailand's rice price, the forecast increases from 381.66 USD/ton in April 2026 to 388.89 USD/ton in March 2027, corresponding to an increase of about 7.23 USD/ton (approximately 1.9%). Meanwhile, Vietnam's rice price increases from 346.34 USD/ton to 352.50 USD/ton, corresponding to an increase of 6.16 USD/ton (about 1.78%). The increases in both price series occur relatively evenly across months, indicating that price fluctuations

during the forecast period do not exhibit abrupt changes but instead maintain a stable upward trend.

The gap between Thailand's and Vietnam's rice prices remains relatively stable, fluctuating around 35-36 USD/ton throughout the forecast period. This indicates that the rice prices of the two countries tend to move in the same direction, reflecting a close relationship between the two major rice-exporting markets in the region. This result is also consistent with the study's underlying assumption in selecting Thailand's rice price as the exogenous variable in the ARIMAX model, given its ability to reflect fluctuations in the international rice market and their influence on Vietnam's rice export price.

Furthermore, the stable upward trend of Vietnam's rice price during the forecast period suggests that the Hybrid ARIMAX-LSTM model not only preserves the long-term trend identified by the ARIMAX model but also helps adjust for nonlinear fluctuations through the LSTM network, resulting in smoother and more stable forecast outcomes. From a practical perspective, this implies that, absent major fluctuations in international market conditions and trade policy, Vietnam's rice export price is likely to continue its mild upward trend in 2026-2027, creating favorable conditions for export activities and strategic business planning within the industry.

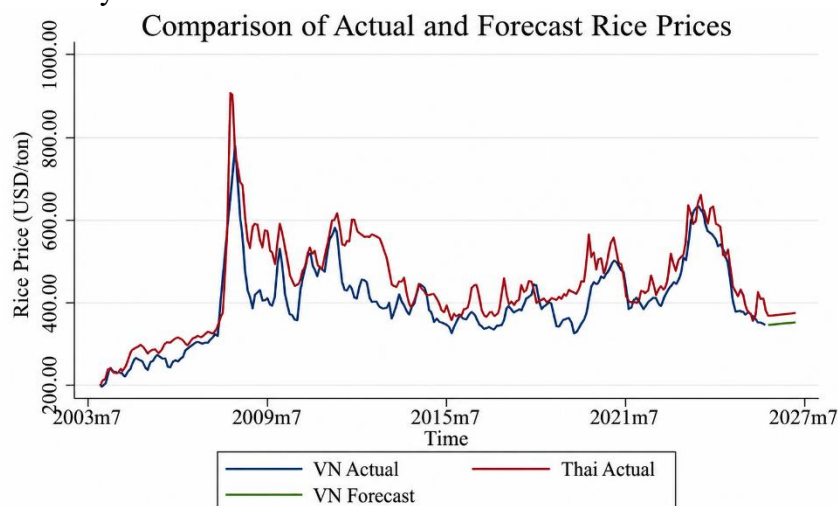


Figure 4.8. Actual and ARIMAX Forecasted Vietnam Rice Export Prices with Thailand Rice Prices

Figure 4.8 presents the movements of three data series: Vietnam's actual rice export price (VN Actual), Thailand's actual rice price (Thai Actual), and Vietnam's rice price forecast using the ARIMAX model (VN Forecast) over the study and forecast periods.

The chart shows that Vietnam's and Thailand's rice prices exhibit fairly similar movement patterns throughout the study period. Both series recorded sharp increases during the

2008 global food crisis, followed by a gradual decline and subsequent fluctuations in line with market cycles. From 2022 onward, rice prices in both countries rose again before adjusting to lower levels toward the end of the study period. Although Thailand's rice price generally remains higher than Vietnam's, the two series still move in the same direction during most periods.

This result indicates that Thailand's rice price has a close relationship with Vietnam's rice export price, further supporting the choice of Thailand's rice price as the exogenous variable in the ARIMAX model. The use of this exogenous variable helps the model better reflect the impact of international rice market developments on Vietnam's rice export price.

In the forecast period from April 2026 to March 2027, the ARIMAX forecast line shows that Vietnam's rice export price is expected to increase mildly and maintain a stable trend, rising from about 346 USD/ton to about 352 USD/ton. The forecast line shows no abrupt fluctuations and evolves relatively smoothly, reflecting the ARIMAX model's characteristic of describing the linear component and the long-term trend of the time series.

However, it can also be observed that at certain points with very large fluctuations - particularly sudden price peaks - the forecast line does not fully track the amplitude of the actual series. This indicates that the ARIMAX model primarily reflects linear relationships and cannot fully describe the nonlinear characteristics or unusual market shocks. Accordingly, the study proceeds to apply a Long Short-Term Memory (LSTM) neural network to learn the ARIMAX residual series, thereby constructing a Hybrid ARIMAX-LSTM model that combines the strengths of econometric and deep learning approaches. According to Zhang (2003), Khashei and Bijari (2011), and Babu and Reddy (2014), combining a linear model with a neural network helps improve the ability to capture complex data structures, thereby enhancing the accuracy and stability of forecast results.

e. Discussion of ARIMAX result

The empirical results show that the ARIMAX model provides a suitable framework for describing and forecasting Vietnam's rice export price during both the estimation and forecasting periods. The forecast results indicate that the price series is expected to maintain a relatively stable upward trend in the short term. This suggests that the dynamic patterns observed in the past still contain useful information for forecasting over the coming period. However, this trend should be interpreted within the scope of the model and the forecasting horizon, rather than as evidence of a sustained upward trend in the long term.

One notable finding is that Thailand's rice export price is statistically significant in the ARIMAX model. This finding is consistent with Chen and Saghaian (2016), who examined market integration and price transmission across rice export markets and identified price linkages among exporting countries. Therefore, developments in the Thai market may contain useful signals for explaining and forecasting Vietnam's rice price. This finding is also consistent with the approach of Anggraeni et al. (2019), who incorporated exogenous variables into a Hybrid NNs-ARIMAX model to forecast rice prices, showing that external factors can provide additional information for the forecasting process. However, in the present study, the statistical significance of Thailand's rice export price indicates that this variable provides useful information for explaining and forecasting Vietnam's rice export price; it does not provide sufficient evidence to establish a causal relationship or the existence of a volatility spillover mechanism between the two markets.

However, Vietnam's rice export price is not determined solely by developments in the Thai market. The estimated coefficient suggests that other factors may also contribute to price formation, including domestic supply and demand conditions, product quality, trade policies, exchange rates, and international market conditions. Therefore, the Thai market can be viewed as an important external signal, while price movements in Vietnam still reflect the specific characteristics of its domestic market. This interpretation is also consistent with Nhung and Wang (2022), whose study highlights the opportunities and challenges facing the competitiveness of Vietnamese rice in international trade.

Another notable finding is that the price gap between Vietnam and Thailand is forecast to change relatively little throughout the forecasting period. Although both countries are expected to experience an upward price trend, the difference between the two price series remains relatively stable. This suggests that Vietnam's relative position compared with its main competitor may not change substantially in the short term. However, such stability does not necessarily mean that price-based competitive advantages will be maintained in the long term. As international buyers increasingly focus on quality, food safety, traceability, sustainability, and value added, competing mainly on price may become less sustainable. Therefore, improving product quality, building rice brands, expanding geographical indications, and developing higher value-added rice products are important directions for strengthening Vietnam's export competitiveness in the long term.

Overall, ARIMAX not only captures the systematic dynamics of the price series but also incorporates signals from the Thai market into the forecasting process. However, the remaining residuals indicate that some variations are not fully explained by the linear component and the exogenous variable. This provides a basis for the next stage of the study, which examines whether the nonlinear characteristics remaining in the residuals can be exploited using LSTM. Accordingly, ARIMAX serves both as the baseline forecasting model and as the foundation for developing the Hybrid ARIMAX-LSTM model.

4.2.2. Hybrid ARIMAX-LSTM Model Configuration and Architecture

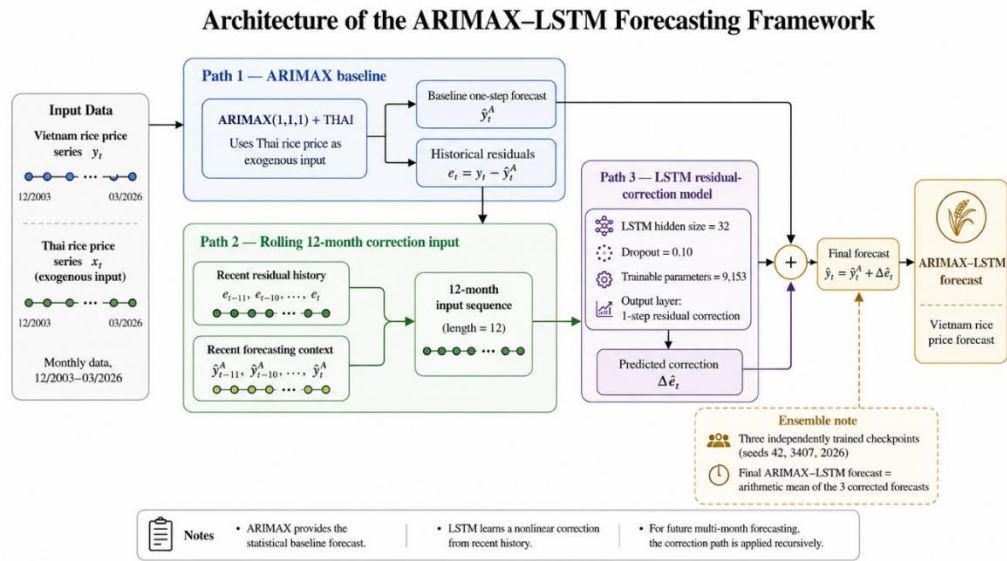


Figure 4.9. Architecture of the ARIMAX-LSTM Model Used in the Study.

Source: Author's compilation

Figure 4.9 illustrates the overall architecture of the Hybrid ARIMAX-LSTM model used in this study. The diagram is read from left to right. The two input series are Vietnam's rice price and Thailand's rice price. ARIMAX generates the baseline forecast; the historical ARIMAX errors, together with the recent forecast context, are organized into a 12-month sequence; LSTM uses this sequence to estimate an adjustment value; finally, this adjustment is added to the ARIMAX forecast to produce the ARIMAX-LSTM forecast. From an econometric perspective, this structure divides the task between the two components. ARIMAX is responsible for describing the time-varying dynamics of Vietnam's rice price and the concurrent effect of Thailand's rice price. LSTM does not replace ARIMAX but instead only learns the remaining error pattern over the most recent 12 months. The final forecast can

therefore be understood as ARIMAX's interpretable baseline forecast plus a nonlinear correction component learned from the data.

Specifically, Path 1 is responsible for extracting the linear and interpretable component of Vietnam's rice price series, through the ARIMAX (1,1,1) model with Thailand's rice price serving as the exogenous variable, reflecting the price spillover effect between the two competing rice-export markets. The output of this branch is not only the baseline forecast $\hat{y}^A_t$ but also the historical residual series $e_t = y_t - \hat{y}^A_t$, which serves as the information "bridge" connecting the statistical and machine-learning components within the hybrid model.

In Path 2, the study employs a sliding-window mechanism with a length of 12 months. For each forecast point, the 12 most recent residual values, combined with the 12 most recent forecast-context values, are organized into an input sequence for the LSTM. Using a 12-month window allows the model to access information over a relatively long data cycle, while also being consistent with the characteristics of monthly price data. Rather than relying solely on the error at a single point in time, the LSTM can observe the continuous variation in residuals over preceding months to determine whether systematic patterns exist that ARIMAX has not yet captured.

In Path 3, the input sequence is fed into an LSTM model with 32 hidden units and a dropout rate of 0.10. This configuration allows the model to learn temporal-sequence relationships within the residuals, while dropout is used to limit the risk of overfitting to the training data. The LSTM's output is a one-step correction value, denoted $\Delta\hat{e}_t$. In essence, this value does not represent a complete re-forecast of Vietnam's rice price, but rather an additional adjustment based on the structure that LSTM has detected in the recent error history.

The output of the LSTM is the forecast correction component. Another notable feature of architecture is the use of three independent initializations with different random seeds. Because the training process of a neural network can be affected by the initial weight initialization, a single training run may produce results influenced to some extent by randomness. Therefore, the study uses three independently trained LSTM models and averages their three correction forecasts to produce the final forecast value. This approach helps reduce dependence on any single specific initialization, while also contributing to greater stability in the forecast results.

In summary, the ARIMAX-LSTM architecture is designed in line with the spirit of modern statistical-machine learning hybrid models: it clearly separates the roles of the interpretable

linear component and the corrective nonlinear component, while incorporating a stability-control mechanism through an ensemble of multiple initializations. This provides a sound basis for expecting the model to achieve improved forecasting accuracy compared with using ARIMAX or LSTM independently.

4.2.3. Model Performance Evaluation

Table 4.9. Comparison of Forecasting Performance Across Models

Model	MAE	RMSE	MAPE (%)	SMAPE (%)
Random Walk (without drift)	13.878	19.533	2.846	2.829
Random Walk (with drift)	14.235	19.692	2.929	2.905
ARIMAX	11.873	15.759	2.488	2.483
LSTM	11.744	15.821	2.429	2.427
ARIMAX-LSTM	11.352	15.611	2.381	2.388

Source: Author's calculation.

The ARIMAX-LSTM model ranks first in terms of MAE, RMSE, MAPE, and SMAPE among the five models. Compared with the Random Walk without drift, ARIMAX-LSTM reduces MAE by 18.20%, RMSE by 20.08%, and MAPE by 16.36%. This addresses the first part of the assessment: the model not only outperforms the other forecasting models but also surpasses the simple forecasting baseline. However, to better understand whether this superior performance comes from the hybrid structure itself, it is also important to examine the performance of the standalone LSTM model.

The standalone LSTM model performs competitively with ARIMAX. LSTM achieves a lower MAE (11.744) and MAPE (2.429%) than ARIMAX (11.873 and 2.488%, respectively), although its RMSE (15.821) is slightly higher than that of ARIMAX (15.759). This indicates that LSTM can capture certain nonlinear patterns in the rice price series, but its performance is not consistently superior across all error metrics. In contrast, the ARIMAX-LSTM hybrid achieves the lowest values across all four evaluation metrics. This consistent improvement suggests that the hybrid structure can combine the strengths of ARIMAX and LSTM, integrating the linear forecasting structure captured by ARIMAX with the nonlinear learning capability of LSTM, and thereby providing more balanced forecasting performance than either model used independently.

Average-based metrics, however, can mask cases where the model outperforms in some months but underperforms in others. It is therefore necessary to further examine the forecast plot and the month-by-month errors.

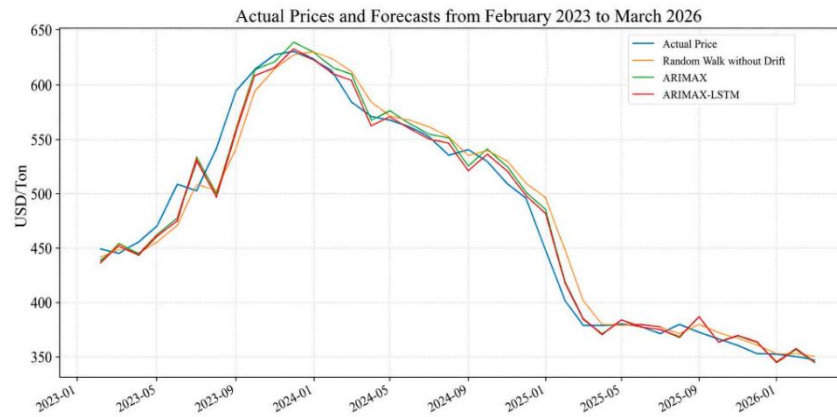


Figure 4.10. Actual Values and Forecast Lines over the 38-Month Evaluation Period.

Source: Author's compilation.

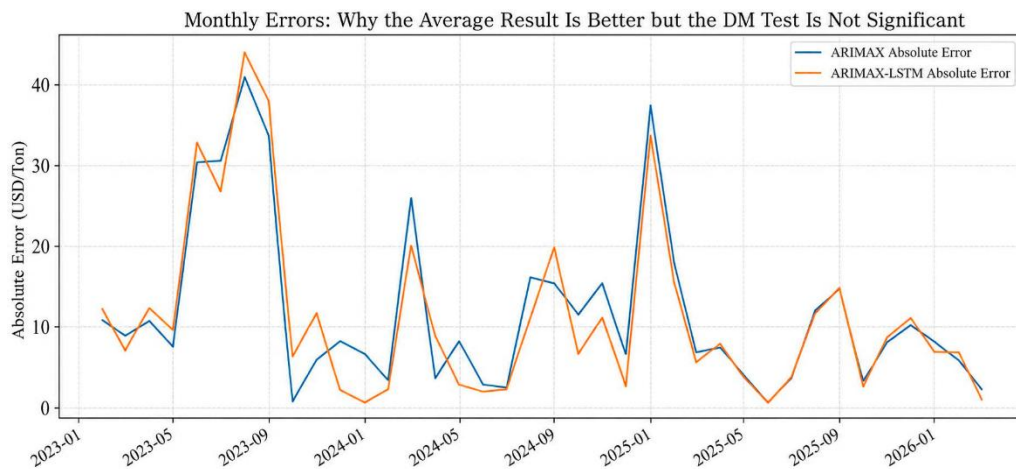


Figure 4.11. Monthly Absolute Errors of the ARIMAX and ARIMAX-LSTM Models.

Source: Author's compilation.

HLN refers to the Harvey-Leybourne-Newbold correction for small samples. Given the sample size of only 38 months, this correction provides a more conservative DM statistic than applying the large-sample formula mechanically.

Table 4.10. Diebold-Mariano Test Results (HLN Correction)

Metric	DM-HLN	p-value	Conclusion at 5% Significance Level
MAE	0.882	0.384	Not statistically significant
RMSE/MSE	0.227	0.822	Not statistically significant
MAPE	0.993	0.327	Not statistically significant

Source: Author's calculation.

All three p-values are greater than 0.05. Therefore, the appropriate conclusion is that ARIMAX-LSTM achieves lower average forecasting errors numerically; however, the 38-month evaluation period does not provide sufficient evidence to establish that this advantage is statistically significant at the 5% level. The ARIMAX-LSTM model achieves lower absolute errors than ARIMAX in 21 out of 38 months (55.3%), while ARIMAX performs better in the remaining 17 months. Thus, ARIMAX-LSTM performs better in aggregate, but its advantage is not observed consistently in every month. This is an important reason why the statistical test does not yield a small p-value.

From an applied economic perspective, the results can be interpreted at three levels. First, the Random Walk benchmark indicates that using only the previous month's price is insufficient for accurate forecasting; both ARIMAX and ARIMAX-LSTM achieve substantial improvements over this benchmark. Second, ARIMAX captures the dynamics of Vietnam's rice prices as well as information from Thailand's rice prices, thereby substantially reducing forecast errors relative to the Random Walk benchmark. Third, LSTM further adjusts the remaining errors from ARIMAX, leading to reductions in MAE, MAPE, and SMAPE. However, the improvement in RMSE is relatively small, and the performance gains are not observed consistently across all months.

Therefore, the appropriate interpretation is not that “ARIMAX-LSTM is overwhelmingly superior,” but rather that “ARIMAX-LSTM provides a moderate but relatively consistent improvement across the aggregate performance metrics; however, the current 38-month evaluation sample does not provide sufficiently strong statistical evidence to confirm its superiority at the 5% significance level.”

Over the 38-month evaluation period, both ARIMAX and ARIMAX-LSTM outperform the two Random Walk benchmark models. Compared with the Random Walk without drift, ARIMAX-LSTM reduces MAE by 18.20%, RMSE by 20.08%, and MAPE by 16.36%. Compared with ARIMAX, ARIMAX-LSTM reduces MAE by 4.38%, RMSE by 0.94%,

MAPE by 4.32%, and SMAPE by 3.86%. However, the Diebold-Mariano test with the Harvey-Leybourne-Newbold correction does not indicate a statistically significant difference at the 5% level, with p-values of 0.3835 for MAE, 0.8216 for the squared-error loss associated with RMSE, and 0.3269 for MAPE. Therefore, ARIMAX-LSTM demonstrates an improvement in average forecasting accuracy, but the evidence from the current 38 observations is insufficient to establish statistically significant superiority over ARIMAX at the 5% level.

Note: For RMSE, the DM test uses the monthly squared errors (MSE loss), because RMSE is the square root of the mean squared error and there is no separate “monthly RMSE” that can be directly used as the loss series in the DM test.

4.2.4. Hybrid ARIMAX-LSTM Model Results

a. Forecasting Results for the 4/2026 - 3/2027 Period

Table 4.11. ARIMAX and Hybrid ARIMAX-LSTM Forecasts

Month	ARIMAX	ARIMAX-LSTM	LSTM Adjustment
Apr-26	345.765	346.945	1.180
May-26	347.137	348.319	1.182
Jun-26	348.314	349.486	1.172
Jul-26	349.407	350.561	1.154
Aug-26	350.442	351.575	1.133
Sep-26	351.430	352.539	1.109
Oct-26	352.375	353.459	1.084
Nov-26	353.282	354.340	1.059
Dec-26	354.152	355.186	1.034
Jan-27	354.989	355.999	1.010
Feb-27	355.796	356.782	0.986
Mar-27	356.575	357.538	0.964

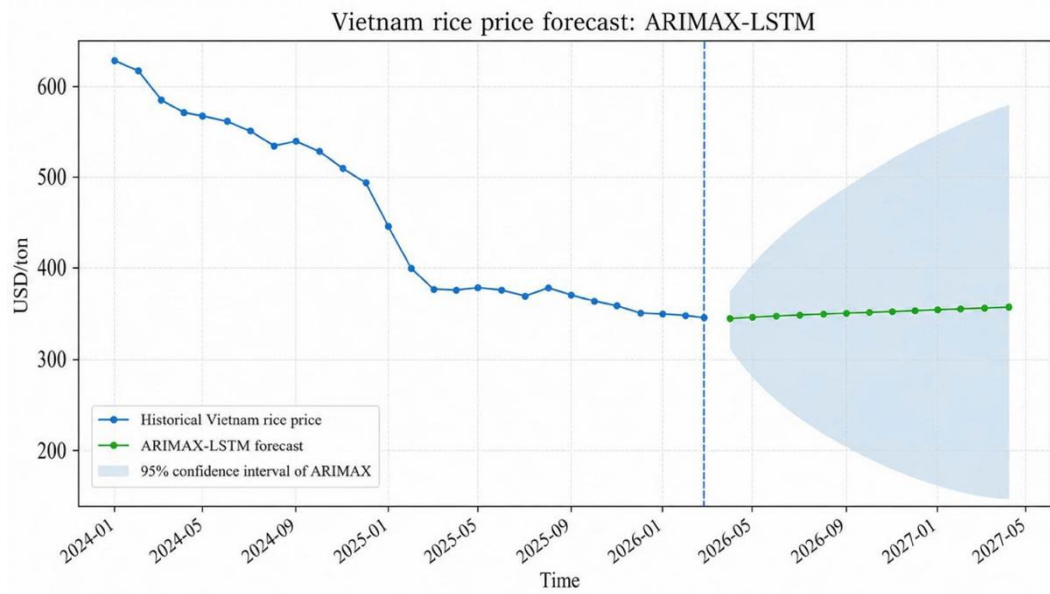


Figure 4.12. Vietnam Rice Price Forecast Using the Hybrid ARIMAX-LSTM Model

The forecasting results show that both the ARIMAX and Hybrid ARIMAX-LSTM models predict a gradual upward trend in Vietnam's rice export prices from April 2026 to March 2027. The ARIMAX forecast increases from 345.765 USD/ton in April 2026 to 356.575 USD/ton in March 2027. Meanwhile, the Hybrid ARIMAX-LSTM forecast is consistently higher, increasing from 346.945 USD/ton to 357.538 USD/ton over the same period.

The difference between the two models is reflected in the LSTM adjustment. The adjustment remains positive throughout the forecasting period, reaching its highest level of 1.182 USD/ton in May 2026 and gradually decreasing to 0.964 USD/ton in March 2027. This indicates that the LSTM component provides an upward adjustment to the baseline ARIMAX forecasts. However, the decreasing magnitude of the adjustment over time suggests that the contribution of the nonlinear residual component becomes less pronounced as the forecasting horizon extends.

In addition to the upward trend in the forecasted prices, the 95% confidence interval gradually widens over time. This indicates that the level of forecast uncertainty increases as the forecasting horizon becomes longer.

b. Discussion of ARIMAX-LSTM forecast results

The empirical results show that the Hybrid ARIMAX-LSTM model provides a moderate improvement in forecasting Vietnam's rice export price compared with the ARIMAX model alone. The Hybrid model achieves lower MAE, RMSE, MAPE, and SMAPE values

during the evaluation period. Rather than simply reflecting the addition of a more complex model, this result highlights the complementary roles of ARIMAX and LSTM. ARIMAX captures the systematic time dynamics of Vietnam's rice export price together with information from Thailand's rice export price included as an exogenous variable, while LSTM is applied to the residual component to further capture nonlinear relationships over time. Therefore, the Hybrid framework allows different sources of forecasting information to be considered within the same forecasting process.

The relative improvement also provides an important interpretation of the structure of the rice export price series. The results suggest that the residuals may contain some additional predictive information that can be captured by LSTM. At the same time, the limited improvement indicates that ARIMAX has already captured a substantial part of the predictable structure of the series. Therefore, the contribution of LSTM should be understood as an additional forecasting component rather than a replacement for the statistical model. This finding is particularly relevant to agricultural commodity price forecasting, where price movements may contain both systematic time-dependent characteristics and more complex fluctuations. The Hybrid approach therefore provides a framework for examining whether the information remaining after the linear component has been modeled can continue to contribute to forecasting performance.

This interpretation is generally consistent with previous studies on Hybrid forecasting methods. For example, Mahawan et al. (2022) applied a Hybrid ARIMAX-LSTM framework to forecast Thailand's rice export price and found that the proposed Hybrid model performed better than the individual ARIMA, ARIMAX, and LSTM models. Similarly, Anggraeni et al. (2019) applied a Hybrid NNs-ARIMAX model with exogenous variables to forecast rice prices in Indonesia, where evaluation measures such as MAPE and RMSE indicated an improvement over the individual models, although the improvement was relatively limited. Dave et al. (2021) also applied a Hybrid ARIMA-LSTM model to forecast Indonesia's exports, with MAE, RMSE, MAPE, and SMAPE indicating improved forecasting performance when the statistical model was combined with LSTM, although the differences in forecasting accuracy between the models were not large. These findings support the general argument that statistical models and deep learning models can provide complementary forecasting information. However, the degree of improvement varies across markets and datasets, indicating that the usefulness of the Hybrid framework depends on the amount and

characteristics of the additional information remaining after the statistical component has been modeled.

From a practical perspective, the results suggest that the Hybrid model can be viewed as an extension of ARIMAX rather than a complete replacement for ARIMAX. The relatively small improvement indicates that analysts do not always need a highly complex model to generate useful forecasts, while the Hybrid framework can still be valuable when additional predictive information exists in the residual series. For Vietnam's rice export market, this approach may support short-term monitoring of price movements by combining information on price dynamics captured by ARIMAX with additional patterns identified by LSTM.

For future research, the Hybrid framework could be extended by incorporating additional market and macroeconomic variables, such as exchange rates, transportation costs, global rice supply conditions, or other related commodity prices. Future studies could also compare the proposed framework with nonlinear models and other models focusing on price volatility to determine whether additional information can be exploited more effectively. These extensions would help assess whether the modest contribution of LSTM observed in this study is specific to the selected variables and forecasting period or reflects a broader characteristic of Vietnam's rice export price dynamics.

Overall, the main message of the study is that the Hybrid ARIMAX-LSTM framework provides a useful complementary approach for forecasting Vietnam's rice export price. The results show that ARIMAX can capture a substantial part of the systematic price dynamics and information from the external market, while LSTM can further exploit additional predictive information from the remaining residuals. Although the relative improvement in forecasting accuracy is modest, the findings indicate that combining econometric and deep learning approaches can provide additional information for the forecasting process. Therefore, the value of the Hybrid framework does not necessarily lie in replacing ARIMAX, but in extending its forecasting capability by exploiting information that is not fully explained by the linear component.

CHAPTER 5. CONCLUSION AND RECOMMENDATIONS

This chapter concludes the study by synthesizing the major empirical findings and assessing the extent to which the research objectives and questions have been addressed. It summarizes the forecasting performance of the ARIMAX and Hybrid ARIMAX-LSTM models and highlights the contribution of integrating linear, exogenous, and nonlinear information in forecasting Vietnam's 5% broken white rice export prices. The chapter then discusses the theoretical and practical contributions of the study and proposes policy recommendations related to production management, export activities, forecasting systems, and strategic planning. Finally, the research limitations are acknowledged and directions for future research are suggested, particularly regarding data expansion, additional exogenous variables, alternative deep learning architectures, and further improvements in forecasting robustness and practical applicability.

5.1. Conclusion

This study was conducted to analyze the historical fluctuations of Vietnam's 5% broken white rice Free on Board (FOB) export price, develop and evaluate the ARIMAX and Hybrid ARIMAX-LSTM forecasting models, and assess their applicability to short-term rice export price forecasting. The empirical analysis employed 268 monthly observations from December 2003 to March 2026, with Vietnam's 5% broken white rice FOB price (VNRICE) serving as the target variable and Thailand's 5% broken white rice FOB price (THRICE) incorporated as the sole exogenous variable.

Regarding the first research objective, the descriptive analysis confirms that Vietnam's benchmark rice export price experienced substantial fluctuations over the study period rather than following a stable or uniform pattern. VNRICE recorded an average price of 399.0793 USD/ton, with a standard deviation of 97.5302 USD/ton and values ranging from 197.00 to 780.00 USD/ton. By comparison, THRICE recorded a higher average price of 446.0526 USD/ton and greater dispersion, with a standard deviation of 112.8916 USD/ton and a range from 198.25 to 907.00 USD/ton. The time-series analysis further indicates broadly similar movements in the two benchmark price series during several periods, including pronounced price increases around the 2007-2008 global food crisis and the 2023-2024 period of international rice-market disruption. These historical episodes provide relevant market context for interpreting the observed price movements; however, the present forecasting design does

not empirically identify such events as causal determinants of Vietnam's rice export price. Similarly, the observed co-movement between VNRICE and THRICE should be interpreted as descriptive evidence of closely related regional price dynamics rather than as proof of a formally identified price-transmission mechanism.

Regarding the second research objective, the Augmented Dickey-Fuller tests indicate that both price series are non-stationary at their original levels. At level, the p-values are 0.1244 for VNRICE and 0.0531 for THRICE, both exceeding the 5% significance threshold. After first-order differencing, the ADF statistics are -8.473 for VNRICE and -11.698 for THRICE, with p-values of 0.0000 for both series. Accordingly, both variables are integrated of order one, $I(1)$, and their first-differenced forms were employed in the subsequent ARIMAX estimation.

Based on the model-identification and selection procedure, ARIMAX (1,1,1) was selected as the baseline econometric specification. Although ARIMAX (0,1,1) produced the lowest BIC, ARIMAX (1,1,1) achieved the lowest AIC value of 2320.708 and contained statistically significant AR and MA components, making it the preferred specification under the forecasting-oriented selection criterion adopted in this study. Within the selected model, the coefficient of the first-differenced Thailand rice price variable is 0.1162 and is statistically significant at the 1% level ($p < 0.001$). The AR (1) coefficient is 0.2262 ($p = 0.008$), while the MA (1) coefficient is also statistically significant. In contrast, the constant term is not statistically significant ($p = 0.827$). These estimation results indicate that movements in THRICE contain statistically relevant information for modeling changes in VNRICE within the ARIMAX specification. The significance of THRICE should nevertheless be interpreted as statistical and predictive relevance within the forecasting model rather than as evidence of a causal effect.

The adequacy of the selected ARIMAX specification is further supported by the residual diagnostic results. At lag 24, the Ljung-Box statistic is 21.8843 with a p-value of 0.5862, indicating that the null hypothesis of no residual serial autocorrelation cannot be rejected. The residual correlogram similarly shows no persistent autocorrelation pattern across the examined lags. These findings suggest that ARIMAX (1,1,1) captures most of the systematic linear dependence represented by the selected specification and provides an appropriate baseline for the subsequent hybrid forecasting procedure.

The Hybrid ARIMAX-LSTM model was then constructed using a sequential residual-learning framework. In this structure, ARIMAX generates the baseline forecast using the historical dynamics of Vietnam's rice price together with information from Thailand's benchmark price, while the LSTM component is applied to the ARIMAX residual series to generate an additional forecast correction. Therefore, the LSTM component functions as a complementary residual-learning mechanism rather than as a standalone replacement for the econometric model.

The updated out-of-sample evaluation over 38 monthly observations shows that both ARIMAX and Hybrid ARIMAX-LSTM outperform the two Random Walk benchmark specifications and that the Hybrid model achieves the lowest average forecast errors among the four evaluated models. The Random Walk without drift records an MAE of 13.878 USD/ton, an RMSE of 19.533 USD/ton, a MAPE of 2.846%, and an SMAPE of 2.829%, while the Random Walk with drift records corresponding values of 14.235 USD/ton, 19.692 USD/ton, 2.929%, and 2.905%. The standalone ARIMAX model improves substantially on these benchmarks, producing an MAE of 11.873 USD/ton, an RMSE of 15.759 USD/ton, a MAPE of 2.488%, and an SMAPE of 2.483%.

The Hybrid ARIMAX-LSTM model provides a further, although relatively modest, improvement over ARIMAX, with an MAE of 11.352 USD/ton, an RMSE of 15.611 USD/ton, a MAPE of 2.381%, and an SMAPE of 2.388%. Relative to the standalone ARIMAX model, these values represent reductions of approximately 4.38% in MAE, 0.94% in RMSE, 4.32% in MAPE, and 3.86% in SMAPE. The relatively larger improvements in MAE, MAPE, and SMAPE, together with the smaller reduction in RMSE, indicate that the benefit of the LSTM correction is not uniform across evaluation criteria. Therefore, the empirical evidence supports the interpretation that residual learning provides an incremental improvement in average point-forecast accuracy rather than a substantial transformation of the forecasting performance established by ARIMAX.

The month-by-month comparison provides additional evidence for this interpretation. Hybrid ARIMAX-LSTM generates a lower absolute forecast error than ARIMAX in 21 of the 38 evaluation months, equivalent to 55.3% of the test observations, whereas ARIMAX performs better in the remaining 17 months. Thus, the Hybrid model performs better in aggregate, but its advantage is not consistently observed in every individual period.

More importantly, the Diebold-Mariano test with the Harvey-Leybourne-Newbold small-sample correction does not establish a statistically significant difference in forecasting accuracy between ARIMAX and Hybrid ARIMAX-LSTM at the 5% significance level. The reported p-values are 0.383514 for the absolute-error loss associated with MAE, 0.821597 for the squared-error loss associated with RMSE, and 0.326945 for the percentage-error loss associated with MAPE. All three p-values exceed 0.05. Consequently, although Hybrid ARIMAX-LSTM produces lower average forecast-error metrics, the appropriate statistical conclusion is that the current 38-month evaluation sample does not provide sufficient evidence to establish statistically significant superiority of the Hybrid model over ARIMAX. The results should therefore be described as better observed average forecasting performance within the current evaluation sample rather than as conclusive evidence that the Hybrid model is universally or statistically superior.

Compared with the Random Walk without drift benchmark, the Hybrid ARIMAX-LSTM model reduces MAE by 18.20%, RMSE by 20.08%, and MAPE by 16.36%. This benchmark comparison provides additional evidence that incorporating historical price dynamics and Thailand's benchmark rice price contains meaningful forecasting information beyond a simple persistence-based forecast. However, the relatively small incremental gain of Hybrid ARIMAX-LSTM over the already competitive ARIMAX model indicates that the additional residual correction contributes only moderately within the present dataset and model configuration.

For the genuine future forecasting horizon extending from April 2026 through March 2027, the results indicate a gradual upward trajectory in Vietnam's benchmark rice export price. Table 4.11 reports April 2026 as the first forecast month, with ARIMAX and Hybrid ARIMAX-LSTM forecasts of 345.765 and 346.945 USD/ton, respectively. Over the full forecast horizon, the Hybrid ARIMAX-LSTM forecast increases gradually from 346.945 USD/ton in April 2026 to 357.538 USD/ton in March 2027. Over the same period, the ARIMAX forecast increases from 345.765 USD/ton to 356.575 USD/ton. The Hybrid forecasts remain slightly above the corresponding ARIMAX forecasts because the LSTM adjustment is positive throughout the reported period. This adjustment reaches 1.182 USD/ton in May 2026 and gradually declines to 0.964 USD/ton by March 2027, indicating that the magnitude of the residual correction becomes smaller as the forecasting horizon extends. At the same time, the widening forecast interval over the horizon reflects increasing uncertainty associated with longer-range recursive forecasting.

Accordingly, the future forecast should be interpreted as a model-based short-term outlook rather than as a deterministic prediction of future rice prices. The projected gradual increase is conditional on the information contained in the historical data and on the forecasting structure employed in the study. Unexpected developments in trade policy, weather conditions, production, logistics, international supply and demand, or other market conditions may generate actual price paths that differ from the model forecasts. Therefore, the forecast is most appropriately used as supplementary quantitative information for market monitoring and short-term decision support.

Regarding the third research objective, the empirical findings provide a basis for recommendations aimed at improving forecasting effectiveness and supporting decision-making in Vietnam's rice export market. For exporting enterprises, model-generated forecasts may provide additional quantitative information for short-term procurement assessment, inventory management, pricing discussions, contract preparation, and price-risk management. The statistical relevance of THRICE within the ARIMAX specification also suggests that movements in Thailand's 5% broken white rice benchmark price should be included in regular market-monitoring activities as a relevant predictive indicator. For production and management decisions, however, forecasts should be treated as supplementary information rather than as deterministic signals for expanding or reducing production. Likewise, government agencies and other stakeholders should combine quantitative forecasts with current information on domestic production, international supply and demand, trade policies, logistics, and broader market conditions.

The results also support the establishment of a periodically updated forecasting process. As new monthly observations become available, the dataset should be extended, model parameters re-estimated where appropriate, forecast errors reassessed, and alternative forecasting models compared using consistent out-of-sample evaluation procedures. Such periodic reassessment is particularly important because the relative performance observed over the current 38-month evaluation period may change under different market conditions or evaluation windows.

Overall, the study achieved its research objectives by documenting the historical fluctuations of Vietnam's 5% broken white rice FOB price, developing an ARIMAX framework that incorporates Thailand's benchmark rice price as the sole exogenous variable, construct-

ing a sequential residual-based Hybrid ARIMAX-LSTM model, and evaluating these models against both each other and simple Random Walk benchmarks. The empirical results show that ARIMAX provides a substantial improvement over the Random Walk benchmarks and that adding the LSTM residual-learning component generates a further reduction in average forecast errors. Nevertheless, the incremental improvement over ARIMAX is relatively modest, particularly for RMSE, and the Diebold-Mariano tests do not establish a statistically significant difference between the two models at the 5% level.

Therefore, the principal conclusion of this study is that Hybrid ARIMAX-LSTM demonstrates better observed average point-forecast performance than standalone ARIMAX within the current dataset and 38-month evaluation sample, but the available statistical evidence does not support a claim of significant or universal superiority. The Hybrid framework should consequently be regarded as a feasible short-term forecasting and decision-support approach whose value lies in providing an additional residual correction to an already competitive econometric baseline. Its forecasting performance should continue to be evaluated as new data become available and interpreted within the limitations of the dataset, model specification, evaluation design, forecast horizon, and prevailing market conditions.

5.2. Research Contributions

5.2.1. Theoretical Contributions

From a theoretical perspective, this study contributes to the empirical literature on time-series forecasting in agricultural economics and international trade by examining the application of a hybrid econometric-deep learning framework to forecasting Vietnam's benchmark rice export price.

First, the study provides additional empirical evidence on the applicability of the Hybrid ARIMAX-LSTM model for agricultural commodity price forecasting. The study adopts a sequential residual-based framework in which ARIMAX serves as the baseline econometric model, capturing the historical dynamics of the target series while incorporating Thailand's rice price as the sole exogenous variable. The residuals generated by ARIMAX are subsequently used as inputs to the LSTM network to model remaining patterns in the residual series and generate an additional forecast correction. Thus, LSTM functions as a complementary residual-learning component within the hybrid framework rather than as a separate standalone forecasting model. This structure contributes context-specific empirical evidence on how econometric modeling and deep learning can be integrated sequentially, with the

LSTM component providing an additional residual correction to the ARIMAX baseline rather than replacing the econometric component.

Second, the study contributes additional empirical evidence to the literature on Vietnam's rice export market by incorporating Thailand's 5% broken white rice FOB price (THRICE) as the sole exogenous variable in the ARIMAX model. The findings indicate that THRICE provides statistically relevant information within the estimated ARIMAX specification and contains predictive information relevant to modeling Vietnam's 5% broken white rice FOB price (VNRICE). This finding supports the relevance of incorporating information from a closely related regional benchmark market into agricultural commodity price forecasting. However, the role of THRICE identified in this study is limited to its statistical and predictive relevance within the forecasting framework and does not establish a causal relationship, long-run market integration, or a formally identified price-transmission mechanism between the Thai and Vietnamese rice markets.

Third, the study contributes to the empirical evaluation of hybrid forecasting models by applying a structured assessment framework rather than relying solely on whether the Hybrid model reports lower average forecast errors. The empirical procedure encompasses data preprocessing, stationarity testing, ARIMAX model selection and diagnostic checking, residual-based Hybrid ARIMAX-LSTM construction, and out-of-sample forecast evaluation using MAE, RMSE, MAPE, and SMAPE. The evaluation further incorporates Random Walk benchmark models, month-by-month forecast-error comparison, and the Diebold-Mariano test with the Harvey-Leybourne-Newbold correction to assess whether the observed differences in forecasting accuracy are statistically significant. By combining multiple error measures, simple forecasting benchmarks, and formal statistical comparison within the same empirical setting, the study provides a more comprehensive basis for evaluating whether numerical improvements from a hybrid model are also sufficiently robust to support claims of forecasting superiority.

The empirical findings reinforce the importance of this distinction. Although the Hybrid ARIMAX-LSTM model achieves lower average forecast errors than the standalone ARIMAX model across the reported evaluation metrics, the Diebold-Mariano tests do not establish a statistically significant difference in forecasting accuracy at the 5% level. Accordingly, the contribution of the Hybrid framework in this study should be understood as evidence of incremental improvement in observed forecasting performance rather than evidence

that the addition of a deep-learning component necessarily produces statistically significant or universally superior forecasts. This finding provides a more cautious perspective on hybrid-model performance and highlights the importance of distinguishing improvements in point-forecast metrics from statistically established differences in predictive accuracy.

Overall, the theoretical contribution of this study lies not in proposing a new hybrid forecasting architecture or claiming the universal superiority of Hybrid ARIMAX-LSTM, but in providing context-specific empirical evidence on the application and rigorous evaluation of a sequential, residual-based ARIMAX-LSTM framework to Vietnam's benchmark rice export price forecasting. By integrating information from a relevant regional benchmark market with residual-based deep learning and evaluating the resulting forecasts through multiple accuracy measures, benchmark models, and formal statistical comparison, the study contributes an empirically grounded perspective on both the potential and the limitations of hybrid econometric-deep learning approaches in agricultural commodity price forecasting.

5.2.2. Practical Contributions

From a practical perspective, this study provides a quantitative reference for government agencies, rice-exporting enterprises, and research institutions involved in monitoring and assessing short-term movements in Vietnam's rice export prices. In particular, the forecasts for April 2026 to March 2027 provide an empirical basis for interpreting the short-term outlook of Vietnam's 5% broken white rice FOB price, while the forecasting framework demonstrates how historical price information and information from a relevant regional benchmark market can be incorporated into a systematic forecasting process. For government agencies, the forecasting results can provide an additional quantitative input for short-term market monitoring and the assessment of alternative market conditions. Rather than serving as a direct basis for policy intervention, model-based forecasts can complement updated information on domestic production, international supply and demand, trade-policy developments, and other relevant market conditions. The gradual upward price trajectory projected through March 2027 should therefore be interpreted as a model-based short-term outlook under the information and forecasting structure employed in this study, rather than as a guarantee that prices will continue to increase. This distinction is particularly important in a market where unexpected changes in production, weather conditions, trade policies, logistics, or international supply and demand may alter price conditions beyond those reflected in historical price data.

For rice-exporting enterprises, the forecasting results can serve as one quantitative input for short-term procurement and business planning, pricing discussions, inventory management, contract preparation, and price-risk assessment. A particularly relevant practical finding is that Thailand's 5% broken white rice FOB price (THRICE), which is incorporated as the sole exogenous variable in the ARIMAX model, provides useful predictive information for forecasting Vietnam's benchmark rice export price within the empirical specification employed in this study. Accordingly, exporters may benefit from incorporating movements in the Thai benchmark price into their regular market-monitoring activities rather than relying exclusively on domestic price information. Other information concerning Thailand's production conditions, inventories, export policies, and broader market developments may also be monitored as contextual market information; however, these indicators are not separately included as explanatory variables in the empirical model developed in this study. Therefore, their relevance should be distinguished from the model-based predictive role identified for THRICE.

The study also provides practical insight into how forecasting models should be evaluated before being incorporated into decision-support processes. Over the 38-month out-of-sample evaluation period, the Hybrid ARIMAX-LSTM model achieved lower average forecast errors than the standalone ARIMAX model across MAE, RMSE, MAPE, and SMAPE. Relative to ARIMAX, the Hybrid model reduced MAE by 4.38%, RMSE by 0.94%, MAPE by 4.32%, and SMAPE by 3.86%. However, the Diebold-Mariano tests with the Harvey-Leybourne-Newbold correction did not indicate statistically significant differences in forecast accuracy at the 5% level. Therefore, the practical implication is not that the Hybrid ARIMAX-LSTM model should automatically replace ARIMAX, but that forecasting models should be selected and monitored on the basis of multiple accuracy measures, formal forecast-comparison tests, and performance across different evaluation periods rather than on numerical error reductions or model complexity alone.

Furthermore, the forecasting procedure developed in this study provides a practical reference for establishing a periodically updated forecasting process. As new monthly observations become available, the dataset can be updated, the models can be re-estimated where appropriate, forecast errors can be reassessed, and model performance can be compared over time. Such periodic evaluation is important because the forecasting performance observed in the current dataset may not remain unchanged under different market conditions. The inclusion of benchmark models and formal forecast-accuracy comparison also provides

a practical basis for assessing whether a more complex forecasting approach continues to offer sufficient incremental value as new observations become available. Accordingly, the Hybrid ARIMAX-LSTM model should be regarded as a decision-support approach whose performance requires continued empirical evaluation rather than as a permanently superior forecasting method.

Overall, the practical contribution of this study lies in translating the empirical forecasting results into a structured source of quantitative information for short-term market monitoring and decision support. For government agencies, the results may complement broader market information used in monitoring Vietnam's rice export market. For exporting enterprises, the forecasts and the predictive information contained in the Thai benchmark price may support short-term commercial planning and market assessment. For research institutions and analysts, the forecasting procedure provides a reproducible framework that can be periodically updated and re-evaluated as new market observations become available. However, because the model forecasts the benchmark FOB price of Vietnam's 5% broken white rice rather than firm-specific transaction prices, production volumes, or individual contract prices, its results should be combined with current commercial and market information before being applied to specific operational decisions.

5.3. Policy Recommendations

The findings of this study provide a quantitative basis for several practical recommendations related to production coordination, export activities, and forecasting practices in Vietnam's rice sector. These recommendations are derived from the empirical forecasting results together with the market characteristics and practical issues discussed in the preceding chapters. Since the empirical model is designed primarily for price forecasting rather than causal analysis, the recommendations should not be interpreted as direct policy effects inferred from the estimated model. Instead, the forecasting results are intended to complement current market, production, and commercial information in supporting short-term planning and decision-making. In particular, the recommendations should be interpreted within the empirical scope of the study, which forecasts the benchmark FOB price of Vietnam's 5% broken white rice rather than production volumes, farm-gate prices, firm-specific transaction prices, or broader market outcomes.

5.3.1. Recommendations for Production Management

From a production-management perspective, the forecast of Vietnam's 5% broken white rice FOB price should be used primarily as supplementary market information for coordinating production-related and procurement planning rather than as a direct signal for expanding or reducing rice production. The model forecasts a benchmark export price and does not directly forecast farm-gate prices, production volumes, cultivated areas, or yields. Therefore, production decisions should not be determined solely by movements in the forecasted FOB price.

For government authorities and organizations involved in rice-sector planning, short-term export price forecasts can provide an additional market reference when assessing expected export conditions and coordinating information among production, procurement, processing, and export activities. However, these forecasts should be considered together with current information on crop conditions, expected harvest volumes, domestic supply, export demand, trade-policy developments, and other relevant market developments. This combined approach can help reduce the risk of production and procurement decisions being based solely on historical price movements or a single forecast indicator. Given that the forecasting model does not directly estimate the effects of these production and market factors, such information should be treated as complementary inputs to the planning process rather than as variables whose effects have been established by the empirical model.

For rice-processing and exporting enterprises, short-term price forecasts may also support the coordination of raw-material procurement, storage requirements, and processing schedules with expected export-market conditions. In particular, the gradual upward price trajectory projected through March 2027 should not automatically be interpreted as a signal to expand procurement, storage, or production-related activities, because actual procurement decisions remain dependent on available supply, product quality, contractual demand, storage capacity, and prevailing commercial conditions. Moreover, the projected price path represents a model-based short-term outlook and may change if new market information or unexpected shocks emerge. Forecast information should therefore function as one component of the planning process rather than as a deterministic trading or production signal.

In addition, production management should remain aligned with the broader development of Vietnam's rice export sector toward quality, product differentiation, traceability, and compliance with destination-market requirements. Improving the consistency of rice quality

and strengthening coordination between production, processing, and export activities can help enterprises respond more effectively to market requirements while reducing excessive dependence on short-term price competition. These considerations extend beyond the variables directly estimated in the forecasting model and should therefore be understood as broader management implications rather than direct empirical effects established by the model. Accordingly, the practical role of the forecasting model at the production-management level is to improve market awareness and coordination rather than to prescribe specific production decisions.

5.3.2. Recommendations for Export Activities

For rice-exporting enterprises, the forecasting results can be used as one quantitative decision-support input for short-term business planning, procurement assessment, pricing discussions, inventory management, export-contract preparation, and price-risk assessment. The forecasts should not be interpreted as predetermined transaction prices or direct trading signals, because actual export prices may differ from the benchmark FOB series according to rice quality, contract specifications, delivery conditions, destination markets, exchange-rate movements, logistics conditions, and other commercial factors. In particular, the gradual upward price trajectory projected from April 2026 to March 2027 should be interpreted as a model-based short-term outlook rather than as a predetermined path for actual transaction prices.

Accordingly, enterprises should combine model-based forecasts with current quotations, customer demand, contract conditions, inventory positions, logistics information, and developments in destination markets before making operational decisions. Scenario-based planning may also be useful when market uncertainty is elevated. Instead of relying exclusively on a single point forecast, firms can consider alternative conditions in which market prices remain stable, increase, or decline, and assess the implications of these scenarios for procurement, inventory, pricing, and contract negotiations. This approach is particularly relevant because the forecast path is conditional on the information and model structure available at the time of forecasting, while unexpected market developments may cause realized prices to deviate from the projected trajectory.

A particularly relevant finding of this study is the predictive role of Thailand's 5% broken white rice FOB price (THRICE) within the ARIMAX specification. This finding suggests that movements in the Thai benchmark price should form part of the regular market

information monitored by Vietnamese rice exporters when assessing the short-term outlook for Vietnam's benchmark export price. The role identified in this study is predictive rather than causal; therefore, changes in THRICE should be treated as a market reference rather than as evidence that Thai price movements directly cause changes in Vietnamese prices. Accordingly, THRICE may be incorporated into regular market-monitoring systems as one relevant predictive indicator, but exporters should avoid using movements in the Thai benchmark as an isolated basis for pricing, procurement, or contracting decisions.

Market monitoring may extend beyond the Thai benchmark price to information concerning production conditions, harvest progress, inventories, export-policy developments, and demand conditions in major competing and destination markets. However, these indicators should be distinguished from THRICE: they provide broader contextual information for commercial assessment but are not separately included as explanatory variables in the empirical model developed in this study. This distinction is important to avoid attributing to the forecasting model information that it does not explicitly estimate. Exchange-rate movements, freight and logistics costs, and changes in contract or destination-market conditions should likewise be monitored separately when enterprises translate benchmark-price forecasts into operational decisions, since these factors may affect actual commercial outcomes but are not explicitly modeled in the forecasting specification.

From a forecasting-use perspective, enterprises should also periodically reassess the performance of the forecasting model rather than assume that the Hybrid ARIMAX-LSTM specification will remain the preferred model under all market conditions. Although the Hybrid model achieves lower average forecast errors than ARIMAX in the current evaluation sample, the Diebold-Mariano tests with the Harvey-Leybourne-Newbold correction do not establish a statistically significant difference in forecasting accuracy at the 5% level. Therefore, the practical choice of forecasting model should be based on updated out-of-sample performance, forecast stability, and operational usefulness rather than on model complexity alone.

In addition to short-term price monitoring, Vietnamese rice exporters should continue to strengthen quality- and value-oriented competitiveness rather than relying predominantly on price-based competition. This direction is consistent with the market developments discussed in Chapter 2, particularly the increasing importance of rice quality, food safety, traceability, product differentiation, and compliance with importing-market requirements. For

enterprises, this implies strengthening quality control, traceability, product positioning, and the ability to meet the specifications of different destination markets. Price forecasting can support the timing and planning of commercial activities, but sustainable export competitiveness ultimately depends on factors extending beyond the benchmark FOB price itself. Accordingly, the forecasting framework should be viewed as a complementary commercial-planning tool within a broader export strategy that integrates price information with product quality, market requirements, contractual conditions, and destination-market developments.

5.3.3. Forecasting System and Strategic Planning

Based on the findings of this study, rice export price forecasting should be treated as a periodically updated decision-support process rather than a one-time forecasting exercise. Since the empirical dataset is constructed at a monthly frequency, new observations can be incorporated as they become available, after which the models can be re-estimated and their forecasting performance reassessed. A practical forecasting process should therefore follow a recurring sequence of data updating, model re-estimation, forecast generation, forecast-error monitoring, and model comparison. This rolling evaluation process is particularly important because the relative performance of forecasting models may change as new observations are introduced and market conditions evolve.

Model selection should be based primarily on out-of-sample forecasting performance rather than model complexity alone. In the current 38-month out-of-sample evaluation period, the Hybrid ARIMAX-LSTM model achieved lower average forecast errors than the standalone ARIMAX model across MAE, RMSE, MAPE, and SMAPE. However, the improvements were relatively modest, particularly for RMSE, and the Diebold-Mariano tests with the Harvey-Leybourne-Newbold correction did not establish a statistically significant difference in forecasting accuracy at the 5% level. This result indicates that a more complex hybrid model should not automatically be assumed to provide substantially better forecasts under all evaluation criteria or future market conditions. Forecasting systems should therefore retain appropriate benchmark models and periodically compare their performance using consistent datasets, forecast horizons, and evaluation measures. Simple benchmark models, such as the Random Walk specifications employed in this study, should also be retained where appropriate to determine whether more complex models continue to provide meaningful forecasting gains. Formal forecast-accuracy comparison should complement conventional error metrics when sufficient out-of-sample observations are available.

Where prediction intervals are incorporated into the final forecasting framework, point forecasts should be reported together with measures of forecast uncertainty. This is particularly important because uncertainty generally increases as the forecasting horizon extends. In the present forecasting framework, the widening prediction intervals over the future horizon illustrate the increasing uncertainty associated with multi-step recursive forecasting. Prediction intervals can help users distinguish between the central forecast and the range of plausible future outcomes, thereby reducing the risk of treating a single forecast value as certain. The usefulness of interval forecasts should also be evaluated using appropriate measures of interval quality, rather than relying solely on point-forecast error metrics.

For strategic planning, forecast outputs should be translated into alternative planning scenarios rather than treated as deterministic projections. The gradual upward price trajectory projected from April 2026 to March 2027 can serve as a baseline model-based outlook, while alternative scenarios may be considered to reflect possible deviations arising from changing market conditions. Such scenario-based use is more appropriate for strategic planning than relying exclusively on a single forecast path, particularly when decisions involve procurement, inventory, contracting, or other activities that may be affected by forecast uncertainty.

Future forecasting systems may also consider a broader set of candidate exogenous variables, such as exchange rates, logistics indicators, weather-related information, production and inventory measures, or relevant policy indicators. However, these variables should not be included merely because they are theoretically related to rice prices. Each candidate variable should first be evaluated to determine whether it provides incremental predictive information beyond the variables already included in the model, particularly THRICE. Variable selection should therefore be guided by both economic relevance and demonstrated out-of-sample forecasting value, while also considering multicollinearity, data availability, and the risk of overfitting. This approach would help avoid unnecessary model complexity and reduce the risk of adding variables that do not improve out-of-sample forecasting performance.

Data development should therefore focus not simply on constructing the largest possible database, but on improving data consistency, timeliness, comparability, and relevance for forecasting purposes. Cooperation among government agencies, research institutions,

universities, and enterprises may facilitate the regular collection and sharing of market information and support more systematic evaluation of forecasting models. Where sufficient data and institutional capacity are available, forecasting outputs may subsequently be incorporated into market-monitoring or early-warning systems, with model-based signals interpreted alongside current market intelligence and expert assessment. Any such system should clearly distinguish between variables formally incorporated into the forecasting model and broader contextual indicators used only for market interpretation, thereby avoiding the attribution of predictive effects to information that has not been empirically evaluated within the model.

Finally, forecasting tools should be regarded as complements to, rather than substitutes for, market expertise and managerial judgment. Unexpected policy changes, climatic events, geopolitical disruptions, or other structural shocks may create market conditions that differ substantially from historical patterns. A practical forecasting system should therefore combine quantitative forecasts with updated qualitative and market information and should be capable of reassessment when new conditions emerge. Accordingly, the practical value of the forecasting framework lies not in identifying a permanently superior model, but in establishing a transparent and adaptable process through which competing models can be continuously evaluated, updated, and used alongside current market information to support short-term monitoring and strategic planning.

5.4. Research Limitations

Although this study achieves its research objectives, several limitations should be acknowledged when interpreting the findings. First, the study uses 268 monthly observations from December 2003 to March 2026, which provides a consistent time series but remains relatively limited for the LSTM component. In addition, Thailand's 5% broken white rice FOB price (THRICE) is employed as the sole exogenous variable; therefore, the model does not separately evaluate the predictive contribution of other potentially relevant market information. Second, the Hybrid ARIMAX-LSTM model adopts a sequential residual-learning structure in which LSTM learns from the residuals generated by ARIMAX; however, the residual component may not represent a complete separation between linear and nonlinear information, and forecasting performance remains dependent on model configuration, hyperparameter selection, and the treatment of future values of the exogenous variable. Third, the out-of-sample evaluation is based on only 38 monthly observations. Although the Hybrid

ARIMAX-LSTM model produced lower MAE, RMSE, MAPE, and SMAPE values than the standalone ARIMAX model, the Diebold-Mariano tests with the Harvey-Leybourne-Newbold correction yielded p-values above 0.05 across the evaluated loss functions and therefore did not establish a statistically significant difference in forecasting accuracy between the two models at the 5% level. Accordingly, the lower forecast errors should be interpreted as better observed average forecasting performance within the current evaluation sample rather than as conclusive evidence of statistical superiority. Furthermore, the use of a single train-test design may make the results sensitive to the specific evaluation period and prevailing market conditions. Fourth, the study is designed primarily for forecasting rather than causal inference; accordingly, the predictive relevance of THRICE should not be interpreted as evidence of causality, long-run market integration, or a formally identified price-transmission mechanism between Thailand and Vietnam. Finally, the empirical scope is limited to Vietnam's 5% broken white rice benchmark FOB price and a short-term forecasting horizon extending through March 2027; therefore, the findings should not be directly generalized to other rice varieties, transaction-level prices, other markets, or longer-term forecasting horizons. Overall, these limitations define the scope within which the forecasting results should be interpreted and provide a basis for the future research directions discussed in the following section.

5.5. Directions for Future Research

Based on the findings and limitations of this study, future research should focus on improving the predictive robustness, empirical reliability, and practical applicability of rice export price forecasting models while remaining consistent with the scope of benchmark FOB price forecasting.

First, future studies may consider incorporating a broader set of exogenous variables beyond Thailand's 5% broken white rice FOB price. Potential candidates may include exchange rates, logistics costs, production and inventory indicators, import demand from major markets, weather-related variables, and trade-policy indicators. However, these variables should not be included solely because they are theoretically related to rice prices. Their inclusion should be empirically evaluated to determine whether they provide additional predictive information beyond the existing model specification, particularly beyond the information already captured by THRICE. Appropriate variable-selection procedures should also be employed to limit multicollinearity, redundant information, and overfitting.

Second, future research should consider improving data coverage and frequency where reliable and consistent information is available. The present study is based on monthly benchmark FOB price data, which supports long-term consistency but may not fully capture short-lived market adjustments. Future studies may therefore evaluate whether higher-frequency data, such as weekly observations, provide additional forecasting value without sacrificing data quality and comparability. Expanding the number of observations may also provide a stronger empirical basis for training and evaluating the LSTM component, particularly when more complex model configurations are considered.

Third, future studies should broaden the comparison of forecasting methods under a consistent empirical framework. In addition to ARIMAX and Hybrid ARIMAX-LSTM, alternative approaches such as SARIMAX, GRU, XGBoost, VAR/VECM, GARCH, or ensemble methods may be evaluated. The purpose of such comparisons should not be to identify the most complex model, but to determine which model performs most reliably under the same dataset, forecast horizon, and evaluation design. The present study shows that the Hybrid ARIMAX-LSTM model achieved lower average forecast-error metrics than the standalone ARIMAX model within the current evaluation sample, although the difference was not statistically significant at the 5% level according to the Diebold-Mariano tests with the Harvey-Leybourne-Newbold correction. Future comparisons could therefore examine whether this observed numerical advantage persists across alternative models, datasets, forecast horizons, and evaluation periods.

Fourth, future research should strengthen the robustness and statistical assessment of forecast evaluation. Given that the present study is based on a 38-month out-of-sample testing period, future studies should consider longer testing periods and multiple forecast windows where sufficient data are available. Rather than relying on a single train-test split, researchers may employ rolling-origin evaluation and repeated out-of-sample testing to assess whether the observed performance differences remain stable across different periods and market conditions. Since the present study already applies the Diebold-Mariano test with the Harvey-Leybourne-Newbold correction, future research should extend this statistical assessment across longer out-of-sample periods, multiple forecast windows, and alternative evaluation designs. Such extensions would help determine whether the statistically non-significant difference observed in the current 38-month evaluation period remains consistent when a larger and more diverse set of forecast errors becomes available. These approaches would

provide stronger evidence regarding the robustness and stability of the observed forecasting performance differences.

Fifth, future studies should investigate the relationship between Vietnam's rice prices and those of competing exporting countries more explicitly. The present study uses Thailand's benchmark rice price as an exogenous predictor within a forecasting framework and does not attempt to establish causal relationships or formal price-transmission mechanisms. Future research could therefore apply cointegration analysis, Granger causality testing, and Vector Error Correction Models (VECM) to examine long-run relationships, short-run adjustment processes, and price transmission among major rice-exporting markets. Such analyses should be treated as extensions with an explanatory focus distinct from the forecasting objective of the present study. This would complement the predictive focus of the present study with greater explanatory insight.

Future research may also extend the empirical scope to other benchmark rice categories or exporting countries where sufficiently consistent data are available. For example, incorporating additional benchmark price series from major rice-exporting countries could help evaluate whether the predictive role observed for THRICE remains robust in a broader international market setting. Similarly, extending the analysis to other standardized rice categories may help assess whether the forecasting framework performs consistently across different market segments. Any such extension should preserve comparability in price definitions, product quality, quotation conditions, data frequency, and forecast horizons to ensure that differences in model performance are not driven primarily by inconsistencies in the underlying data.

Finally, future research should move toward periodically updated and empirically validated forecasting systems rather than one-time model estimation. As new market data become available, models should be re-estimated, forecast errors monitored, and performance reassessed. Where prediction intervals are included in the final forecasting framework, future operational applications may also report these intervals together with point forecasts to communicate forecast uncertainty more clearly. The quality and calibration of these intervals should also be evaluated over repeated forecasting periods where sufficient realized observations become available. A simple decision-support interface or dashboard may be considered as a later-stage application, but only after the forecasting framework has demonstrated stable performance across repeated real-time evaluations.

Overall, future research should prioritize forecasting reliability, transparency, and empirical validation rather than methodological complexity alone. Expanding relevant predictive information, strengthening robustness testing, comparing alternative models under consistent conditions, and distinguishing forecasting relationships from causal market relationships will provide a more solid basis for advancing research on Vietnam's rice export price forecasting.

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APPENDICES

APPENDIX A

Stata Commands Used for Data Preparation

```
</> stata
```

```
gen time = monthly(Date,"YM")
```

```
format time %tm
```

```
tsset time
```

```
. gen time = monthly(Date,"YM")
```

```
. 
```

```
. format time %tm
```

```
. 
```

```
. tsset time
```

```
Time variable: time, 2003m12 to 2026m3  
Delta: 1 month
```

Figure A.1. Time-series declaration in Stata

Source: Author's computation using Stata 17.

APPENDIX B

Augmented Dickey-Fuller (ADF) Unit Root Test

Appendix B.1 Vietnam Rice Price

`</> stata`

dfuller VN

`. dfuller VN`

Dickey-Fuller test for unit root Number of obs = **267**
Variable: VN Number of lags = **0**

H0: Random walk without drift, d = 0

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-2.464	-3.459	-2.879	-2.570

Mackinnon approximate *p*-value for Z(t) = **0.1244**.

Figure B.1. Augmented Dickey-Fuller test for Vietnam rice price

Source: Author's computation using Stata 17.

Appendix B.2 Thailand Rice Price

`</> stata`

dfuller Thai

```
. dfuller Thai
```

Dickey-Fuller test for unit root
Variable: **Thai**

Number of obs = **267**
Number of lags = **0**

H0: Random walk without drift, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-2.837	-3.459	-2.879	-2.570

Mackinnon approximate p -value for Z(t) = **0.0531**.

Figure B.2. Augmented Dickey-Fuller test for Thailand rice price

Source: Author's computation using Stata 17.

Appendix B.3 First Difference of Vietnam Rice Price

```
</> stata
```

```
gen d_VN=D.VN
```

```
dfuller d_VN
```

```
. dfuller d_VN
```

Dickey-Fuller test for unit root
Variable: **d_VN**

Number of obs = **266**
Number of lags = **0**

H0: Random walk without drift, $d = 0$

	Test statistic	Dickey-Fuller critical value		
		1%	5%	10%
Z(t)	-8.473	-3.459	-2.879	-2.570

Mackinnon approximate p -value for Z(t) = **0.0000**.

Figure B.3. ADF test after first differencing of Vietnam rice price

Source: Author's computation using Stata 17.

Appendix B.4 First Difference of Thailand Rice Price

```
</> stata

gen d_Thai=D.Thai

dfuller d_Thai

. dfuller d_Thai

Dickey-Fuller test for unit root          Number of obs  = 266
Variable: d_Thai                          Number of lags  =  0

H0: Random walk without drift, d = 0

          Test          _____ Dickey-Fuller
          statistic          1%      critical value      5%      10%
-----
Z(t)      -11.698      -3.459      -2.879      -2.570
-----

MacKinnon approximate p-value for Z(t) = 0.0000.
```

Figure B.4. ADF test after first differencing of Thailand rice price

Source: Author's computation using Stata 17.

APPENDIX C

ARIMA Model Estimation

Appendix C.1 Thailand Rice Price ARIMA(0,1,1)

```
</> stata
```

```
arima Thai, arima (0,1,1)
```

```
estat ic
```

```

. . arima Thai, arima(0,1,1)

(setting optimization to BHHH)
Iteration 0: log likelihood = -1287.3201
Iteration 1: log likelihood = -1286.508
Iteration 2: log likelihood = -1286.2829
Iteration 3: log likelihood = -1286.1671
Iteration 4: log likelihood = -1286.0813
(switching optimization to BFGS)
Iteration 5: log likelihood = -1286.016
Iteration 6: log likelihood = -1285.7729
Iteration 7: log likelihood = -1285.749
Iteration 8: log likelihood = -1285.7262
Iteration 9: log likelihood = -1285.7253
Iteration 10: log likelihood = -1285.7253

ARIMA regression

Sample: 2004m1 thru 2026m3      Number of obs   =      267
                                Wald chi2(1)          =     327.38
Log likelihood = -1285.725      Prob > chi2       =     0.0000


```

D.Thai	Coefficient	OPG std. err.	z	P> z	[95% conf. interval]	
Thai						
_cons	.6575567	3.044199	0.22	0.829	-5.308963	6.624076
ARMA						
ma						
L1.	.4114565	.0227405	18.09	0.000	.3668859	.456027
/sigma	29.84947	.3971951	75.15	0.000	29.07098	30.62795

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

```

. .
. . estat ic

Akaike's information criterion and Bayesian information criterion

```

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	267	.	-1285.725	3	2577.451	2588.212

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

Figure C.1. ARIMA estimation results for Thailand rice price

Source: Author's computation using Stata 17.

Appendix C.2 Vietnam Rice Price ARIMAX(1,1,1)

`</> stata`

`arima VN Thai, arima (1,1,1)`

`estat ic`

```

. . arima VN Thai, arima(1,1,1)

```

```

(setting optimization to BHHH)
Iteration 0: log likelihood = -1167.8268
Iteration 1: log likelihood = -1159.9593
Iteration 2: log likelihood = -1157.9531
Iteration 3: log likelihood = -1157.0208
Iteration 4: log likelihood = -1156.3533
(switching optimization to BFGS)
Iteration 5: log likelihood = -1156.0368
Iteration 6: log likelihood = -1155.4583
Iteration 7: log likelihood = -1155.3884
Iteration 8: log likelihood = -1155.3573
Iteration 9: log likelihood = -1155.3541
Iteration 10: log likelihood = -1155.354

```

ARIMA regression

Sample: **2004m1** thru **2026m3**

Log likelihood = **-1155.354**

```

Number of obs      =      267
Wald chi2(3)       =      237.74
Prob > chi2        =      0.0000

```

		OPG		z	P> z	[95% conf. interval]	
D.VN	Coefficient	std. err.					
VN	Thai						
	D1.	.116168	.0232337	5.00	0.000	.0706307	.1617053
	_cons	.4836698	2.210587	0.22	0.827	-3.849001	4.816341
ARMA	ar						
	L1.	.2262003	.0853413	2.65	0.008	.0589344	.3934661
	ma						
	L1.	.3900288	.0899096	4.34	0.000	.2138091	.5662484

/sigma	18.31134	.4821727	37.98	0.000	17.3663	19.25638
--------	----------	----------	-------	-------	---------	----------

Note: The test of the variance against zero is one sided, and the two-sided confidence interval is truncated at zero.

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	267	.	-1155.354	5	2320.708	2338.644

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

Figure C.2. ARIMAX estimation results for Vietnam rice price

Source: Author's computation using Stata 17.

APPENDIX D

Model Selection Criteria

Appendix D.1 Information Criteria

```
. . estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	267	.	-1285.725	3	2577.451	2588.212

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

Figure D.1. AIC and BIC values for the ARIMA(0,1,1) model of Thailand rice price

Appendix D.2 Information Criteria

```
. estat ic
```

Akaike's information criterion and Bayesian information criterion

Model	N	ll(null)	ll(model)	df	AIC	BIC
.	267	.	-1155.354	5	2320.708	2338.644

Note: BIC uses N = number of observations. See [\[R\] BIC note](#).

Figure D.2. AIC and BIC values for the ARIMAX(1,1,1) model

Source: Author's computation using Stata 17.

APPENDIX E

Residual Diagnostic Test

```
</> stata
```

```
wntestq residVN
```

```
. wntestq residVN
```

Portmanteau test for white noise

Portmanteau (Q) statistic =	50.9357
Prob > chi2(40) =	0.1152

Figure E.1. Portmanteau test for residual white noise

Source: Author's computation using Stata 17.